CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

REVISION FLASH CARDS &
MANY MORE <<< CLICK

Exercise - I
The Living World

1. Match column-I with column-II and select the correct option :-

Column-I	Column-II
(a) Family	(i) *tuberosum*
(b) Kingdom	(ii) Polymoniales
(c) Order	(iii) *Solanum*
(d) Specific epithet	(iv) Plantae
(e) Genus	(v) Solanaceae

(1) (a)–v, (b)–iv, (c)–ii, (d)–i, (e)–iii
(2) (a)–v, (b)–iv, (c)–i, (d)–iii, (e)–ii
(3) (a)–i, (b)–ii, (c)–iii, (d)–v, (e)–iv
(4) (a)–ii, (b)–i, (c)–iii, (d)–iv, (e)–v

BLW169

2. System of classification of angiosperms or plants which is based on maximum number of existing phenotypic characters is :-
(1) Artificial system
(2) Natural nonphylogenetic system
(3) nonphylogenetic system
(4) Natural sexual system

BLW170

3. Choose the correct option from following :-
(1) *Mangifera*, is a name of species
(2) Polymoniales, is a name of family
(3) Dicotyledonae, is a name of division
(4) *Triticum*, is a name of genus

BLW171

4. If in a taxonomic hierarchy, if *Triticum aestivum* is placed in Poales then what would be the correct taxonomic category for *Panthera tigris* at the same level of poales :
(1) Mammalia (2) Chordata
(3) Carnivora (4) Felidae

BLW172

5. Intrinsic growth is not a characteristic feature of :
(1) Plants (2) Animals
(3) Coral reef (4) Bryophytes

BLW173

6. Which set of organisms multiply by fragmentation?
(1) Fungi, all algae, protonema of mosses
(2) Fungi, filamentous algae, protonema of mosses
(3) Prothallus, fungi, *Planaria*
(4) All algae, fungi, bacteria

BLW174

7. Growth cannot be taken as a defining property of living organisms because
(1) All living organism do not show growth.
(2) Non-iving things also grow from inside.
(3) Non-living things also grow from outside.
(4) Some living organism does not exhibit the process of reproduction.

BLW175

8. The isolated metabolic reaction outside the body of an organism, performed in a test tube (in-vitro) is :-
(1) Living
(2) Non-living
(3) Neither living nor non-living
(4) Either living or non-living

BLW176

9. Which of the following is a defining property of all living organisms :-
(1) Reproduction (2) Growth
(3) Consciousness (4) Self consciousness

BLW177

10. From the given below list, how many are taxa. Dog, Mammals, Cat, Wheat, Rice, Wood, Rock, Branches, Human, Insect
(1) Six (2) Three (3) Ten (4) Seven

BLW178

11. Read the following statements carefully about binomial nomenclature and identify the incorrect statement: -
(1) Biological names are generally in Latin and printed in italics.
(2) The first component in a biological name represents the species while second component denotes the genus.
(3) Both the words in a biological name, when hand written, are separately underlined or printed in italics to indicate their Latin origin
(4) The first word denoting genus starts with a capital letter while specific epithet starts with a small letter

BLW179

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

12. Which of the following statements are correct?
(A) Genera are aggregates of closely related species.
(B) Families are characterised on the basis of vegetative characters only.
(C) Order and other higher categories are identified on the basis of aggregates of characters.
(1) A and B (2) B and C
(3) A, B, C (4) A and C
BLW180

13. According to hierarchical system of classification which of the following is correct
(A) Similarities decrease from species to kingdom
(B) Similarities increase from kingdom to species
(C) Dissimilarities decrease from division to species
(D) Similarities decrease from species to class
(1) Only A, B, C (2) Only B, C, D
(3) Only B, C (4) A, B, C, D all
BLW181

14. In taxonomic hierarchy, given below organisms are representatives of how many families
Wheat, Potato, Brinjal, Mango
(1) Two (2) Three
(3) Four (4) One
BLW182

15. Which of the following statement is not true with respect to binomial nomenclature?
(1) No names are recognised prior to those used by Linnaeus in 1785 in the 9th edition of Systema Naturae
(2) Generic name, species name and author citation collectively form binomial epithet
(3) Principle of priority is the most important of all the rules of ICBN
(4) If a species name has two or more words in its name, a hyphen is put between these
BLW183

16. Mark the **correct** statement with respect to living organisms
(1) Growth and cellular organisation are considered as defining properties of living organisms
(2) Reproduction can be an all inclusive defining characteristic of living organisms
(3) Living organisms are self replicating, self regulating and evolving interacting systems
(4) Isolated metabolic reactions *in vitro* are living things
BLW184

17. What is **incorrect** for taxonomic hierarchy?
(1) There are 7 obligate categories
(2) Higher the category, fewer the number of organism
(3) Lower the category, more the number of common features
(4) Greatest similarity in structure is found in between members of a species
BLW185

18. Select incorrect statement
(1) Growth occurs by synthesis of protoplasmic and apoplasmic substances
(2) Reproduction is synonymous with growth in unicellular organisms
(3) Isolated metabolic reactions *in-vitro* are surely living things
(4) Consciousness is the most obvious and technically complicated feature of all living organisms
BLW186

19. Select **correct** statement with respect to binomial nomenclature -
(1) Biological names are generally in italics
(2) ltalics printing of scientific names indicate their Latin origin
(3) This system was given by father of taxonomy in Historia Plantarum
(4) Binomial epithet consists of three Latin words
BLW187

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	4	3	3	2	3	3	3	4	2	4	4	2	1
Question	16	17	18	19											
Answer	3	2	3	2											

Exercise - II
The Living World

1. Nomenclature is must in taxonomy, which of the following is not correct about scientific nomenclature ?
(1) They ensure that each organism has only one name with two component
(2) They also ensure that such a name has not been used for any other known organism
(3) Scientific nomenclature is a standardised naming system
(4) Different countries of the world use different kinds of scientific nomenclature system

BLW188

2. Find the correct statements from the followings:-
(A) Each rank or taxon, in fact represents a unit of classification
(B) Every step in taxonomic hierarchy represents a rank or category
(C) A taxonomic category is a part of overall taxonomic hierarchy
(D) Taxonomic categories are not distinct biological entities, they are only morphological aggregates
Options :-
(1) A, B & D are correct
(2) A, C & D are correct
(3) A, B & C are correct
(4) Only A and B are correct

BLW189

3. Read the following statements.
(A) Each genus may have one or more than one specific epithet.
(B) Genera are aggregates of closely related species.
(C) Family has a group of related genera with still more number of similarities as compared to genus and species.
How many statements are correct and incorrect among these?
(1) 3 correct, 0 incorrect
(2) 2 correct, 1 incorrect
(3) 1 correct, 2 incorrect
(4) 0 correct, 3 incorrect

BLW190

4. Read the following statements :-
(i) Ernst Mayr, has been called "The Mendel of the 20th century.
(ii) All living organisms grow
(iii) Unicellular organisms multiply by cell division.
(iv) Non–living object also exhibits metabolism
How many statements is/are correct ?
Option :-
(1) Four (2) Three (3) Two (4) One

BLW191

5. **Assertion :** Growth can not be taken as defining property of living organisms.
Reason : Non living objects also grow by accumulation of materials on surface.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) (A) is correct but (R) is not correct.
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

BLW192

6. **Assertion :** There are seven obligate categories in hierarchy of taxonomy.
Reason : Other categories of similar type can be called as sub-categories.
(1) If both statements A and B arc true and statement B is the correct explanation of statement A.
(2) If both statements A and B are true but statement B is not the correct explanation of statement A.
(3) If statement A is true but statement B is false.
(4) If both statements A and B are false.

BLW193

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

7. **Statement A :** Metabolism is the sum total of anabolism and catabolism inside the cell/body.

Statement B : Diverse types of metabolic reactions occur simultaneously in a living organism.

(1) If both statements A and B are true and statement B is the correct explanation of statement A .

(2) If both statements A and B are true but statement B is not the correct explanation of statement A.

(3) If statement A is true but statement B is false.

(4) If both statements A and B are false.

BLW194

8. Match the columns :

	Column-I		Column-II
(A)	*Yeast*	(i)	Asexual spore and fragmentation
(B)	*Amoeba*	(ii)	Binary fission
(C)	Fungi	(iii)	Budding
(D)	*Planaria*	(iv)	True regeneration

(1) A-(i), B-(ii), C-(iii), D-(iv)
(2) A-(ii), B-(iii), C-(iv), D-(i)
(3) A-(i), B-(iv), C-(iii), D-(ii)
(4) A-(iii), B-(ii), C-(i), D-(iv)

BLW195

9. For taxonomic hierarchy given; select the correct match :

	Taxonomic Category	Examples
(1)	Class	Sapindales, Insecta
(2)	Order	Primata, Diptera
(3)	Genus	*Musca*, Poales
(4)	Family	*Triticum*, Muscidae

BLW196

10. Match column-I with column-II and select the correct option :-

	Column-I		Column-II
(A)	Cell wall absent	(i)	A species
(B)	Dorsal hollow neural system	(ii)	Animalia
(C)	Mango	(iii)	Chordata

(1) A-i, B-ii, C-iii (2) A-ii, B-iii, C-i
(3) A-ii, B-i, C-iii (4) A-iii, B-ii, C-i

BLW197

11. Identify the correct option having correct match of the column-I, II and III.

Column-I		Column-II		Column-III	
(A)	Wheat	(a)	*Homo*	(i)	Monocotyledonae
(B)	Man	(b)	*Musca*	(ii)	Dicotyledonae
(C)	Mango	(c)	*Mangifera*	(iii)	Insecta
(D)	Housefly	(d)	*Triticum*	(iv)	Mammalia

(1) A-d-i, B-a-iv, C-c-ii, D-b-iii
(2) A-d-i, B-c-ii, C-a-iv, D-b-iii
(3) A-d-ii, B-c-i, C-a-iv, D-b-iii
(4) A-a-iv, B-d-i, C-b-iii, D-c-ii

BLW198

12. **Assertion :** Numerical taxonomy which is carried out using computers is based on all observable characters.

Reason : Number and codes are assigned to all the characters and data are then processed.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BLW199

13. **Assertion :** *Panthera leo* is the scientific name of lion.

Reason : In this name *Panthera* indicates generic name while *leo* indicates specific epithet.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BLW200

AIM : FREE EDUCATION TO ALL APUL

14. Which of the following option is correct?

(A) Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(B) Classes with a few similar characters are assigned to a higher category called order.

(C) As we go higher from species to kingdom, the number of common characters goes on decreasing.

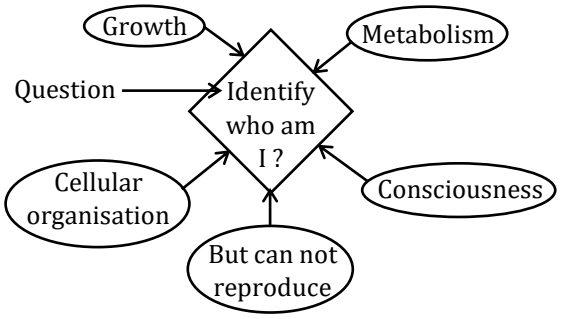
(D) Animals, mammals and dogs are different taxa at the same level

(1) A and B (2) B and C

(3) A and C (4) A and D

BLW201

15. I can perform :-



(1) A virus

(2) A living bacterium only

(3) A living animal only

(4) May be any living entity

BLW202

16. **Assertion :** Characterisation, identification, classification & nomenclature are the processes that are basic to taxonomy.

Reason : Systematics takes into account evolutionary relationships between organisms.

(1) Both Assertion & Reason are correct and the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are correct but Reason is not a correct explanation of the Assertion.

(3) Assertion is correct but the Reason is incorrect.

(4) Both Assertion & Reason are incorrect.

BLW203

17. Read the following statements and select the correct ones:

(i) Genera such as *Petunia* and *Datura* are placed in the family Brassicaceae.

(ii) *Solanum* includes species like potato and brinjal.

(iii) The similar characters are less in number in families as compared to different genera included in a family.

(1) (i) & (ii) (2) (ii) & (iii)

(3) (i) & (iii) (4) (i), (ii) & (iii)

BLW204

18. In the following statements how many are incorrect?

(i) Protonema of mosses multiply by fragmentation.

(ii) In unicellular organisms' reproduction is synonymous with growth.

(iii) Metabolic reactions cannot be demonstrated outside the body in cell free systems.

(iv) Isolated metabolic reactions are living things but not living reactions.

(1) Only (i) & (ii)

(2) Only (ii) & (iii)

(3) Only (iii) & (iv)

(4) Only (i) & (iv)

BLW205

19. **Statement I :** All living organisms present, past & future are linked to one other by the sharing of the common genetic material, but to varying degrees.

Statement II : All plants, animals, fungi and microbes exhibit metabolism.

(1) Statements-I and statement-II, both are correct.

(2) Statements-I and statement-II, both are incorrect.

(3) Only statements-I is correct.

(4) Only statements-II is correct.

BLW206

AIM : FREE EDUCATION TO ALL APUL

20. Which one of the following combination of taxa represent same category of hierarchy?
(1) Lion, leopard, tiger
(2) Potato, brinjal, grasses
(3) Petunia, Datura, Solanaceae
(4) Solanaceae, Convolvulaceae, Polymoniales

BLW207

21. Which of the given scientific name is written, as mentioned in the rules of binomial nomenclature?
(1) Rosa indica *Linn.*
(2) *Rosa Indica Linnaeus*
(3) *Rosa Indica L.*
(4) *Rosa indica* Linn.

BLW208

22. Find the odd one with respect to taxonomic hierarchy of mango
(1) Sapindaceae (2) Sapindales
(3) Dicotyledonae (4) Angiospermae

BLW209

23. The scientific nomenclature representing the species of wheat is -
(1) *Triticum* (2) *Aestivum*
(3) *aestivum* (4) *Triticum aestivum*

BLW210

24. Consider the following statements and find the **correct** one -
a. Living organisms are self replicating, evolving and self regulating interactive systems capable of responding to external stimuli
b. Wheat, monocots and plants represent taxa of the same level
c. Higher the category, greater is the difficulty of determining the relation to other taxa of same level
d. Lower the taxa, more are the characteristics that the members within the taxon share
(1) All are correct, except 'b'
(2) All are correct
(3) All are correct, except 'c'
(4) All are correct, except 'd'

BLW211

25. Fundamental elements of taxonomy are :
(1) Characterisation and classification only
(2) Characterisation, identification and classification only
(3) Characterisation, classification and nomenclature only
(4) Characterisation, identification, classification and nomenclature

BLW212

26. Which of the following statement is **incorrect**?
(1) Family is a group of related genera with still less number of similarities as compared to genus and species
(2) Families are characterised on the basis of vegetative characters only
(3) Each genus may have one or more than one specific epithets representing different organisms, but having morphological similarities
(4) Genera *Solanum*, *Datura* and *Petunia* are placed in the family Solanaceae

BLW213

27. Select the correct combination of the statements (a-d) regarding taxonomic hierarchy.
a. As we go higher from species to kingdom, the number of common characters goes on decreasing
b. Lower the taxa, more are the characteristics that the members within the taxon share
c. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level
d. Kingdom plantae comprises all plants from various phyla

The correct statements are :
(1) Only c & d (2) a, b, c
(3) b, c d (4) only a

BLW214

28. Select the incorrect statement.
(1) Isolated metabolic reactions in vitro are living reactions
(2) Human being is the only organism who has self-consciousness
(3) Reproduction is an defining property of living beings
(4) Unicellular organisms also grow by cell-division

BLW215

29. Select the **incorrect** statement
(1) A multicellular organism grows by cell division
(2) In plants growth by cell division occurs only up to a certain age
(3) In animals growth is definite
(4) Cell division occurs in certain tissues of animals to replace lost cells

BLW216

AIM : FREE EDUCATION TO ALL APUL

Hint :

Q.3 A, B are correct, C is incorrect.

Q.4 Statement ii & iii are correct.

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	3	4	2	2	4	2	2	1	1	1	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	2	2	3	1	1	4	1	4	1	4	2	2	3	2	

Exercise - I
Biological Classification

1. Which group of protists is responsible for producing neurotoxin that become concentrated in the aquatic food chain ?
(1) Chlorophyta (2) Chrysophyta
(3) Dinoflagellates (4) Slime moulds
BBC566

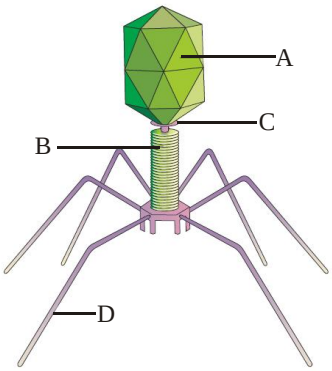
2. Choose the incorrect match :
(1) Ascomycetes - Used in genetic work
(2) Deuteromycetes - Rust and smut fungi
(3) Basidiomycetes-Sex organs are generally not formed
(4) Phycomycetes - Aquatic habitat also
BBC567

3. Rust, smut and mushroom, all the three have common characters, that they :-
(1) are pathogen
(2) are saprobes
(3) produce basidiospore
(4) bear basidiocarp
BBC568

4. Which of following microbes serve as an important biofertiliser in paddy fields ?
(1) *Glomus* (2) *Azotobacter*
(3) *Anabaena* (4) *Rhizobium*
BBC569

5. Which lichen is used for making litmus paper?
(1) *Roccella* (2) *Cetraria*
(3) *Cladonia* (4) *Usnea*
BBC570

6. Given below in the digaram of bacteriophage. In which one of the options all the four parts A, B C and D are correct ?



Options :-

	A	B	C	D
(1)	Sheath	Collar	Head	Tail fibres
(2)	Head	Sheath	Collar	Tail fibres
(3)	Collar	Tail fibres	Head	Sheath
(4)	Tail fibres	Head	Sheath	Collar

BBC571

7. Which of the following statements is wrong about lichen ?
(1) Lichens are sensitive to air pollution
(2) Algal and fungal partner live mutually
(3) Algae prepare food and fungi provide shelter, absorb water and minerals for algal partner
(4) Alage live as master & fungi live as slave
BBC572

8. Select the correct pair :-
(I) Methanogen → occurs in swampy ares
(II) Halophiles → some have pigments but they are not photosynthetic
(III) Thermoacidophiles → Mostly parasitic
Options :-
(1) I and II (2) II and III
(3) I and III (4) All of the above
BBC573

9. (A) It causes potato spindle tuber disease.
(B) It is free RNA.
(C) Molecular weight of RNA is low.
(D) Infectious agent, smaller than viruses.
The above statements are assigned to :-
(1) Viruses (2) Viroids
(3) Virulent virus (4) Mycoplasma
BBC574

10. Pili do not play a role in motility because :-
(1) They are small bristle like fibres.
(2) They help bacteria to attach rocks and host tissues.
(3) They are made up of non-contractile protein.
(4) They are found inside the surface of bacteria.
BBC575

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

11. Which one of the following pairs is mismatched :-
(1) *Penicillium* - Source of antibiotics
(2) *Albugo* - Parasitic fungi on legume
(3) *Neurospora* - Used in biochemical and genetic work
(4) *Agaricus* - Edible fungus

BBC576

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Cyanobacteria that are referred to as blue green algae are not "algae" anymore.
Reason (R) : All unicellular prokaryotes are placed in kingdom Monera and algae are placed in kingdom Plantae
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBC577

13. List some of the fungi is given below: -
Mucor, Rhizopus, Albugo, Aspergillus, Penicillium, Puccinia, Ustilago.
In how many fungi, plasmogamy is followed by karyogamy, immediately: -
(1) Two (2) Three
(3) Four (4) Five

BBC578

14. How many of the following spores are exogenous?
Zoospores, Aplanospore, Conidia, Ascospore, Basidiospore.
(1) Four (2) Three
(3) Two (4) One

BBC579

15. How many of the organisms present below are the members of the kingdom plantae according to two kingdom system of classification?
Chara, Ficus, Triticum, Mangifera, Homo sapiens, E. Coli, Salmonella
(1) 5 (2) 7 (3) 3 (4) 6

BBC580

16. Which of the following is not a feature of Bentham and Hookers classification ?
(1) It was based on actual observations
(2) It was a natural classification
(3) It was based on floral features
(4) It was a phylogenetic classification

BBC581

17. Earlier classification system include bacteria, blue green algae, fungi, mosses, ferns, gymnosperms and angiosperms together in 'Plant' because they all have :-
(1) Eukaryotic cells
(2) Nucleus
(3) Cell wall
(4) Photosynthetic ability

BBC582

18. In five kingdom system bacteria are included in Monera but acellular organism like viruses and viroid's, and lichens are included in :-
(1) Kingdom - Fungi
(2) Kingdom - Animalia
(3) Kingdom - Protista
(4) In no any kingdom

BBC583

19. Plant groups under sub kingdom embryophyta can be :
(1) Bryophyta & pteridophyta
(2) Pteridophyta & gymnosperm
(3) Gymnosperm & angiosperm
(4) Bryophyta & tracheophyta

BBC584

20. In which book Linnaeus gave the codes of nomenclature?
(1) *Species plantarum*
(2) *Philosophia botanica*
(3) *Genera plantarum*
(4) *Hortus uplandicus*

BBC585

AIM : FREE EDUCATION TO ALL APUL

21. The demerit of Bentham & Hooker's classification is :-
(1) Position of gymnospermae
(2) Position of dicotyledonae
(3) Position of monocotyledonae
(4) Both (1) & (2)
BBC586

22. Which of the following statement is correct for both blue-green algae and bacteria?
(1) Both show anaerobic respiration.
(2) Both have chlorophyll pigment.
(3) Both are devoid of true nucleus.
(4) None of the above
BBC587

23. Which group of fungi is commonly known as imperfect fungi?
(1) Phycomycetes
(2) Ascomycetes
(3) Basidiomycetes
(4) Deuteromycetes
BBC588

24. The subunit of capsid is called
(1) Core (2) Nucleotide
(3) Amino acid (4) Capsomere
BBC589

25. Which of the following is not a viral disease?
(1) AIDS and mumps
(2) Small pox and herpes
(3) Influenza
(4) Cholera
BBC590

26. Which of the following processes are involved in the reproduction of protists?
(1) Binary fission and budding
(2) Cell fusion and zygote formation
(3) Spore formation and cyst formation
(4) All of the above
BBC591

27. Prions are/have-
(1) DNA as genetic material
(2) RNA as genetic material
(3) Non-infectious
(4) Abnormal folded protein
BBC592

28. Animal like feature of slime moulds is–
(1) Formation of fruiting bodies
(2) Cellulosic cell wall around spores
(3) Absence of cell wall in plasmodium
(4) Formation of spores
BBC593

29. An archaeon that lives in extremely salty condition is referred to as a -
(1) Thermophiles
(2) Halophiles
(3) Thermo acidophiles
(4) Methanogen
BBC594

30. Eubacteria include -
(1) Bacteria and Archaebacteria.
(2) Cyano bacteria + Archaebacteria.
(3) Bacteria and Blue green algae.
(4) Bacteria + Eukaryote.
BBC595

31. All of the following diseases are caused by bacteria except :-
(1) Flu (2) Cholera
(3) Tetanus (4) Typhoid
BBC596

32. Citrus canker is a ____ disease.
(1) Bacterial (2) Viral
(3) Fungal (4) Mycoplasmal
BBC597

33. Cyanobacteria have :-
(1) Chl a similar to plant including green algae.
(2) Distinct from chlorophyll of plants.
(3) Bacteriochlorophyll.
(4) Bacteriorhodopsin.
BBC598

34. In which of the following the cell wall has stiff outer surface :-
(1) Dinoflagellates (2) Desmids
(3) Diatoms (4) Euglenoids
BBC599

35. Which of the following releases toxins that may even kill other marine animal like fishes:
(1) *Gonyaulax* (2) *Paramecium*
(3) Euglenoids (4) Sporozoans
BBC600

36. The slimy mass of protoplasm with nuclei forms the body of slime moulds is called as :-
(1) Plasmodium
(2) Myxamoeba
(3) Sporocytes
(4) Periplasmodium
BBC601

37. The body of a multicellular fungus are organized into a rapidly growing individual filaments called :-
(1) Mycelium (2) Hyphae
(3) Rhizoids (4) Dikaryon
BBC602

38. Which of the following statement is correct?
(1) *Albugo* belongs to group Basideomycetes.
(2) *Aspergillus* belongs to group Ascomycetes.
(3) Mushroom belongs to group Plantae.
(4) *Trichoderma* belongs to group Euglenoids.
BBC603

39. Which of the following statement is false?
(1) *Euglena* shows mixotrophic nutrition.
(2) Spores of slime moulds have true cell walls.
(3) Prions cause Potato spindle tuber disease.
(4) Mad cow disease is caused by Prions.
BBC604

40. **Assertion (A) :** Viruses can have either DNA or RNA.
Reason (R) : Viriods have low molecule weight RNA.
(1) Both **(A)** and **(R)** are true and **(R)** is correct explanation of (A).
(2) Both **(A)** and **(R)** are true but **(R)** is not correct explanation of **(A).**
(3) **(A)** is true but **(R)** is false.
(4) Both **(A)** and **(R)** are false.
BBC605

Hint :
Q.13 *Mucor*, *Rhizopus* & *Albugo* (In members of phycomycetes)
Q.14 Conidia & Basidiospores
Q.15 Only Homo sapiens do not have cell wall, rest others have the cell wall.

Exercise - I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	3	3	1	2	4	1	2	3	2	4	2	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	4	4	1	4	3	4	4	4	4	4	3	2	3
Question	31	32	33	34	35	36	37	38	39	40					
Answer	1	1	1	1	1	1	2	2	3	2					

AIM : FREE EDUCATION TO ALL APUL

Exercise - II
Biological Classification

1. Which of the following statements is incorrect?
(1) Fungi prefer to grow in cold and humid places.
(2) Karyogamy in fungi occurs after plasmogamy.
(3) Yeasts are used to make bread and beer.
(4) Body of fungi consists of long, slender thread like structures called mycelium.

BBC606

2.

Column-I	**Column-II**
I. *Streptomyces*	A. Nitrogen fixation
II. *Rhizobium*	B. Source of antibiotics
III. *Nitrosomonas*	C. Vinegar synthesis
IV. *Acetobacter*	D. Nitrification

Which of the following combination is correct?

	I	**II**	**III**	**IV**
(1)	B	A	D	C
(2)	C	D	A	B
(3)	A	B	C	D
(4)	D	B	A	C

BBC607

3. Which one of the following pairs is correctly matched?

(A)	*Rhizobium*	(i)	Parasite in the root of legume plants.
(B)	Mycorrhizae	(ii)	Symbiotic association between roots of higher plants and fungi
(C)	Yeast	(iii)	Production of biogas
(D)	Slime moulds	(iv)	are mostly parasitic

BBC608

4. Select the incorrect statement among the following
(1) Chrysophytes are found in fresh water as well as in marine environment.
(2) Dinoflagellates are mostly fresh water organisms and are photosynthetic.
(3) Majority of euglenoids are fresh water organisms.
(4) Slime moulds are saprophytic protist

BBC609

5. Find out the incorrect statement from the followings: -
(1) Heterotrophic bacteria are most abundant in nature.
(2) Bacterial structure is very complex and they have very simple behaviour.
(3) Bacteria reproduce mainly by fission.
(4) Archaebacteria differ from other bacteria in having different cell wall structure.

BBC610

6. Consider the following statements: -
(i) Bacteria have rigid cell wall made up of cellulose
(ii) Bacteria reproduce by sexual reproduction.
(iii) Ammonia is oxidised to nitrite by the bacteria *Nitrosomonas.*
(iv) Viruses are obligate parasites, cannot be living without host cell.
Which of the above statements are correct and incorrect?
(1) i, ii correct and iii, iv incorrect
(2) ii, iii correct and i, iv incorrect
(3) iii, iv correct and i, ii incorrect
(4) iv correct and i, ii, iii, incorrect

BBC611

7. Which of the following statements are correct?
(A) Mycoplasma has no cell wall
(B) Viruses can multiply only when they are present inside the living cell.
(C) True sexuality is not found in bacteria.
(D) Mycoplasma cannot survive without O_2.
Options: -
(1) A, B, C & D (2) A, B & D
(3) A, B & C (4) C & D

BBC612

8. How many organisms in the list given below are of ascomycetes?
Aspergillus, Rhizopus, Yeast, Claviceps, Morels, Agaricus, Trichoderma, Puccinia
(1) Six (2) Three
(3) Four (4) Five

BBC613

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

9. Study the five statements (A–D) given below and select the correct statements: -

(A) The cyanobacteria are unicellular, colonial or filamentous algae.

(B) Euglenoids are photosynthetic in the presence of sunlight when deprived of sunlight they behave like heterotrophs.

(C) In phycomycetes mycelium is aseptate and coenocytic.

(D) Protista are primarily aquatic organisms which make link with plants, animals and fungi.

(1) A, B & C

(2) C, D & A

(3) A & B only

(4) All statements are correct

BBC614

10. Match of the following columns correctly: -

Column-I		Column-II	
(i)	Smut fungi	(a)	*Sphagnum*
(ii)	White rust	(b)	*Ustilago*
(iii)	Rust fungi	(c)	*Albugo candida*
(iv)	Peat moss	(d)	*Puccinia*

Options :-

(1) i–b, ii–a, iii–d, iv–c

(2) i–b, ii–c, iii–d, iv–a

(3) i–d, ii–c, iii–b, iv–a

(4) i–d, ii–b, iii–c, iv–a

BBC615

11. Choose the incorrect statement from the following:

(1) Dinoflagellates have stiff cellulosic plates on the outer surface.

(2) Euglenoids have two flagella one lies longitudinally and the other transversely.

(3) Slime mould's spores are dispersed by air current.

(4) In diatoms the cell wall forms two thin overlapping shells.

BBC616

12. Consider the following four statements (a-d) and select the option which includes all the correct ones :-

(a) Unicellular member *Spirulina* can produce large quantities of food rich in proteins, minerals and vitamins.

(b) Diatoms are the chief producers of ocean.

(c) Slime moulds are saprophytic protists.

(d) Spores of moulds are devoid of cell wall.

Options:-

(1) b, c and d (2) a, b

(3) b, c (4) a, b and c

BBC617

13. Given below are two statements:

Statement-I : Fungi and members of the Monera and Protista having cell walls have now been excluded from Plantae, although according to earlier classifications they were placed in the same kingdom.

Statement-II : The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves etc.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are correct.

(2) Both Statement-I and II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BBC618

14. Given below are two statements:

Statement-I : Mosses along with lichens are the first organisms to colonise rocks and hence are of great ecological importance.

Statement-II: Gammae are unicellular, asexual buds which develop in small receptacles called gemma cups located on the thalli.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are correct.

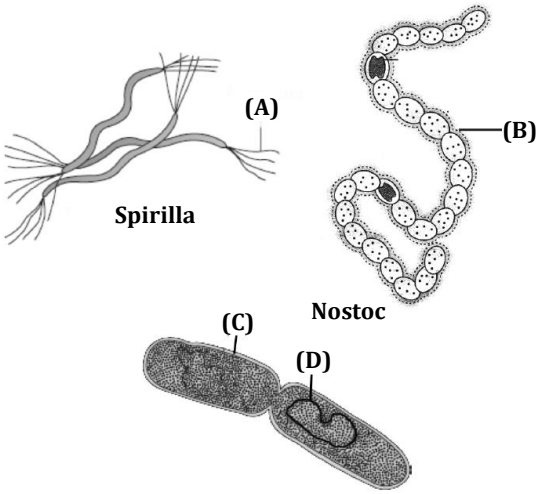
(2) Both Statement-I and II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BBC619

15. Identify A, B, C & D from the given diagram:-

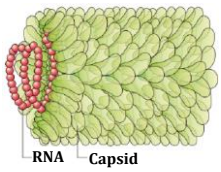
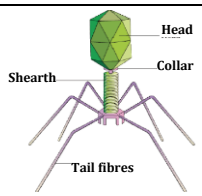
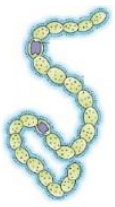


A dividing bacterial cell

	A	B	C	D
(1)	Pili	Cell wall	Cell wall	Cell membrane
(2)	Flagellum	Mucilagenous sheath	Cell membrane	DNA
(3)	Flagellum	Slime layer	DNA	RNA
(4)	Pili	Cell membrane	RNA	Nuclear membrane

BBC620

16. Match the Description in Column A with Diagrams in Column B ?

	Description		Diagrams
(a)	Tobacco Mosaic Virus (TMV)	(i)	
(b)	Bacteriophage	(ii)	
(c)	Dinoflagellates	(iii)	
(d)	*Nostoc*	(iv)	

Choose the correct answer from the options given below:

(1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(2) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(3) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

(4) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

BBC621

17. Which one of the following matches is correct?

(1)	*Alternaria*	Sexual reproduction absent	Deuteromycetes
(2)	*Mucor*	Reproduction by Conjugation	Ascomycetes
(3)	*Agaricus*	Parasitic fungus	Basidiomycetes
(4)	*Phytophthora*	Aseptate mycelium	Basidiomycetes

BBC622

18. Archaebacteria differ from other bacteria in having a different ....(a).... structure. Methanogens are present in ....(b)..... of serveral ruminant animals and are responsible for the production of ....C..... from the dung of these animals. Archaebacteria are special since they live in some of the .....d.... habitats.

What are the most suitable words in place of a, b, c and d respectively from the following option: -

	(a)	(b)	(c)	(d)
(1)	Cell membrane	Head	Biogas	Most harsh
(2)	Cell wall	Tongue	DDT	Most favourable
(3)	Cell membrane	Heart	Biogas	Most common
(4)	Cell wall	Gut	Methane	Most harsh

BBC623

19. Match column I (Kingdom) with column II (Class) and select the correct options

	Column-I (Kingdom)		Column-II (Class)
(A)	Plantae	(I)	Archaebacteria
(B)	Fungi	(II)	Euglenoids
(C)	Protista	(III)	Phycomycetes
(D)	Monera	(IV)	Algae

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

(1) A–IV, B–III, C–II, D–I
(2) A–I, B–II, C–III, D–IV
(3) A–III, B–IV, C–II, D–I
(4) A–IV, B–II, C–III, D–I

BBC624

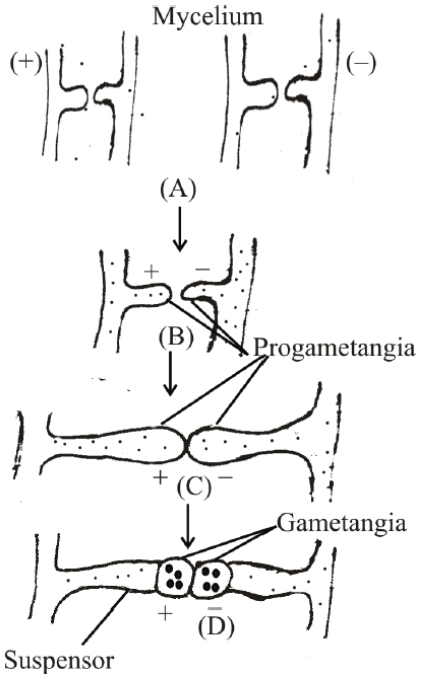
20. **Statement I :** Characterisation, identification, classification and nomenclature are the processes that are basic to taxonomy.
Statement II : External and internal structure, along with the structure of cell, development process and ecological information of organisms are essential and form the basis of modern taxonomic studies.
(1) Both statements are correct
(2) Statement I is correct and II is incorrect
(3) Statement I is incorrect and II is correct
(4) Both statements are incorrect

BBC625

21. In Phycomycetes sexual reproduction occurs by:
(1) Isogamy or anisogamy only
(2) Isogamy or oogamy only
(3) Isogamy, anisogamy or oogamy
(4) Anisogamy or oogamy only

BBC626

22. The figures represent stages in the sexual reproduction in *Rhizopus*. If gametangia are fused, then what is the fate of cells ?



(1) They will develop into a zoospore
(2) They will develop into a prothallus
(3) They will develop into a Parthenos pore.
(4) They will develop into a zygospore

BBC627

23. Read the following statement, how many are not incorrect for Euglenoids?
(A) Are usually found in running fresh water
(B) Have pellicle which makes their flagella flexible
(C) Have pigments identical to higher plants
(D) Cannot behave like predator
(E) Do not have nuclear membrane
(F) Well defined nucleus
(1) Two (2) Three
(3) All are correct (4) Four

BBC628

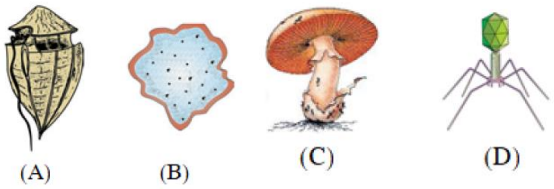
24. Read the following statements.
(A) Lichens do not grow in polluted areas.
(B) Prions are made of proteins only
(C) Viruses are inert outside their specific host cell.
(D) Mumps, small pox and influenza disease are caused by bacteria.
Which statements are correct.
(1) A and B (2) A and C
(3) A, B and C (4) A, B, C and D

BBC629

25. How many organisms in the list given below are photosynthetic: -
Chlorobium, Clostridium, Mycobacterium, Nitrobacter, Nitrococcus, Euglena, Albugo, E. coli, Nostoc, Anabaena.
(1) Six (2) Eight
(3) Four (4) Five

BBC630

26. Identify the structure of following:



AIM : FREE EDUCATION TO ALL APUL

	A	B	C	D
(1)	Dinoflagellates	Euglenoids	*Rhizopus*	T.M.V
(2)	Chrysophytes	Slime moulds	*Mucor*	T.M.V
(3)	Dinoflagellates	Slime moulds	*Agaricus*	Bacteriophage
(4)	T.M.V	Euglenodis	*Agaricus*	*Mucor*

BBC631

27. Match the column A and column B and select the correct option:

	Column-I		Column-II
(A)	Red tide	(i)	Diatom
(B)	Diatomaceous earth	(ii)	Slime mould
(C)	Plasmodium	(iii)	Prions
(D)	Unicellular fungi	(iv)	Dinoflagellates
(E)	Infectious proteins	(v)	Yeast

(1) A-i, B-iii, C-iv, D-v, E-ii
(2) A-iv, B-ii, C-i, D-iii, E-v
(3) A-iv, B-ii, C-i, D-v, E-iii
(4) A-iv, B-i, C-ii, D-v, E-iii

BBC632

28. **Assertion:** Boundaries of kingdom protista are not well defined.
Reason: Kingdom protista includes all single celled eukaryotes.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BBC633

29. **Assertion:** The cell wall of diatoms are indestructible.
Reason: The cell walls of diatoms are embedded with silica.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BBC634

30. How many of the following are examples of class Ascomycetes of fungi?
Albugo, *Alternaria*, *Neurospora*, *Agaricus*, *Penicillium*.
(1) Two only (2) Three only
(3) Four only (4) One only

BBC635

31. Among the following which statement is correct:
(1) Red sea is due to Dinoflagellates.
(2) *Aspergillus* reproduce asexually by conidia formation.
(3) Deuteromycetes do not produce conidia.
(4) *Trichoderma* belongs to Basideomycetes group.

BBC636

32. How many statements are correct among the following :-
(i) Plasmodium is formed in life cycle of Slime moulds.
(ii) Large number of Deuteromycetes are decomposers of litter.
(iii) Dikaryon stage is found in life cycle of *Rhizopus*.
(iv) Bacteriophage have ss-RNA.
(1) Two only (2) Three only
(3) All four (4) One only

BBC637

33. Red sea is due to :-
(1) Diatoms (2) *Gonyaulax*
(3) *Trichodesmium* (4) *Alternaria*

BBC638

34. Which among the following causes disease mumps.
(1) Viroids (2) Viruses
(3) Fungi (4) Prions

BBC639

35. Among the following which is false statement.
(1) Diatoms are the chief producers of the ocean.
(2) *Euglena* have chlorophylls a and b like that of higher plants.
(3) *Colletotrichum* causes disease Mad cow disease.
(4) Deuteromycetes help in mineral recycling.
BBC640

36. Bracket fungi belong to class :-
(1) Ascomycetes
(2) Basideomycetes
(3) Phycomycetes
(4) Deuteromycetes
BBC641

37. **Assertion :** Spores of slime moulds are dispersed by air currents.
Reason : Spores of slime moulds are formed during unfavourable conditions.
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A).**
(2) **(A)** is true but **(R)** is false.
(3) Both **(A)** and **(R)** are true but **(R)** is not correct explanation of **(A).**
(4) Both **(A)** and **(R)** are false.
BBC642

38. TMV have genetic material :-
(1) ss-RNA (2) ds-RNA
(3) ss-DNA (4) ds-DNA
BBC643

39. Which are edible fungi :-
(1) Morels, truffles only
(2) Morels, truffles, *Agaricus* only
(3) Puff balls, morels, truffles only
(4) *Agaricus*, truffles, bracket fungi only
BBC644

40. Which among the following statement is correct?
(1) *Puccinia* causes Smut disease in plants.
(2) Citrus canker in plants is caused by bacteria.
(3) *Colletotrichum* causes black rust in wheat.
(4) Late blight in potato is caused by *Albugo*.
BBC645

Hint :
Q.8 *Aspergillus*, Yeast, *Claviceps*, morels are Ascomycetes

Q.28 *Chlorobium*, *Euglena*, *Nostoc*, *Anabaena*.

Exercise - II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	2	3	2	3	3	3	4	2	2	4	1	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	4	1	1	3	4	1	3	3	3	3	2	1	1
Question	31	32	33	34	35	36	37	38	39	40					
Answer	2	1	3	2	3	2	1	1	2	2					

AIM : FREE EDUCATION TO ALL APUL

Exercise - I
Plant Kingdom

1. Stored food of red algae is similar to :-
(1) Chitin
(2) Mannitol
(3) Amylopectin and glycogen
(4) Glycogen and chitin
BPK518

2. Complex post fertilization development is seen in which algae ?
(1) Green algae
(2) Brown algae
(3) Yellow green algae
(4) Red algae
BPK519

3. Pyrenoids are made up of –
(1) Core of starch surrounded by sheath of protein
(2) Core of protein surrounded by fatty sheath.
(3) Proteinaceous centre and starchy sheath
(4) Core of nucleic acid surrounded by protein
BPK520

4. Study the given diagram and give answer of questions a, b and c :-



(a) Name of plant ?
(b) Class of plant ?
(c) Ploidy of rhizoids ?

	(a)	(b)	(c)
(1)	*Sphagnum*	Lycopsida	Haploid
(2)	*Funaria*	Bryopsida	Haploid
(3)	*Funaria*	Hepaticopsida	Diploid
(4)	*Marchantia*	Bryopsida	Haploid

BPK521

5. Asexual reproduction in most brown algae is by zoospores, these zoospores are :-
(1) Biflagellate : Pear-shaped : Unequal flagella
(2) Triflagellate : Pear-shaped : Equal flagella
(3) Biflagellate : Spiral-shaped : Unequal flagella
(4) Biflagellate : Oval-shaped : Equal flagella
BPK522

6. In the members of green algae the cell wall is double layered and outer cell wall is made up of pectose while inner cell wall composed of :
(1) Cutin (2) Pectose
(3) Cellulose (4) Lignin
BPK523

7. The needle like leaves, reduced surface area, thick cuticle and sunken stomata are the characteristic features of the which of the following groups of plants :
(1) Pteridophyte
(2) All bryophytes
(3) Gymnosperm (conifers)
(4) Angiosperm
BPK524

8. In pteridophyte, gametophyte (prothallus) prefers to grow in moist shady places because:
(1) In these conditions, it survive very well
(2) It represents the ancestory of bryophyte
(3) It need water for fertilization
(4) It is a aquatic structure
BPK525

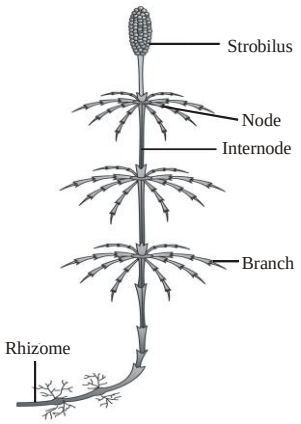
9. Cytotaxonomy is based on :
(1) Cytological information
(2) Chromosome number
(3) Chromosome structure and behaviour
(4) All of the above
BPK526

10. Agar, used to grow microbes and in preparations of ice creams and jellies, obtained from :-
(1) *Gelidium* and *Gracilaria*
(2) *Spirulina* and *Oscillatoria*
(3) *Fucus* and *Polysiphonia*
(4) *Vaucheria* and Brown algae
BPK527

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

11. Identify the given figure :-



(1) *Selaginella* (2) *Equisetum*
(3) *Marsilea* (4) *Lycopodium*

BPK528

12. Stored food in brown algae is/are :-
(1) Glycogen and oil
(2) Laminarin and mannitol
(3) Floridean starch
(4) Cellulose

BPK529

13. Which of the following show association with sloth bear?
(1) Algae/B.G.A
(2) Bryophyta
(3) Pteridophyta
(4) Gymnosperm

BPK530

14. Unbranched stem is characteristic of which gymnosperm?
(1) *Pinus* (2) *Cycas*
(3) *Cedrus* (4) *Lycopodium*

BPK531

15. Match the content of column-I with those of column-II :-

Column-I		Column-II	
(A)	Fungi	(i)	Chitinase
(B)	Bacteria	(ii)	Cellulase
(C)	Plant cell	(iii)	Lysozymes

(1) (a)-(i) , (b)-(ii) , (c)-(iii)
(2) (a)-(iii) , (b)-(ii) , (c)-(i)
(3) (a)-(ii) , (b)-(iii) , (c)-(i)
(4) (a)-(i) , (b)-(iii) , (c)-(ii)

BPK532

16. Both thallophyta and bryophyta are most correctly termed as :-
(1) Phanerogames
(2) Cryptogames
(3) Non-vascular cryptogames
(4) Vascular cryptogames

BPK533

17. Given below are two statements:
Statement-I : The form and size of algae is highly variable, ranging from colonial forms like *Ulothrix* and *Spirogyra* and to the filamentous forms like *Volvox*.
Statement-II: Aniosogamous type of reproduction is found in *Eudorina*.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK534

18. Given below are two statements:
Statement-I : Agar is one of the commercial products obtained from members of phaeophyceae.
Statement-II : Major pigments chlorophyll a, d and phycoerythrin is found in the members of phaeophyceae.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK535

AIM : FREE EDUCATION TO ALL APUL

19. Given below are two statements:

Statement-I : Most of the members of Chlorophyceae have one or more storage bodies called pyrenoids located in the chloroplasts.

Statement-II : Cell wall of members of rhodophyceae is made up of cellulose, pectin and polysulphate esters.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK536

20. Given below are two statements:

Statement-I: The red thalli of most of the red algae are unicellular.

Statement-II: Sexual reproduction in red algae is isogamous and accompanied by complex post fertilisation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK537

21. Given below are two statements:

Statement-I: Bryophytes are also called amphibians of plant kingdom because these plants can live in soil but are dependent on water for sexual reproduction.

Statement-II: The leaves in pteridophyta are small (microphylls) as in ferns or large (macrophylls) as in *Selaginella*.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK538

22. Given below are two statements:

Statement-I : Bryophytes are atracheophyte, embryophytes and non-spermatophyte cryptogams.

Statement-II : Protonema is creeping, green, branched and frequently filamentous stage.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK539

23. Given below are two statements:

Statement-I: Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophytes do not have an independent free-living existence.

Statement-II: *Pinus* is monoecious and *Cycas* is dioecious.

In the light of the above statements, choose the most appropriate answer from the options given below:

AIM : FREE EDUCATION TO ALL APUL

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK540

24. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Artificial system of classification separated closely related species.

Reason (R): Artificial system of classification was based on a few characters.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK541

25. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Most algal genera are haplontic.

Reason (R): In algae, mitosis in zygote results in the formation of diploid spores, the diploid spores divide meiotically to form gametophyte.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK542

26. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Mosses are of great ecological importance.

Reason (R) : Mosses along with lichens are the first organisms to colonise rocks. They decompose rocks making the substrate suitable for the growth of higher plants.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK543

27. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Bryophytes are also called amphibians of plant kingdom.

Reason (R): Bryophytes can live in soil but are dependent on water for sexual reproduction.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK544

AIM : FREE EDUCATION TO ALL APUL

28. Identify the correct set of statements:
(a) The main plant body of the bryophytes is haploid.
(b) In bryophytes, zygotes do not undergo reduction division immediately and they produce a multicellular body called sporophyte.
(c) In *Selaginella* and *Equisetum*, sporophylls may form distinct compact structures called strobili or cones.
(d) *Funaria*, *Polytrichum* and *Sphagnum* are liverworts.
(e) In mosses rhizoids are unbranched and unicellular.
Choose the correct answer from options given below:
(1) (a), (b) and (c) only
(2) (a), (b), (c) and (d) only
(3) (d) and (e) only
(4) (a), (b) and (d) only

BPK545

29. Identify the correct set of statements:
(a) One of the gymnosperms, the Giant red wood tree *sequoia* is one of the tallest tree species.
(b) Gymnosperms are heterosporous cryptogams.
(c) Gymnosperms are evolutionarily last archegoniate plant group.
(d) Pteridophytes are evolutionarily last plant group having antheridia.
(e) The stems are unbranched in *Cycas* and branched in *Pinus* and *Cedrus*.
Choose the correct answer from options given below:
(1) (a), (b) and (d) only
(2) (c), (d) and (e) only
(3) (a), (b), (c) and (d) only
(4) (a), (c), (d) and (e) only

BPK546

30. Identify the correct set of statements:
(a) In *spirogyra* sporophytic generation is represented only by one called zygote.
(b) *Fucus* is a brown alga shows diplontic life cycle.
(c) Bryophytes and pteridophytes show diplontic life cycle.
(d) *Ectocarpus*, *Polysiphonia*, kelps are haplodiplontic.
(e) Gymnosperms and angiosperms show haplontic life cycle.
Choose the correct answer from options given below:
(1) (a), (b) and (e) only
(2) (a), (b), (d) and (e) only
(3) (a), (b) and (d) only
(4) (b) and (d) only

BPK547

31. Which of the following is correct for algae ?
(1) Possess vascular bundles.
(2) Lack chlorophyll.
(3) They are mainly aquatic.
(4) They cannot grow on moist stones, soil and wood.

BPK548

32. Formation of asexual spores, autotrophic nutrition, no vascular tissue, thallus body are characters of :-
(1) Gymnosperm (2) Pteridophyta
(3) Fungi (4) Algae

BPK549

33. Which one of the following statement is wrong?
(1) Algin and Carrageenan are products of algae.
(2) Agar-agar is obtained from *Gelidium* and *Gracilaria.*
(3) *Chlorella* used as space food.
(4) Mannitol is stored food in Rhodophyceae.

BPK550

34. Which of the following match is incorrect?
(1) Microscopic angiosperm – *Wolfia*.
(2) Tallest gymnosperm – *Sequoia.*
(3) Persist pinnate leaves – *Pinus.*
(4) Coralloid roots – *Cycas.*

BPK551

AIM : FREE EDUCATION TO ALL APUL

35. Algae have cell wall made up of :
(1) Cellulose, hemicellulose and pectins.
(2) Cellulose, galactans and mannans.
(3) Hemicellulose, pectins and proteins.
(4) Pectins, cellulose and proteins.
BPK552

36. Bryophyta are divided into :-
(1) Pteropsida and *Gnetum*.
(2) Sphenopsida and Liverworts.
(3) Liverworts and Mosses.
(4) Mosses and Cycadophyta.
BPK553

37. Nearly how many species of marine algae are used as food?
(1) 60 (2) 70 (3) 50 (4) 80
BPK554

38. Monoecious plant of *Chara* shows occurrence of :
(1) Upper oogonium and lower antheridium on the same plant.
(2) Antheridiophore and archegoniophore on the same plant.
(3) Stamen and carpel on the same plant.
(4) Upper antheridium and lower oogonium on the same plant.
BPK555

39. **Assertion:** Phaeophyceae are brown algae.
Reason: Phaeophyceae contain large amount of xanthophyll pigment called fucoxanthin which gives brown colour.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BPK556

40. Read Assertion and Reason carefully to mark the correct option out of the options given below.
Assertion : Artificial systems of classification separate closely related species.
Reason : Artificial systems of classification are based on a few superficial morphological characteristics.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.
BPK557

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	3	2	1	3	3	3	4	1	2	2	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	2	1	2	3	1	1	4	2	4	4	1	4	3
Question	31	32	33	34	35	36	37	38	39	40					
Answer	3	4	4	3	2	3	2	1	1	1					

Exercise - II
Plant Kingdom

1. Some characters/structures are given below. How many of them are found in both bryophyta and pteridophyta?

(A) Archegonium (B) Protonema
(C) Embryo (D) Ovule
(E) Vascular tissue (F) Antheridium

Option :-

(1) Two (2) Three
(3) Four (4) Five

BPK558

2. Read the following statements regarding bryophytes and do as asked next to them: -

(A) All cells of sporophyte undergo reduction division to produce haploid spores.
(B) They play an important role in plant succession on barren rocks.
(C) Plant body is thallus like and prostate or erect and attached to the substratum by unicellular or multicellular rhizoids.
(D) They prevent soil erosion.
(E) Sex organs are multicellular. Male and female sex organs are called archegonium and antheridium respectively.

How many of the above statements is/are not correct?

(1) 2 (2) 3
(3) 1 (4) 4

BPK559

3. Which of the following statements is not correct?

(1) In gymnosperms the sporophyte and the gametophyte do not have an independent free-living existence.
(2) The leaves of gymnosperms are well-adapted to withstand extremes of temperature, humidity and wind.
(3) Pollination in gymnosperms is mainly carried by air current.
(4) The multicellular female gametophyte of gymnosperm is retained within mega-sporangium.

BPK560

4. Read the following paragraph and fill in the blanks with appropriate words: -

In pteridophytes the main plant body is ___(A)___ which bears sporangia that are subtended by leaf-like appendages called ___(B)___. The sporangia produce spores by ___(C)___ and after germination of spores, they give rise to photosynthetic gametophytes called ___(D)___.

Choose the correct option: -

	A	B	C	D
(1)	Sporophyte	Sporangia	Meiosis	Protonema
(2)	Gametophyte	Sporophylls	Mitosis	Sporophyte
(3)	Sporophyte	Sporophylls	Meiosis	Prothallus
(4)	Gametophyte	Sporophylls	Mitosis	Protonema

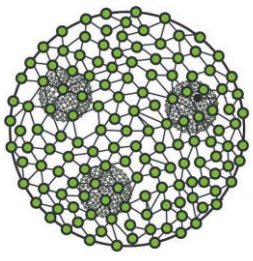
BPK561

5. Read the following statements regarding Gymnosperm: -

(A) The main plant body is sporophyte and heterosporous in nature.
(B) Roots of some gymnosperm have symbiotic association (blue green algae).
(C) The multicellular female gametophyte bears two or more archegonia.
(D) The male and female cones may be borne on the same tree or on different trees.
(1) Statement A, C and D are correct and statement B is incorrect.
(2) Statement A, B and D are correct and statement C is incorrect.
(3) Statement A and D are correct and statement B and C are incorrect.
(4) All statement are correct.

BPK562

6. Study the given algae and answer the following questions: -



AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

(a) Name of algae ?
(b) Type of sexual reproduction?
(c) Name of division?
(d) Stored food?
(1) *Laminaria:* Isogamous: Chlorophyta: Starch.
(2) *Volvox*: Oogamous: Chlorophyta: Starch.
(3) *Focus*: Isogamous: Rhodophyta: Floridian starch.
(4) *Chara* : Oogamous : Rhodophyta : Glycogen.

BPK563

7. Which of the following statement is correct?
(1) In all bryophytes asexual reproduction takes place by green, multicellular buds called gemmae.
(2) The sporophyte of bryophytes is differentiated into foot, seta and capsule and it is a free-living generation.
(3) Protonema is a green, creeping, branched and filamentous stage in mosses, which develops directly from a spore.
(4) The liverworts have an elaborate mechanism of seed dispersal.

BPK564

8. **Assertion :** In Gymnosperm, the male and female gametophytes do not have an independent free living existence.
Reason : The microspores develop in to a male gametophytic generation which is highly reduced.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(4) (A) is not correct but (R) is correct.

BPK565

9. **Assertion:** Mosses reduce the impact of falling rain and prevent soil erosion.
Reason: Mosses form dense mats on soil.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) (A) is not correct but (R) is correct.
(2) (A) is correct but (R) is not correct.
(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

BPK566

10. **Statement-I :** *Gracilaria* and *Porphyra* are members of Rhodophyceae.
Statement-II : Floridean starch is similar to amylopectin and glycogen in structure.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is incorrect but Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and Statement-II are incorrect.
(4) Both Statement-I and statement-II are correct.

BPK567

11. **Statement-I :** In bryophytes, the dominant phase in the life cycle is the gametophyte.
Statement-II : In pteridophytes the main plant body is sporophytic.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is incorrect but Statement-II are correct.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Both Statement-I and II are correct.

BPK568

AIM : FREE EDUCATION TO ALL APUL

12. **Statement-I :** Bladderwort is an insectivorous plants.
Statement-II : Venus fly trap and Cuscuta both are parasite plants.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are correct.
(4) Statement-I is incorrect but Statement-II are correct.

BPK569

13. Given below are two statements:
Statement-I: Natural phylogenetic classification systems are also based on evolutionary relationships between the various organisms.
Statement-II : In numerical taxonomy each character is given equal importance and at the same time hundreds of characters can be considered.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK570

14. Given below are two statements:
Statement-I : Algae are non-embryophytes, atracheophytes, spermatophytes and homosporous.
Statement-II : Sex organs of algae are multicellular and jacketed.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK571

15. Given below are two statements:
Statement-I : Most of the Pteridophytes are tracheophytes, embryophytes, heterosporous, spermatophytes cryptogams.
Statement-II : Prothallus is inconspicuous, small but multicellular, free living, mostly photosynthetic thalloid gametophyte.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK572

16. Given below are two statements:
Statement-I: *Selaginella* and *Salvinia* are homosporous pteridophytes.
Statement-II: Gymnosperms are tracheophytes, embryophytes, heterosporous, spermatophytes phanerogams.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK573

AIM : FREE EDUCATION TO ALL APUL

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Majority of pteriodophytes are heterosporous.

Reason (R) : Genera like *Selaginella* and *Salvinia* produce two kinds of spores, macro (large) and micro (small).

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK574

18. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Sexual reproduction in *Ulothrix* and *Spirogyra* is oogamous.

Reason (R) : Male and female gametes are flagellated and similar in size (as in *Ulothrix*) or non-flagellated (non motile) but similar in size (as in *Spirogyra*).

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK575

19. Identify the correct set of statements:

(a) Natural classification systems were not based on natural affinities among the organisms and evolutionary relationship between various organisms.

(b) In *Volvox*, *Fucus* sexual reproduction is oogamous.

(c) In numerical taxonomy, each character is given equal importance and at the same time hundreds of characters can be considered.

(d) In algae, asexual reproduction is by production of different types of spores, the most common being the conidia.

(e) Atleast half of the total carbon di-oxide fixation on earth is carried out by algae through photosynthesis.

Choose the correct answer from the options given below :

(1) (a) and (b) only

(2) (b), (c), (d) and (e) only

(3) (a), (b), (d) and (e) only

(4) (b), (c) and (e) only

BPK576

20. Identify the correct set of statements:

(a) Red colour of red algae is due to predominance of the red pigment fucoxanthin.

(b) *Chlorella* is a multicellular alga rich in proteins is used as food supplement even by space travallers.

(c) Green algae usually have a rigid cell wall made of an outer layer of cellulose and an inner layer of pectose.

(d) In red algae food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.

(e) *Fucus* is a brown alga shows haplo-diplontic life cycle.

Choose the correct answer from options given below :

(1) (a), (b) and (c)

(2) (a), (b), (c) and (e)

(3) only (d)

(4) (c) and (d) only

BPK577

AIM : FREE EDUCATION TO ALL APUL

21. Identify the correct set of statements:

(a) *Polysiphonia*, *Porphyra* are red algae, and *Laminaria*, *Sargassum* are brown algae.

(b) Asexual reproduction in most brown algae is by biflagellated zoospores that are pear shaped and have two equal laterally attached flagella.

(c) The plant body of bryophytes is less differentiated than pteriodophytes.

(d) Species of *Gelidium*, a moss provide peat that have long been used as fuel.

(e) *Marchantia*, is the leafy member of liverworts has tiny leaf like appendages in two rows on stem like structure.

Choose the correct answer from options given below:

(1) (a) and (c) only

(2) (b), (d) and (e) only

(3) (a), (b) and (c) only

(4) (c) only

BPK578

22. Identify the correct set of statements:

(a) Pteridophytes are first spermatophytes evolutionarily.

(b) In homosporous pteridophytes, spores germinate to give rise to inconspicuous, small but multicellular, free living mostly photosynthetic thalloid gameteophytes called prothallus.

(c) Majority of pteridophytes are homosporous.

(d) Homosporous pteridophyes show event precursor to the seed habit.

(e) *Equisetum* belongs to class lycopsida.

Choose the correct answer from options given below :

(1) (a), (d) and (e) only

(2) (b) and (c) only

(3) (a), (b), (c) and (d) only

(4) (b), (c) and (d) only

BPK579

23. Identify the correct set of statements:

(a) Roots in some genera have fungal association in the form of mycorrhiza (like *Cycas*), while in some others (like *Pinus*) small specialised roots called coralloid roots are associated with N_2–fixing cyanobacteria.

(b) In gymnosperms the multicellular female gametophytes is also retained within megasporangium.

(c) In gymnosperms endoseprm represents female gametophyte.

(d) Gymnosperms show anemophilous pollination.

(e) In *Cycas* the pinnate leaves persist for few year.

Choose the correct answer from options given below:

(1) (a), (b), (c) and (e) only

(2) (a), (c), (d) and (e) only

(3) (b), (c), (d) and (e) only

(4) (a), (b), (c), (d) and (e)

BPK580

24. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): In heterosporous pteridophytes, gametophyte is dioecious.

Reason (R): The female gametophytes in heterosporous pteridophytes, are retained on parent sporophyte for variable periods.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK581

AIM : FREE EDUCATION TO ALL APUL

25. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In gymnosperms endosperm is haploid.

Reason (R) : In gymnosperms within the megasporangium, megaspore mother cell divides meiotically to form four haploid megaspores, one of the megaspore develops into a multicellular female gametophyte.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK582

26. Match the column-I with column-II and choose the correct option.

	Column-I		Column-II
(A)	Smallest flowering plant	I.	*Eucalyptus*
(B)	Male sex organ in flowering plant	II.	*Wolffia*
(C)	Female sex organ in flowering plant	III.	Stamen
(D)	Tallest tree	IV.	Pistil

(1) A-I; B-II; C-III; D-IV
(2) A-IV; B-III; C-II; D-I
(3) A-II; B-III; C-IV; D-I
(4) A-II; B-IV; C-III; D-I

BPK583

27. Match the following columns correctly :-

Column-I	Column-II	Column-III
(i) *Volvox* & *Spirogyra*	(a) Floridean starch	(x) Chlorophyll-a, c
(ii) *Fucus* & *Laminaria*	(b) Starch	(y) Chlorophyll-a, d
(iii) *Porphyra* & *Polysiphonia*	(c) Mannitol & laminarin	(z) Chlorophyll-a, b

(1) i-c-z, ii-b-x, iii-a-y
(2) i-b-z, ii-c-x, iii-a-y
(3) i-b-y, ii-c-x, iii-a-z
(4) i-a-x, ii-b-y, iii-c-z

BPK584

28. Match the following:

	List-I		List-II
(A)	Pteropsida	(i)	*Equisetum*
(B)	Sphenopsida	(ii)	*Selaginella*
(C)	Psilopsida	(iii)	*Dryopteris*
(D)	Lycopsida	(iv)	*Psilotum*

(1) A–i; B–ii; C–iii; D–iv
(2) A–iii; B–iv; C–ii; D–i
(3) A–iv; B–iii; C–ii; D–i
(4) A–iii; B–i; C–iv; D–ii

BPK585

29. Match the following:

	List-I		List-II
(A)	Haplontic life cycle	(i)	*Ectocarpus*
(B)	Haplo-diplontic lifecycle	(ii)	*Spirogyra*
(C)	Diplontic life cycle	(iii)	*Cycas*
(D)	Haplo-diplontic lifecycle	(iv)	*Marchantia*

(1) A-(ii); B-(iv), C-(iii); D-(i)
(2) A-(i); B-(ii), C-(iii); D-(iv)
(3) A-(iv); B-(iii), C-(ii); D-(i)
(4) A-(ii); B-(iii), C-(iv); D-(i)

BPK586

30. Match the column [A] with column [B] :-

	Column-A		Column-B
(a)	Gemmae	(i)	Protonema
(b)	Mosses	(ii)	Male gametophyte
(c)	Megaspore	(iii)	Female gametophyte
(d)	Microspore	(iv)	Liver wort

(1) a-i, b-ii, c-iii; d-iv
(2) a-iv, b-ii, c-i; d-iii
(3) a-ii, b-iv, c-i; d-iii
(4) a-iv, b-i, c-iii; d-ii

BPK587

AIM : FREE EDUCATION TO ALL ♥ APUL

31. Read the following statements about pteridophytes :-

(i) The main plant body is sporophyte.

(ii) The sporophyte is differentiated into true roots, stem and leaves.

(iii) Their gametophyte lack vascular bundle.

(iv) Microphylls are found in *Selaginella* and macrophylls in the ferns like *Dryopteris.*

(v) Strobili or cones are found in *Selaginella* and *Equisetum.*

How many of the above statements are correct ?

(1) Four (2) Five

(3) Two (4) Three

BPK588

32. Bryophyte → Pteridophyte → Gymnosperm

(A) (B) (C)

Choose the ploidy of gametophytic generation of A, B and C :-

(1) A-2n, B-2n, C-2n

(2) A-n, B-2n, C-n

(3) A-2n, B-n, C-2n

(4) A-n, B-n, C-n

BPK589

33. The spread of living bryophytes and pteridophytes are restricted to moist, damp and shady places because:-

(1) They lack roots and vascular tissues

(2) Not able to tolerate high light intensity

(3) They require water for fertilization

(4) One generation dependent upon another for water.

BPK590

34.

Classes	Major Pigments	Stored food
Chlorophyceae	chl-a,b	A
Phaeophyceae	B	Mannitol
Rhodophyceae	chl-a,d	C

Choose the correct options for A, B and C :

(1)	A-Cellulose	B-chl-a,c	C-Floridean starch
(2)	A-Starch	B-chl-a,e	C-Laminarin
(3)	A-Starch	B-chl-a,c	C-Floridean starch
(4)	A-Cellulose	B-chl-a,e	C-Starch

BPK591

35. Some structures are given below they are produced during life cycle of different plant groups :-

(i) Gemmae (ii) Protonema

(iii) Prothallus (iv) Haploid endosperm

(v) Archegonium (vi) Pollen grains

(vii) Double fertilization

(viii) Fruit formation

(ix) Ovule formation

(x) Vascular tissues

What structures can be observed in the life cycle of Gymnosperm :-

(1) Gemmae, Prothallus, Archegonium.

(2) Vascular tissue, Prothallus, Ovule formation.

(3) Archagonium, Haploid endosperm, Vascular tissue.

(4) Prothallus, Archegonium, Vascular tissue.

BPK592

36. Arrange the following events of life cycle of Bryophyte in correct sequence :-

I. Germination of spore.

II. Gametes formation.

III. Formation of Gametophyte.

IV. Fertilization.

V. Embryo formation.

(1) I, III, IV, V, II (2) II, IV, I, V, III

(3) III, II, IV, V, I (4) V, I, II, III, IV

BPK593

37. Arrange the following plant groups in ascending order with respect to the gradual reduction of gametophytic stage.

(a) Bryophyta (b) Pteridophyta

(c) Angiosperm (d) Gymnosperm

(1) a, b, c, d (2) b, c, d, a

(3) a, b, d, c (4) b, d, c, a

BPK594

38. List some of the plants are given below :-
Cedrus, Funaria, Pinus, Cycas, Ginkgo, Pteridium, Adiantum
How many are 'naked seed' bearing plants -
(1) Three (2) Four
(3) Two (4) Five

BPK595

39. Place the following groups of plants in order, beginning with those that first appeared on the earth and progressing toward those that appeared most recently in time.
(1) Gymnosperms, angiosperms, ferns, moss, algae
(2) Algae, moss, ferns, gymnosperms, angiosperms
(3) Moss, algae, ferns, angiosperms, gymnosperms
(4) Algae, ferns, angiosperms, gymnosperms, moss

BPK596

40. Chemotaxonomy is connected with
(1) Classification of chemicals found in plants.
(2) Use of phytochemical data in systematic botany.
(3) Application of chemicals on herbarium sheets.
(4) Use of statistical methods in chemical yielding plants.

BPK597

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	3	4	2	3	1	4	4	4	2	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	3	4	3	1	2	3	1	4	3	2	4	1	4
Question	31	32	33	34	35	36	37	38	39	40					
Answer	2	4	3	3	3	3	3	2	2	2					

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Exercise – I
Morphology of Flowering Plant

1. Primary root originates from :-
(1) Radicle of embryo
(2) Plumule of embryo
(3) Coleorhiza of embryo
(4) Scutellum of embryo
BMP464

2. Fibrous root system is present in :-
(1) Wheat (2) Mustard
(3) *Monstera* (4) Banyan tree
BMP465

3. Stem originates from which part at the time of seed germination :-
(1) Plumule (2) Radicle
(3) Scutellum (4) Coleorhiza
BMP466

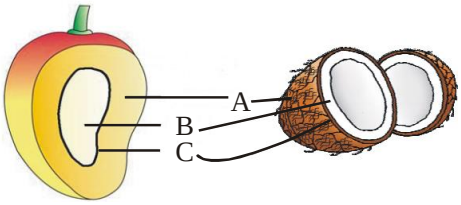
4. Rachis is absent in :-
(1) Simple leaf
(2) Palmately compound leaf
(3) Pinnately compound leaf
(4) Both (1) and (2)
BMP467

5. Which of the following possess simple leaves:
(1) Peepal, Mango, Radish
(2) Silk cotton
(3) *Bombax*
(4) *Azadirachta* (Neem)
BMP468

6. ..... type of inflorescence the main axis terminates in a flower, hence is limited in growth and the flowers are borne in a basipetal order :-
(1) Racemose (2) Cymose
(3) Venation (4) Placentation
BMP469

7. The mode of arrangement of sepals or petals in floral bud with respect to the other members of the same whorl is known as ..... :-
(1) Aestivation (2) Placentation
(3) Phyllotaxy (4) Allogamy
BMP470

8. Identify the components labelled A, B, C in the given diagrams :-



(1) A - mesocarp, B - seed, C - endocarp
(2) A - mesocarp, B - endocarp, C - seed
(3) A - Seed, B - endocarp, C - mesocarp
(4) A - Seed, B - mesocarp, C - endocarp
BMP471

9. Mango and coconut lie in which category of fruit:-
(1) Drupe fruit (2) Pepo fruit
(3) Pome fruit (4) Berry fruit
BMP472

10. If any fruit is produced from ovary without fertilization then it is called ..... :-
(1) Parthenogenesis
(2) Parthenocarpy
(3) Apospory
(4) Diplospory
BMP473

11. Given below are two statements:
Statement-I : In some plants, like grass, monstera and banyan roots arise from radicle.
Statement-II : Epidermal cells of region of elongation in root form very fine and delicate, thread like structures called root hairs.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.
BMP474

AIM : FREE EDUCATION TO ALL APUL

12. Given below are two statements:

Statement-I : Superior ovary is found in mustard, china rose, brinjal and inferior ovary is found in guava, ray florets of sunflower.

Statement-II : Mustard, *Datura*, chilli have actinomorphic flowers and canna has asymmetric flower.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BMP475

13. Given below are two statements:

Statement-I : Epipetalous androecium is found in Belladonna, *Ashwagandha*.

Statement-II : Epitepalous androcium is found in Pea, *Trifolium*.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

BMP476

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The zone, proximal to region of elongation in root, is called the region of maturation.

Reason (R) : The cells of elongation zone gradually differentiate and mature and form region of maturation.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMP477

15. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In Cymose type of inflorescence peduncle shows limited growth.

Reason (R) : In cymose type of inflorescence the main axis terminates into a flower.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMP478

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In mustard, chinarose and brinjal ovary is said to be superior.

Reason (R) : Gynoecium is situated in the centre and other parts of flowers are located on the rim of thalamus almost at the same level.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

AIM : FREE EDUCATION TO ALL APUL

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMP479

17. Identify the correct set of statements :
(a) In plants such as bean, gram and pea, the endosperm is present in mature seeds and such seeds are called endospermous.
(b) Embryo in monocot seed consists of one large and shield shaped cotyledon known as epiblast and a short axis with plumule and a radicle.
(c) The fruit in muliathi is legume and seed is endospermic.
(d) Diadelphous androecium is the feature of brassicaceae family.
(e) After fertilization ovules develop into fruit and ovary matures into seeds.
Choose the correct answer from options given below :
(1) All are correct
(2) All are incorrect
(3) (a), (c), (d) are correct
(4) only (c) is incorrect

BMP480

18. Find out the **correct** one from the following.

	Hypogynous flower	**Perigynous flower**	**Epigynous flower**
(1)	Guava, Cucumber, Sunflower - ray floret	Plum, Rose, Peach	Mustard, Brinjal, China rose
(2)	Plum, Rose, Brinjal	Cucumber, Peach	Guava, Sunflower - ray floret
(3)	Mustard, Guava	Brinjal, Sunflower	Cucumber, Sunflower - ray floret
(4)	Mustard, Brinjal, China rose	Plum, Rose, Peach	Guava, Cucumber, Sunflower - ray floret

BMP481

19. In Cymose, flower are borne in which manner
(1) Acropetal
(2) Basipetal
(3) Both acropetal & basipetal
(4) They do not follow any pattern

BMP482

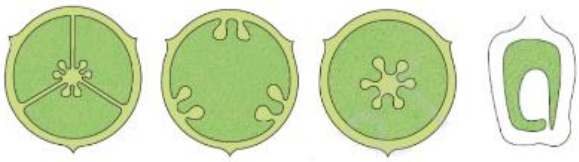
20. Tricarpellary, syncarpous ovary with axile placentation is seen in :-
(1) Solanaceae
(2) Fabaceae
(3) Liliaceae
(4) Brassicaceae

BMP483

21. A monocot can be distinguished from a dicot by:-
(1) Aestivation
(2) Venation
(3) Both (1) & (2)
(4) Placentation

BMP484

22. Select the correct option in which examples are correctly matched with figures of placentation respectively :-



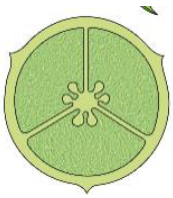
(1) Mustard, Tomato, Sunflower, *Dianthus*
(2) Tomato, Sunflower, Mustard, *Dianthus*
(3) Sunflower, *Dianthus*, *Argemone*, Tomato
(4) Tomato, *Argemone*, *Dianthus*, Sunflower

BMP485

23. Find the incorrect statement :-
(1) In pinnately compound leaf, rachis is present
(2) Rachis, actually represents the midrib of the leaf in a pinnately compound leaf
(3) In palmately compound leaf, rachis is present
(4) Silk cotton has palmately compound leaf

BMP486

24. Answer the following questions (A & B) by using the given below diagram :-



(A) Type of placentation ?
(B) Example of this type of placentation ?
(1) A = Free Central, B = china rose
(2) A = Parietal, B = Mustard
(3) A = Basal, B = Marigold
(4) A= Axile, B = Lemon

BMP487

25. Developing seeds are attached to the fruit with a scar like structure, which can also be seen on seed coat is called as :-
(1) Testa (2) Tegmen
(3) Hilum (4) Micropyle

BMP488

26. In coconut and mango, the fruit is known as :-
(1) Berry (2) Pod
(3) Drupe (4) Samara

BMP489

27. In **__A__** type of inflorescence the main axis terminates in a flower, hence is **__B__** in growth. The flowers are borne in a **__C__** order. **A, B** & **C** are respectively.
(1) Racemose, unlimited, acropetal
(2) Cymose, limited, basipetal
(3) Cymose, unlimited, basipetal
(4) Racemose, limited, acropetal

BMP490

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	1	4	1	2	1	1	1	2	2	1	3	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	2	2	4	2	3	2	4	3	4	3	3	2			

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Exercise – II
Morphology of Flowering Plant

1. The function of stem is :-
(1) Spreading out branches, bearing leaves, flowers and fruits
(2) Conduct water, minerals and photosynthates
(3) Storage of food, support, protection and vegetative propagation
(4) All of the above

BMP491

2. Find out the correct match from given table :-

Column-I		Column-II	
(A)	Alternate phyllotaxy	(i)	Sunflower, china rose, mustard
(B)	Opposite phyllotaxy	(ii)	*Calotropis, guava*
(C)	Whorled phyllotaxy	(iii)	Monocot plants
(D)	Parallel venation	(iv)	*Alstonia* (Devils tree)

(1) A - i, B - iii, C - iv, D - ii
(2) A - i, B - ii, C - iv, D - iii
(3) A - iv, B - iii, C - ii, D - i
(4) A - ii, B - iii, C - iv, D - i

BMP492

3. Find out the correct match :-

Column-I		Column-II	
(A)	Actinomorphic flower	(i)	*Canna*
(B)	Zygomorphic flower	(ii)	Pea Gulmohar, Bean, *Cassia*
(C)	Asymmetric flower	(iii)	*Datura*, Mustard, Chilli
(D)	Underground stems	(iv)	Potato, Ginger

(1) A - i, B - ii, C - iii, D - iv
(2) A - iii, B - ii, C - i, D - iv
(3) A - iv, B - iii, C - ii, D - i
(4) A - ii, B - iv, C - i, D - iii

BMP493

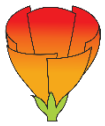
4. Find out the correct match :-

Column-I		Column-II	
(A)	Twisted	(i)	*Calotropis*
(B)	Valvate	(ii)	Chinarose, ladyfinger, cotton
(C)	Vexillary	(iii)	*Cassia,* gulmohur
(D)	Imbricate	(iv)	Pea, bean

(1) A - ii, B - i, C - iv, D - iii
(2) A - iv, B - iii, C - ii, D - i
(3) A - iii, B - iv, C - ii, D - i
(4) A - ii, B - iii, C - iv, D - i

BMP494

5. Find out the correct match :-

	Column-I		Column-II
(A)	Twisted aestivation	(i)	
(B)	Valvate aestivation	(ii)	
(C)	Vexillary aestivation	(iii)	
(D)	Imbricate aestivation	(iv)	

(1) A - ii, B - i, C - iv, D - iii
(2) A - iv, B - iii, C - ii, D - i
(3) A - iii, B - iv, C - ii, D - i
(4) A - iv, B - iii, C - i, D - ii

BMP495

6. Find out the correct match from the table :-

	Column-I		Column-II
(A)	Basal placentation	(i)	Pea
(B)	Free central placentation	(ii)	Chinarose, tomato, lemon
(C)	Parietal placentation	(iii)	*Argemone*, mustard
(D)	Axile placentation	(iv)	*Dianthus, primrose*
(E)	Marginal placentation	(v)	Sunflower, marigold

AIM : FREE EDUCATION TO ALL APUL

(1) A - i, B - ii, C - iii, D - iv, E - v
(2) A - ii, B - iii, C - iv, D - v, E - i
(3) A - v, B - iv, C - iii, D - ii, E - i
(4) A - v, B - iv, C - i, D - ii, E - iii

BMP496

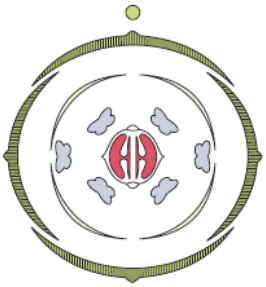
7. Find out the correct match :-

	Column-I		Column-II
(A)	Free cental	(i)	
(B)	Basal	(ii)	
(C)	Parietal	(iii)	
(D)	Axile	(iv)	
(E)	Marginal	(v)	

(1) A - i, B - ii, C - iii, D - iv, E - v
(2) A - i, B - ii, C - v, D - iv, E - iii
(3) A - iv, B - v, C - iii, D - ii, E - i
(4) A - iii, B - ii, C - iv, D - iv, E - i

BMP497

8. Choose the correct floral formula with the help of given floral diagram :-



(1) $\oplus \, ♀ \, K_{2+2}\, C_4\, A_{2+4}\, \underline{G}_{(2)}$

(2) $\oplus \, ♀ \, K_{(2+2)}\, C_{1+2+2}\, A_{2+4}\, \underline{G}_{(2)}$

(3) $\oplus \, ♀ \, K_{2+2}\, C_4\, A_{(2+4)}\, \underline{G}_{(2)}$

(4) $\oplus \, ♀ \, K_{2+2}\, C_4\, A_{2+4}\, \underline{G}_2$

BMP498

9. $\% \, K_{(5)}\, C_{1+1+(2)}\, A_{(9)+1}\, \underline{G}_1$

Select the correct option in respect of given floral formula :-
(1) Posterior petals are fused
(2) Found in flower of tulip plant
(3) Diadelphous condition
(4) Monoadelphous condition

BMP499

10. Identify the correct floral formula on the basis of given diagram ?



(a) (b) (c) (d)

(1) $\% \, ♀ \, K_{(5)}\, C_{1+2+(2)}\, A_{(9)+1}\, \underline{G}_1$

(2) $\oplus \, ♀ \, K_{2+2}\, C_4\, A_{2+4}\, \underline{G}_{(2)}$

(3) $Br \oplus ♀ \, \overset{\frown}{P}_{(3+3)}\, A_{3+3}\, \underline{G}_{(3)}$

(4) $\oplus \, ♀ \, K_{(5)}\, \overset{\frown}{C}_{(5)}\, A_5\, \underline{G}_{(2)}$

BMP500

11. **Statement-I :** The floral features are represented in the summarised form as floral diagrams and floral formula.
Statement-II : Flowering plants exhibit enormous variation in shape, size, structure, mode of nutrition, life span, habit and habitat.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II are correct
(4) Both Statement-I and II are correct

BMP501

12. **Statement-I :** Generally, Monocotyledonous seeds are Non-endospermic but some as in orchids are endospermic.
Statement-II : In some seeds such as castor the endosperm formed as a result of double fertilization, is food storing tissue and called endospermic seeds.

AIM : FREE EDUCATION TO ALL APUL

In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and statement-II are correct

BMP502

13. **Statement-I :** When stamens are attached to the petals they are epiphyllous as in brinjal, or epipetalous when attached to the perianth as in the flowers of lily.
Statement-II : When the ovules are borne on central axis and septa are absent, as in Dianthus and primrose the placentation is called free central.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and II are correct

BMP503

14. Given below are two statements : one is labelled as Assertion(A) and the other is labelled as Reason (R).
Assertion : The stamens of china rose are Monadelphous.
Reason : Filaments of stamens are united into one bunch or one bundle and anthers are free.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A).

BMP504

15. Read the following statements and choose the set of correct statements :-
(a) A few centimetres above the root cap is the region of meristematic activity.
(b) In some plants like monstera, grass, and the mustard plant, root aries from parts of the plant other than the radicle and are called adventitious roots.
(c) Root hairs are generally multicellular and they absorb water and minerals from the soil.
(d) Underground stems of Potato, Ginger, Turmeric, Zaminkand, *Colocassia* are modified store food in them.
(e) The region of the stem where leaves are born are called nodes while internode are the portions between two nodes.
Choose the correct answer from the options given below.
(1) (a), (b), (d) and (e) only
(2) (a), (d) and (e) only
(3) (d) and (e) only
(4) (c), (d) and (e) only

BMP505

16. Read the following statement and choose the set of correct statements :-
(a) The flower is the reproductive unit in the angiosperm.
(b) A flower may be trimerous, tetramerous or pentamerous when the floral appendages are in multiple of 3, 4 or 5 respectively.
(c) The calyx is the innermost whorl of the flower and the members are called sepals.
(d) Each stamen which represents the male reproductive organ consists of stigma, style and ovary.
(e) the arrangement of ovules with in the ovary is known as placentation.
Choose the correct answer from the options given below.
(1) (a), (b) and (c) only
(2) (a), (b) and (e) only
(3) (a), (c) and (d) only
(4) (a) and (b) only

BMP506

AIM : FREE EDUCATION TO ALL APUL

17. Given below are two statements:

Statement-I : Underground stems of some plants such as grass and strawberry etc., spread to new niches and when older parts die new plants are formed.

Statement-II : In *Pistia* and *Eichhornia* the lateral branches originate from the basal and underground portion of the main stem, grows horizontally beneath the soil and the come out obliquely upward giving rise to leafy shoots.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BMP507

18. Given below are two statements:

Statement-I : Micropyle is a scar on the seed coat through which the developing seeds are attached to the fruit, above the micropyle is a small pore called the hilum.

Statement-II : Opposite phyllotaxy is found in *Calotropis* and whorled phyllotaxy is found in *Alstonia*.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BMP508

19. Identify the correct set of statements :

(a) Ovary is half inferior in plum, rose, peach.

(b) If the margins of sepals and petals overlap one another but not in a particular direction as in *Cassia* and gulmohar, the aestivation is called imbricate.

(c) A sterile stamen is called staminate.

(d) In *Dianthus* and *Primrose* the placentation is parietal.

(e) In mango and coconut fruit is known as legume.

Choose the correct answer from options given below :

(1) (a) and (b) only

(2) (a), (b), (c), (d) and (e) only

(3) (b), (c), (d) only

(4) (a) and (e) only

BMP509

20. Axile placentation is found in

(1) Malvaceae (2) Liliaceae

(3) Solanaceae (4) All of the above

BMP510

21. Perianth is reduced and represent as lodicule in the family -

(1) Lilliaceae (2) Graminae

(3) Compositae (4) Malvaceae

BMP511

22. Androecium in Solanaceae is _____

(1) Epipetalous (2) Epiphyllous

(3) Monoadelphous (4) Polyadelphous

BMP512

23. **Assertion :** Syngenesious condition is a characteristic feature of compositae.

Reason : Stamens are united by their anthers only while, filaments remains free.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true and but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BMP513

AIM : FREE EDUCATION TO ALL APUL

24. Floral formula of a plant is

% ♂ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$.

Which of the following feature is correct w.r.t. this plant?

(1) Leaves have parallel venation
(2) Roots are adventitious and fibrous
(3) Seeds are endospermous
(4) Roots may have root nodules

BMP514

25. Plant which to neither belongs solanaceae nor to cruciferae family :

(1) Mustard
(2) *Withania somnifera*
(3) *Aloe vera*
(4) Potato

BMP515

26. Find the incorrect match, w.r.t. the floral formula and its family :

(1) Poaceae – $Br\oplus\male P_2A_3\underline{G}_1$ or $\underline{G}_{(3)}$

(2) Asteraceae – $Br\%\female K_pC_{(5)}A_0\overline{G}_{(2)}$

(3) Malvaceae – $Br\oplus\male Epi_{3\text{-}7}K_{(5)}\overset{\frown}{C_5}\ \overset{\frown}{A}_{(\infty)}\ \underline{G}_{(5\text{-}\infty)}$

(4) Fabaceae – $\%\ \overset{\male}{\underset{\female}{}}K_{(5)}C_{1+2+(2)}A_{(9)+1}\underline{G}_1$

BMP516

27. (I) Aleurone layer
(II) Scutellum
(III) Coleoptile
(IV) Coleorhiza

Find the correct option with respect to above information.

(1) Gram – Monocot seed
(2) Pea – Dicot seed
(3) Maize – Dicot seed
(4) Maize – Monocot seed

BMP517

28. Study carefully the given floral diagram and select the option, which correctly represents the related Floral Formula (FF).



(1) $Br\oplus\male Epi_{3\text{-}7}K_{(5)}\overset{\frown}{C_5}\ \overset{\frown}{A}_{(\infty)}\ \underline{G}_{(5\text{-}\infty)}$

(2) $\oplus\ \male\ P_6A_6\underline{G}_{(3)}$

(3) $\oplus\ \male\ P_{5+5}A_{(5)}\underline{G}_{(2)}$

(4) $\oplus\ \male P_{(5)}\overset{\frown}{C}_{(5)}\overset{\frown}{A}_5\underline{G}_{(2)}$

BMP518

29. $\oplus\ \overset{\male}{\underset{\female}{}}\ K_{(5)}\ \overset{\frown}{C}_{(5)}\ \overset{\frown}{A}_5G_{(2)}$ is the floral formula of :-

(1) *Sesbania*
(2) *Petunia*
(3) *Brassica*
(4) *Allium*

BMP519

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	1	1	3	3	1	3	1	4	3	3	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	2	3	4	1	4	2	1	1	4	3	1	4	1	2	

Exercise – I
Anatomy of Flowering Plants

1. In which part of dicot root, casparian stripes are present?
(1) Hypodermis (2) Endodermis
(3) Pericycle (4) Cortex
BAP376

2. The vascular bundles are surrounded by a layer of thick walled _____ cells in dorsiventral leaf: -
(1) Mesophyll (2) Epidermal
(3) Bundle sheath (4) Hypodermis
BAP377

3. Bulliform cells are present in :
(1) Grass leaf
(2) Mango leaf
(3) Both (1) and (2)
(4) Monocot stem
BAP378

4. The tissues involved in secondary growth can be:
(1) Lateral meristem
(2) Vascular cambium
(3) Cork cambium
(4) All of the above
BAP379

5. The meristem, which occurs between mature tissues and are responsible for primary growth is called:
(1) Intercalary (2) Secondary
(3) Lateral (4) Apical
BAP380

6. Which cambium is primary in origin:
(1) Intrafascicular cambium
(2) Interfascicular cambium
(3) Cork cambium
(4) Vascular cambium
BAP381

7. In the _____, the vascular cambium is completely secondary in origin:
(1) Monocot root (2) Monocot stem
(3) Dicot root (4) Dicot stem
BAP382

8. The endodermis of dicot stem is also called:
(1) Stele (2) Starch sheath
(3) Cortical layers (4) Periderm
BAP383

9. The wood is actually a _____
(1) Secondary phloem
(2) Primary phloem
(3) Primary xylem
(4) Secondary xylem
BAP384

10. The companion cells are specialised parenchymatous cells, which are closely associated with: -
(1) Sieve tube elements
(2) Xylem
(3) Phloem
(4) Subsidiary cells
BAP385

11. The number of xylem bundle in monocot root is:
(1) 4 (2) 6
(3) more than 6 (4) less than 6
BAP386

12. In dicot stem, pericycle is present on the inner side of the endodermis and above the phloem in the form of semi lunar patches, it is made of:
(1) Parenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Prosenchyma
BAP387

13. During the formation of leaves and elongation of stem, some cells left behind from shoot apical meristem, constitute the _____
(1) Axillary bud
(2) Secondary meristem
(3) Lateral meristem
(4) Cambium
BAP388

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

14. Identify the tissue in which cells are much thickened at the corners due to a deposition of cellulose, hemicellulose and pectin: -
(1) Parenchyma (2) Collenchyma
(3) Sclerenchyma (4) Xylem

BAP389

15. _____ provides mechanical support to the growing parts of the plant such as young stem and petiole of a leaf:
(1) Parenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Xylem

BAP390

16. The main mechanical tissue of plant is :
(1) Parenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Xylem

BAP391

17. _____ constitute the complex tissues in plants.
(1) Xylem, phloem
(2) Phloem, metaxylem
(3) Parenchyma, xylem
(4) Xylem, metaxylem

BAP392

18. The cells which are devoid of protoplasm are:
(1) Tracheids (2) Vessels
(3) Xylem fibres (4) All of the above

BAP393

19. Which one of the following is not a part of phloem?
(1) Sieve tubes
(2) Companion cells
(3) Phloem parenchyma
(4) Vessels

BAP394

20. _____ is absent in most of the monocotyledonous, stem:
(1) Sieve tube elements
(2) Phloem parenchyma
(3) Xylem parenchyma
(4) Companion cells

BAP395

21. Phloem fibres (bast fibres) are made up of _____ cells:
(1) Parenchymatous
(2) Sclerenchymatous
(3) Secretory
(4) Collenchymatous

BAP396

22. The stomatal aperture, guard cells and the surrounding subsidiary cells are together called as –
(1) Trichome
(2) Stomata
(3) Stomatal apparatus
(4) Ground tissue

BAP397

23. On stem the epidermal hairs are called:
(1) Trichome
(2) Root hair
(3) Both (1) and (2)
(4) Bundle sheath

BAP398

24. All tissues except epidermis and vascular bundles constitute –
(1) Ground tissue system
(2) Epidermal tissue system
(3) Vascular tissue system
(4) Stele

BAP399

25. The ground tissue system consists of simple tissues such as :
(1) Parenchyma and Collenchyma
(2) Collenchyma and Sclerenchyma
(3) Parenchyma and Sclerenchyma
(4) All of the above

BAP400

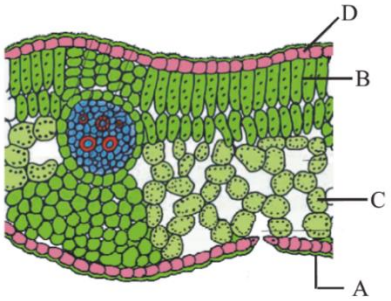
26. In leaves, the ground tissue constitute of thin walled parenchymatous chloroplast containing cells and is called _____
(1) Mesophyll
(2) Epidermis
(3) Stomatal apparatus
(4) Bundle sheath extension

BAP401

27. The innermost layer of the cortex is called:
(1) Endodermis (2) Hypodermis
(3) Pericycle (4) Casparian strips

BAP402

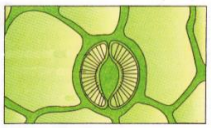
28. Given below is the diagram of a T.S. of dorsiventral leaf. In which one of the options all the four parts A, B, C and D are correct?



	A	B	C	D
(1)	Adaxial epidermis	Palisade mesophyll	Spongy mesophyll	Abaxial epidermis
(2)	Adaxial epidermis	Spongy mesophyll	Palisade mesophyll	Abaxial epidermis
(3)	Abaxial epidermis	Palisade mesophyll	Spongy mesophyll	Adaxial epidermis
(4)	Adaxial epidermis	Bundle sheath	Spongy mesophyll	Abaxial epidermis

BAP403

29. The following figure represent the stomatal apparatus which shows :-



(1) Dumb-bell shaped guard cell and found in grasses
(2) Bean shaped guard cell and found in grasses
(3) Dumb-bell shaped guard cell and found in dicots
(4) Bean shaped guard cell and found in dicots

BAP404

30. Arrange the following cell layers in outer to inner side:
(a) Hypodermis (b) Pericycle
(c) Endodermis (d) Epidermis
(e) Pith
(1) (e), (b), (c), (a), (d)
(2) (e), (c), (b), (a), (d)
(3) (d), (a), (b), (c), (e)
(4) (d), (a), (c), (b), (e)

BAP405

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	1	4	1	1	3	2	4	1	3	3	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	4	4	2	2	3	1	1	4	1	1	3	4	4

AIM : FREE EDUCATION TO ALL APUL

Exercise – II
Anatomy of Flowering Plants

1. The meristem that occurs in the mature regions of roots and shoots of many plants, particularly those that produce woody axis and appear later than primary meristem is called the ____ meristem:
(1) Apical
(2) Secondary
(3) Intercalary
(4) Primary

BAP406

2. Select the wrong statement about parenchyma :
(1) It forms the major component within organs
(2) The cells of the parenchyma are generally non-isodiametric
(3) Its cell wall is thin and made up of cellulose
(4) The parenchyma performs various functions like photosynthesis, storage and secretion

BAP407

3. Select the wrong statement about the collenchyma:
(1) The collenchyma occurs in layers below the epidermis in dicotyledonous stem
(2) They are always found as continuous homogenous layers
(3) These can assimilate food when they contain chloroplasts
(4) Intercellular spaces are generally absent

BAP408

4. Read the following statements and choose the set of correct statements :-
(a) Internal structures also show adaptations to diverse environments.
(b) During the formation of primary plant body, specific regions of the apical meristem produce dermal tissues, ground tissue and vascular tissues.
(c) The first formed primary xylem element is called metaxylem and the later formed primary xylem is called protoxylem.
(d) All tissues except epidermis and vascular bundles constitute the ground tissue.
(e) Gymnosperms have albuminous cells and sieve tubes.
Choose the correct answer from the options given below.
(1) (a) and (c) only
(2) (c), (d) and (e) only
(3) (a), (b) and (d) only
(4) (c), (d) and (e) only

BAP409

5. Read the following statement and choose the set of correct statements :-
(a) Monocotyledonous roots do not undergo any secondary growth.
(b) In dicot root innermost layer of cortex is called pericycle.
(c) In monocot stem cells of endodermis are rich in starch grains and the layer is also referred to as starch sheath.
(d) In grasses certain abaxial epidermal cells along the vein modify themselves into large, empty colourless cells. These are called bulliform cells.
(e) In dorsiventral leaf mesophyll tissue differentiate into palisade and spongy parenchyma.
Choose the correct answer from the options given below.
(1) (a), (d) and (e) only
(2) (a), (b) and (d) only
(3) (a), (c), (d) and (e) only
(4) (a) and (e) only

BAP410

6. Given below are two statements : one is labelled as Assertion(A) and the other is labbelled as Reason (R).
Assertion : Collenchyma, sclerenchyma and xylem provides mechanical support to the plant parts.
Reason : Their cell wall is thick and lignified.

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A).

BAP411

7. Given below are two statements:

Statement-I : In grasses, certain adaxial epidermal cells along with the veins modify themselves into large empty, colourless cells. These are called bulliform cells.

Statement-II : All tissues on the outerside of endodermis such as pericycle, vascular bundles and pith constitute the stele.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BAP412

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The companion cells help in maintaining pressure gradient in sieve tubes.

Reason (R) : The sieve tube elements and companion cells are connected by pit fields present between their common longitudinal walls.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BAP413

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In dicot stems, vascular bundles are said to be open.

Reason (R) : In dicot stems cambium is present between phloem and xylem.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BAP414

10. Identify the correct set of statements :

(a) Both apical meristems and intercalary meristems are primary meristems because they appear early in life of a plant and contribute to the formation of the primary plant body.
(b) Parenchyma consists of cells which are much thickened at the corners due to a deposition of cellulose, hemicellulose and pectin.
(c) The sclerenchyma performs various functions like photosynthesis, storage, secretion.
(d) The sclerenchymatous fibres are thick walled, elongated and pointed cells, generally occurring in group, in various parts of the plant.
(e) Tracheids are living cells with protoplasm.

Choose the correct answer from options given below :

(1) (a), (b) and (d) only
(2) (a) and (d) only
(3) (b), (c) and (e) only
(4) (b), (c) and (d) only

BAP415

AIM : FREE EDUCATION TO ALL APUL

11. Identify the correct set of statements :

(a) Phloem in angiosperms is composed of sieve tube elements, companion cells, phloem parenchyma and phloem fibres.

(b) Phloem parenchyma is made up of elongated, tapering cylindrical cells which have dense cytoplasm and nucleus.

(c) Phloem fibres are generally absent in secondary phloem but present in primary phloem.

(d) Xylem fibres of jute, flax and hemp are used commercially.

(e) The stomatal pore, guard cells and the surrounding subsidiary cells are together called stomatal aperture.

Choose the correct answer from options given below :

(1) (a) and (b) only
(2) (c), (d) and (e) only
(3) (a), (b), (c) and (d) only
(4) (a), (b) and (c) only

BAP416

12. Identify the correct set of statements :

(a) The tangential as well as radial walls of the endodermal cells have a deposition of water impermeable, waxy material lignin in the form of casparian strip.

(b) In dicotyledonous roots there are usually two to four xylem and phloem patches.

(c) In monocot stem the cells of the endodermis are rich in starch grains and the layer is also referred to as the starch sheath.

(d) In dicot leaves, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells.

(e) The activity of cambium is under control of many physiological and environmental factors.

Choose the correct answer from options given below :

(1) (a) only
(2) (b) and (e) only
(3) (a), (b) and (e) only
(4) (a), (c) and (d) only

BAP417

13. In dicot roots, initiation of lateral roots and cambium during the secondary growth takes place by ____ cells.

(1) Parenchyma (2) Pericycle
(3) Endodermis (4) Epidermis

BAP418

14. The parenchymatous cells which lie between the xylem and the phloem in root are called :

(1) Conjuctive tissue (2) Stele
(3) Pith (4) Medullary rays

BAP419

15. Which one of the following is not a part of stele ?

(1) Pith
(2) Vascular bundle
(3) Endodermis
(4) Pericycle

BAP420

16. Given below are two statements:

Statement-I : During the formation of leaves and elongation of stems, some cells left behind from shoot apical meristem, constitute the axillary bud.

Statement-II : Fascicular vascular cambium, inter fascicular cambium and cork cambium are examples of lateral meristems.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BAP421

AIM : FREE EDUCATION TO ALL APUL

17. Given below are two statements:

Statement-I : Collenchyma occurs in layers below the epidermis in most of the monocotyledonous plants.

Statement-II : Sclereids are found in the fruit walls of nuts, pulp of fruits like guava, pear and sapota, seed coats of legumes and leaves of tea.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BAP422

18. Given below are two statements:

Statement-I : In roots the arrangement of primary xylem is endarch and in stem it is exarch.

Statement-II : A mature sieve element possesses a peripheral cytoplasm and a large vacuole and a nucleus.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BAP423

19. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In roots the arrangement of primary xylem is exarch.

Reason (R) : In roots, the protoxylem lies towards the centre (pith) and the metaxylem lies towards the primary of the organ.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

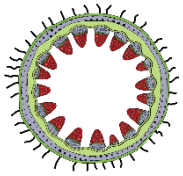
(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

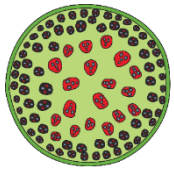
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BAP424

20. Figures X and Y represent the transverse section of ____ and ____ respectively.



X



Y

	A	B
(1)	Dicot root	Dicot stem
(2)	Monocot root	Monocot stem
(3)	Dicot stem	Monocot stem
(4)	Monocot stem	Dicot stem

BAP425

21. Which of the following statement(s) is/are true?

(a) Ground tissue system is fundamental tissue system.

(b) Root hairs are unicellular.

(c) The trichomes help in preventing water loss.

(1) Only a (2) Only a & b

(3) Only b & c (4) All a, b & c

BAP426

22. **Statement-I :** The root hairs are unicellular elongation of the epidermal cells.

Statement-II : The epidermal hairs of stem are called trichomes, which are usually multicellular.

AIM : FREE EDUCATION TO ALL APUL

(1) Both Statement-I & II are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & II is incorrect.

BAP427

23. **Assertion:** Tracheid and vessel elements are found in xylem tissue.
Reason: Tracheids are responsible for only lateral conduction of water, while vessel elements primarily conduct water longitudinally.
Options:
(1) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true, but the Reason is false.
(4) Assertion is false, but the Reason is true.

BAP428

24. **Statement I:** The pith in dicot stems is a region of parenchyma cells located in the center of the stem, primarily involved in storage.
Statement II: Pith cells in dicot stems are lignified and provide structural support to the stem, especially during secondary growth.
Options:
(1) Both statements I & II are correct, and statement II is the correct explanation of statement I.
(2) Both statements I & II are correct, but statement II is NOT the correct explanation of statement I.
(3) Statement I is correct, but statement II is incorrect.
(4) Statement II is correct, but statement I is incorrect.

BAP429

25. Consider the following statements:
A. In a dicot root, the vascular bundles are collateral and end arch.
B. The inner most layer of cortex is endodermis in a dicot root.
C. In a dicot root, the phloem masses are separated from the xylem by parenchymatous cells that are known as the conjunctive tissue.
Of these statements given above
(1) (A) is true but (B) and (C) are false
(2) (B) is true but (A) and (C) are false
(3) (B) and (C) are true but (A) is false
(4) (A) and (C) are true but (B) is false

BAP430

26. **Assertion:** Ground tissue system is the fundamental tissue system of plant body.
Reason: Majority of the plant body consists of ground tissue system.
Options:
(1) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true, but the Reason is false.
(4) Assertion is false, but the Reason is true.

BAP431

27. **Assertion:** The primary function of root hairs is to increase the surface area for absorption of water and minerals.
Reason: Root hairs are extensions of epidermal cells and lack a cuticle layer.
(1) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true, but the Reason is false.
(4) Assertion is false, but the Reason is true.

BAP432

AIM : FREE EDUCATION TO ALL APUL

28. **Statement-I:** Collenchymatous cells provide mechanical support to young plant organs due to their thickened primary cell wall.
Statement-II: Sclerenchyma cells are living cells that undergo secondary thickening and provide additional structural support to the mature plant.
Options:
(1) Both Statement-I and II are correct.
(2) Statement-I is correct, but Statement-II is incorrect.
(3) Statement-II is correct, but Statement-I is incorrect.
(4) Both Statement-I and II are incorrect.
BAP433

29. **Statement-I:** The casparian strip, composed of suberin, is present in the endodermis of plant roots and acts as a barrier to the passive flow of water and solutes into the vascular system.
Statement-II: The pericycle is a layer of cells located between the endodermis and the vascular tissue, responsible for the initiation of lateral root formation.
Options:
(1) Both Statement-I and II are correct.
(2) Statement-I is correct, but Statement-II is incorrect.
(3) Statement-II is correct, but Statement-I is incorrect.
(4) Both Statement-I and II are incorrect.
BAP434

30. **Assertion:** Xylem is conducting as well as mechanical tissue.
Reason: Xylem helps in transportation of water, minerals and show deposition of lignin on wall of their constituents.
Options:
(1) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true, but the Reason is false.
(4) Assertion is false, but the Reason is true.
BAP435

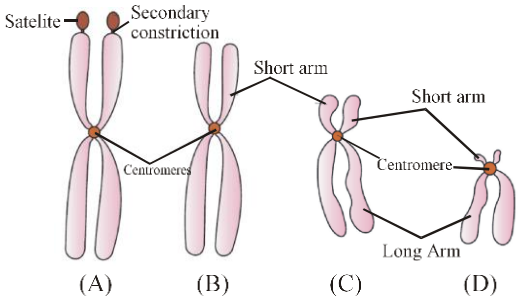
31. **Assertion:** Parenchyma cells are versatile and perform various functions in plant tissues.
Reason: Parenchyma cells have thin cell walls and large central vacuoles, making them suitable for storage, photosynthesis, and secretion.
Options:
(1) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true, but the Reason is false.
(4) Assertion is false, but the Reason is true.
BAP436

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	2	3	4	2	3	1	4	2	1	2	2	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	2	2	3	4	1	3	3	3	1	1	4	1	1
Question	31														
Answer	1														

Exercise - I
Cell : The unit of life

1. The chromosome "B" in this diagram is :-



(1) Metacentric (2) Sub-metacentric
(3) Acrocentric (4) Telocentric

BCL347

2. Normally one cell contains _____ nucleus.
(1) One (2) Two
(3) Three (4) Ten

BCL348

3. The term Chromatin was given by :-
(1) W. Flemming (2) Robert Brown
(3) George Palade (4) Camillo Golgi

BCL349

4. Ribosomal subunits and rRNA are synthesized in which site of nucleus ?
(1) Nucleolus
(2) Cytoplasm
(3) Nucleoplasm
(4) Perinuclear Space

BCL350

5. Telomerase is an enzyme which is a :-
(1) Simple protein (2) RNA
(3) Ribonucleoprotein (4) Repetitive DNA

BCL351

6. Microbodies are present in :-
(1) Both in Plant and Animal cells
(2) Only in Plant cells
(3) Only in Animal cells
(4) Only in bacteria

BCL352

7. Which of the following organelle (s) is/are not include in endomembrane system?
(1) Endoplasmic Reticulum
(2) Golgi body
(3) Lysosome and vacuoles
(4) Mitochondria, chloroplast and ribosome.

BCL353

8. Lysosome arises from which cell organelle?
(1) Nucleus
(2) Endoplasmic reticulum
(3) Golgi Apparatus
(4) Both (2) & (3)

BCL354

9. Detoxification of harmful drugs is the function of :-
(1) RER (2) SER
(3) Lysosome (4) Mitochondria

BCL355

10. The suicidal bag of an animal cell is :-
(1) Sphaerosome (2) Dictyosome
(3) Lysosome (4) Mitochondria

BCL356

11. Mitochondria and chloroplast divides by :-
(1) Meiosis-I (2) Mitosis
(3) Fission (4) Meiosis

BCL357

12. Which of the following cell organelles lack DNA?
(1) Mitochondria (2) Chloroplast
(3) Nucleus (4) Lysosome

BCL358

13. In which cell organelle following reaction takes place?
$6CO_2 + 12\ H_2O \rightarrow 6O_2 + 6H_2O$
(1) Mitochondria
(2) Peroxisome
(3) Chloroplast
(4) Glyoxisome

BCL359

14. Reserve materials in prokaryotic cells are stored in the cytoplasm, related with_____.
(1) Inclusion bodies
(2) Golgi body
(3) ER
(4) Polysome

BCL360

15. The basic unit of life is :-
(1) Cell (2) Tissue
(3) Cell organelle (4) Mitochondria

BCL361

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

16. All living beings are composed of _____?
(1) Cells and product of cells
(2) Cells only
(3) Product of cells only
(4) Nucleus only

BCL362

17. Match the columns I & II and identify the incorrect option :-

Column-I		Column-II	
(1)	Robert brown	(i)	Discovered the Nucleus
(2)	Robert Hooke	(ii)	First observed cell from cork
(3)	Rudolf Virchow	(iii)	Omnis cellula e cellula
(4)	Anton von Leeuwenhoek	(iv)	Discovered cell Theory

BCL363

18. In Human beings, the membrane of the erythrocyte has approximately :-
(1) 52 % Protein and 40 % Lipids
(2) 80 % Protein and 20% Lipids
(3) 40% Protein and 52% Lipids
(4) 52% Carbohydrate and 20% Protein

BCL364

19. The fluid mosaic model of plasma membrane given by Singer and Nicolson is applicable for:
(1) Only for prokaryotic membrane
(2) Only for eukaryotic membrane
(3) Both for prokaryotic and eukaryotic membranes
(4) Only for organellar membranes

BCL365

20. The smooth endoplasmic reticulum is the major site for :-
(1) Synthesis of protein
(2) Synthesis of carbohydrate
(3) Synthesis of lipid
(4) Synthesis of nucleic acid

BCL366

21. Golgi apparatus is the important site for the formation of_____.
(1) Glycoprotein and Glycolipids
(2) Only protein
(3) Lipid formation
(4) Only vitamins

BCL367

22. Which element is required for maintaining the structure of ribosome ?
(1) Mg^{+2} (2) Ca^{+2}
(3) Zn^{+2} (4) Cl^{-}

BCL368

23. Plastids are found in :-
(1) All plant cells and in Euglenoids
(2) All animal cells and in Euglenoids
(3) All bacteria and animal cells
(4) All plants and bacteria

BCL369

24. The three dimensional network of protein filaments with in the cytoplasm of eukaryotic cell is called the :-
(1) Endoplasmic Reticulum
(2) Golgi apparatus
(3) Cytoskeleton
(4) Peroxisome

BCL370

25. Shape of chloroplast found in *Spirogyra* is :-
(1) Lens shaped (2) Oval
(3) Spherical (4) Ribbon like

BCL371

26. Starch is stored in :-
(1) Amyloplasts (2) Elaioplasts
(3) Aleuroplast (4) Cytoskeleton

BCL372

27. Chloroplast, chromoplast, leucoplast are classified under the :-
(1) Plastid (2) Cytoskeleton
(3) Mitochondria (4) Lysosome

BCL373

28. Which cell organelle is related with protein synthesis ?
(1) Cytoskeleton (2) Ribosome
(3) Lysosome (4) Glyoxysome

BCL374

AIM : FREE EDUCATION TO ALL APUL

29. The pattern arrangement of axonemal microtubules is referred to as the ____ array :
(1) 9 + 2 (2) 9 + 0
(3) 27 + 2 (4) 27 + 0
BCL375

30. Which cell organelle is formed by Golgi apparatus.
(1) Lysosome (2) Vacuole
(3) Ribosomes (4) Both (1) and (2)
BCL376

31. Given below are two statements:
Statement-I : Ribosomes are known as organelle within organelle.
Statement-II : Peroxisomes degrade H_2O_2 into H_2O and O_2 with the help of catalase enzyme.
In the light of the above statements, choose the most appropriate answer from the options given below :-
(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II both are incorrect.
BCL377

32. Given below are two statements :-
Statement-I : Nucleus is called to be controller or director of cell.
Statement-II : Golgi complex is endosymbiont in origin.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II are incorrect.
BCL378

33. Given below are two statements :-
Statement-I : Membrane of lysosome is known as tonoplast.
Statement-II : Contractile vacuole helps in osmoregulation.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II both are incorrect.
BCL379

34. Given below are two statements:
Statement-I : Mitochondria and chloroplasts are called semi autonomous cell organelles.
Statement-II : Golgi complex is called suicidal bag of cell.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II both are incorrect.
BCL380

35. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : Middle lamellae is like cementing material between two adjacent plant cells.
Reason (R) : Plasmodesmata traverse cell wall and middle lamellae.

AIM : FREE EDUCATION TO ALL APUL

In the light of the above statements, choose the **most appropriate** answer from the options given below :-

(1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct.

(4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

BCL381

36. Select the incorrect statement among them.

(1) Prokaryotes have cell wall surrounding their plasma membrane.

(2) Genomic DNA provides resistance to antibiotics in bacterial.

(3) Plasmid DNA is used to monitor bacterial transformation with foreign DNA.

(4) Ribosomes is found in prokaryotes also.

BCL382

37. Select the mismatch option :-

(1) Cell wall – Prevent bacteria from bursting

(2) Slime layer – loose sheath

(3) Cell wall – determine the shape of host cell

(4) Capsule -Thick and tough

BCL383

38. The ribosomes which are found in prokaryotes are :-

(1) The site of protein synthesis

(2) 50S and 30S subunits

(3) 70S and form polysome

(4) All of these

BCL384

39. Which properties differentiate animal and plant cell :-

(a) Presence of cell wall

(b) Small size vacuole

(c) Plastid

(d) Centriole

(1) All (2) a, b

(3) a, b, c (4) a, c, d

BCL385

40. Cell membrane of the organism have :-

(1) Phospholipid bilayer.

(2) Hydrophobic tail of unsaturated hydrocarbon.

(3) Maximum amount of cholesterol.

(4) Both (1) and (2)

BCL386

41. Match column I with column II :-

	Column I		Column II
(i)	Algae	(a)	Diminished as the cell membrane mature.
(ii)	Primary wall	(b)	Calcium-pectate
(iii)	Middle lamella	(c)	Capable of growth
(iv)	Secondary wall	(d)	Have minerals like $CaCO_3$
		(e)	Formed towards the membrane

(1) (i)–d, (ii)–a, (iii)–b, (iv)–e

(2) (i)–a, (ii)–d, (iii)–b, (iv)–e

(3) (i)–d, (ii)–c, (iii)–b, (iv)–e

(4) (i)–d, (ii)–e, (iii)–a, (iv)–c

BCL387

42. Read the following properties :-

(i) Network of reticulum of tiny tubular structure.

(ii) Intracellular space has two compartment i.e. luminal and extra luminal.

(iii) If it has ribosome it will synthesis protein.

(iv) Help in nuclear membrane formation.

The above properties is of :-

(1) Golgi body and Endoplasmic reticulum.

(2) Endoplasmic reticulum only.

(3) Mitochondria and Golgi body

(4) Nucleus and Nucleolus

BCL388

43. The organelle which have varied number of cisternae and are concentrically arranged near the nucleus with distinct face is :-
(1) Endoplasmic reticulum
(2) Mitochondria
(3) Golgi body
(4) Rough endoplasmic reticulum

BCL389

44. Which statement is wrong among them :-
(1) Lysosomes are formed by the process of packaging in the Golgi apparatus and are rich in hydrolytic enzyme.
(2) Vacuole contain water sap, excretory product and other useful material of cell.
(3) In amoeba contractile vacuole help in excretion.
(4) Mitochondria unless specifically stained are not easily visible under microscope.

BCL390

45. Read the following statement :
(i) Larger and move numerous nucleoli in cell which actively carrying out protein synthesis.
(ii) Content of nucleolus is discontinuous with the rest of nucleoplasm.
(iii) Chromatin in interphase.
(iv) Outer membrane usually remain continuous with endoplasmic reticulum.

How many statements among them are incorrect.
(1) Two
(2) One
(3) Four
(4) Three

BCL391

Exercise – I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	1	3	1	4	4	2	3	3	4	3	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	1	3	3	1	1	1	3	4	1	1	2	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	3	2	1	2	3	4	4	4	3	2	3	2	2

AIM : FREE EDUCATION TO ALL APUL

Exercise - II
Cell : The unit of life

1. Match the Column-I and Column-II and select the correct answer :-

Column - I	Column - II
(A) Metacentric	(i) At the tip
(B) Sub-Metacentric	(ii) At the middle
(C) Acrocentric	(iii) Almost near the tip
(D) Telocentric	(iv) Slightly away from the middle

(1) A - ii, B - iv, C - i, D - iii
(2) A - ii, B - iv, C - iii, D - i
(3) A - iv, B - ii, C - iii, D - i
(4) A - i, B - iii, C - ii, D - iv

BCL392

2. Function of nuclear pores is .....
(1) Through these pores movement of RNA and protein molecules in both directions between the nucleus and the cytoplasm is performed.
(2) Transport of RNA in both direction between the Nucleus and the cytoplasm is performed.
(3) Only for transport of RNA in unidirectional manner from nucleus to cytoplasm.
(4) 1 and 3 option is correct.

BCL393

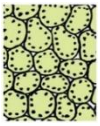
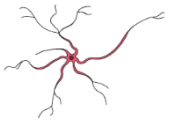
3. Which option provide correct information regarding the nucleus.
(1) Normally, there is only one nucleus per cell, variation in the number of nuclei are also observed.
(2) Normally, there is multiple nucleus per cell.
(3) *Chlamydomonas* contains only one nucleus.
(4) 1 and 3 option is correct.

BCL394

4. Which option is correct regarding eukaryotic Cell:
(1) Cytoplasm of Eukaryotic cell contain 70 S and 80S ribosome.
(2) Eukaryotic cell contain linear ds DNA in chloroplast and mitochondria.
(3) Only 80S ribosomes are found in cytoplasm, mitochondria and chloroplast of eukaryotic cell.
(4) 80S ribosomes are present in cytoplasm of eukaryotic cell only.

BCL395

5. Incorrect match between column -I, II and III:

Column-I		Column-II	
(1)	RBCs		Round/ biconcave
(2)	WBCs		Amoeboid
(3)	Mesophyll		Round and oval
(4)	A Tracheid		Branched & Elongated

BCL396

6. Match between column I and column-II and choose the correct answer ?

Column-I		Column-II	
(A)	Typical Bacteria	(i)	10 - 20 μm
(B)	PPLO	(ii)	0.02 - 0.2 μm
(C)	Viruses	(iii)	0.1 μm
(D)	A Typical eukaryotic cell	(iv)	1 - 2 μm

(1) A-iv B-iii C-ii D-i
(2) A-iv B-ii C-i D-iii
(3) A-i B-ii C-iii D-iv
(4) A-iii B-iv C-ii D-i

BCL397

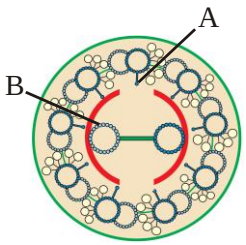
7. In plants, the ___A___ facilitates the transport of a number of ions and other materials against concentration gradient into the ___B___.
(1) A - Tonoplast B - Vacuole
(2) A - Outer membrane B - Mitochondria
(3) A - Outer membrane B - Chloroplast
(4) A - Tonoplast B - Mitochondria

BCL398

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

8. The figure below shows the section of a cilia/flagella showing different parts labelled A and B select the correct labelled name ?



(1) A – Radial spoke
B – Central microtubule
(2) A – Interdoublet Bridge
B – Peripheral microtubule
(3) A – Centeral sheath
B – Plasma membrane
(4) A – Interdoublet Bridge
B – Radial spoke

BCL399

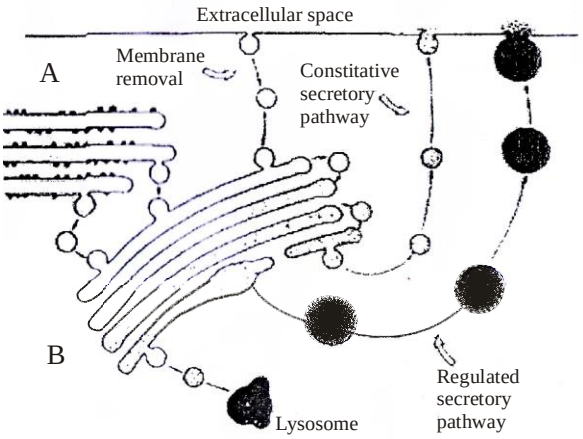
9. Which statement provides correct information regarding prokaryotic and eukaryotic flagella:
(1) The prokaryotic cell also possess flagella but these are structurally different from that of eukaryotic flagella.
(2) Prokaryotic and eukaryotic both flagella are structurally similar.
(3) In case of both microtubule arrangement is 9 + 2.
(4) In case of both flagella is made up of microtubule and flagellin protein.

BCL400

10. Which similarity is found between chloroplast and mitochondria ?
(1) Both are semi-autonomous contain double stranded circular DNA and double membrane bound.
(2) Both are autonomous organelle.
(3) Both are single membrane bounded.
(4) Both contains 80S type of ribosome.

BCL401

11. Select the correct for label "B" in this diagram:



(1) Endoplasmic reticulum
(2) Peroxisome
(3) Golgi apparatus
(4) Glyoxysome

BCL402

12. Secondary cell wall is formed on :-
(1) The inner side of primary cell wall
(2) The outer side of primary cell wall
(3) Middle side of cell
(4) Intercellular space

BCL403

13. The cytoskeleton in a cell are involved in many functions such as :-
(1) Mechanical support
(2) Motility
(3) Maintenance of shape
(4) All of these

BCL404

14. **Statement-I :** Microbodies are present in both animal and plants cells.
Statement-II : Eukaryotic cell contain linear ds DNA in chloroplast and mitochondria.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Both Statement-I and II are correct.
(4) Statement-I is incorrect but Statement-II are correct.

BCL405

AIM : FREE EDUCATION TO ALL APUL

15. **Statement-I :** Smooth endoplasmic reticulum is major site for synthesis of lipid.
Statement-II : In animal cells lipid like steroid hormone are synthesised in SER.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II are correct.
(3) Both Statement-I and II are incorrect.
(4) Both Statement-I and II are correct.
BCL406

16. **Statement-I :** Centrosomes is an organelle usually containing two cylindrical structure called centrioles.
Statement-II : Centrioles in centrosomes lie parallel to each other in which each has an organisation like cart wheel.
In the light of the above statements, choose the most appropriate answer from the options given below :-
(1) Both Statement-I and II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II are correct.
(4) Both Statement-I and II are incorrect.
BCL407

17. **Assertion :** Lysosomes help in digestion of foreign particle in animal cells.
Reason : Lysosome have digestive enzymes.
In the light of the above statements, choose the most appropriate answer from the options given below :-
(1) (A) is correct but (R) is not correct.
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).
BCL408

18. **Assertion :** Chromoplast is coloured plastid in corolla and fruit.
Reason : It has water soluble chlorophyll and carotenoid pigments.
In the light of the above statements, choose the most appropriate answer from the options given below :-
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(4) (A) is not correct but (R) is correct.
BCL409

19. **Assertion :** RER is frequently observed in cell actively involved in protein synthesis and secretion.
Reason : RER have ribosomes attached to its outer surface.
In the light of the above statements, choose the most appropriate answer from the options given below :-
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is not correct but (R) is correct.
(3) (A) is correct but (R) is not correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).
BCL410

20. Given below are two statements:
Statement-I : Oxidative phosphorylation take place in mitochondria.
Statement-II : Calvin cycle take place in chloroplast.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II are incorrect.
BCL411

AIM : FREE EDUCATION TO ALL APUL

21. Given below are two statements :-

Statement-I : Chloroplast is double membranous structure.

Statement-II : Inner membrane of chloroplast is more permeable.

In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Both Statement-I and II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and II are incorrect.

BCL412

22. Given below are two statements:

Statement-I : Periplasmic space in bacteria contain enzyme similar to lysosome.

Statement-II : Heterophagosomes are formed by fusion of primary lysosome and phagosome.

In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Both Statement-I and II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and II are incorrect.

BCL413

23. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : Fluid mosaic model is most widely accepted model of plasma membrane.

Reason (R) : It states that protein icebergs are present in mosaic pattern in the sea of glycolipid.

In the light of the above statements, choose the **most appropriate** answer from the options given below :-

(1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct.

(4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

BCL414

24. Identify for correct set of statements for endoplasmic reticulum :-

(a) ER is known as system of membranes

(b) It has cis-trans face

(c) Ribosomes are present on the surface of RER

(d) SER are the sites of synthesis of lipids and cholesterols

Choose the correct answer from the given options below.

(1) all a, b, c & d are correct

(2) b, c and d are correct

(3) only c and d are correct

(4) a, c and d are correct

BCL415

25. Identify the correct set of statements out of followings :-

(a) Maximum arm ratio exhibited by acrocentric chromosomes

(b) Human cells (2n) has two genomes and each genome consists fourty six chromosomes

(c) Nucleolar organizer regions are not present on each human chromosomes

(d) Secondary constriction is present at the place of centromere

(e) Each centromere posses two protein discs called kinetochores

Choose the correct answer from the given options below.

(1) b, c, d & e (2) a, b, c & d

(3) a, c & e (4) a, b & c

BCL416

AIM : FREE EDUCATION TO ALL APUL

26. Centromere holds two chromatids of a chromosome based on its position the sub-metacentric chromosome has :-
(1) Middle centromere forming two equal arms of chromosome.
(2) Centromere greatly away from middle of chromosome.
(3) One shorter arm and one longer arm.
(4) One extremely short and one very long arm.

BCL417

27. Select the incorrect statement among them for chromosome :-
(1) Mitochondria is present in 80s containing cytoplasm.
(2) Telocentric chromosome has terminal kinetochores.
(3) Chromatin also contain non-histone protein and RNA.
(4) Satellite have not constant location in few chromosomes.

BCL418

28. The organelle in cell which is surrounded by amorphous pericentriolar material has :-
(1) Two centriole which lie perpendicular to each other.
(2) Same arrangement of tubules as in cilia or flagella.
(3) Peripheral fibril which is double as compare to cilia.
(4) Important role during generative and vegetative cell formation by division in microspore.

BCL419

29. How many points are related to nucleus are true :-
(i) Nucleus was first discovered by Robert Brown.
(ii) Chromatin was stained by acidic dyes.
(iii) Chromatin discovered earlier than nucleus.
(iv) Nucleoli is visible in any stage of cell.
(v) Nuclear envelope have two parallel membrane with space between them called perinuclear space.
(1) (i), (ii), (iv) (2) (i), (v)
(3) (i), (iii), (iv), (v) (4) (i), (iii), (v)

BCL420

30. Cytoskeleton of cell have :-
(1) An elaborate network of filamentous proteinaceous structure.
(2) Element that maintains cyclosis.
(3) It helps in cell recognition.
(4) Both option (1) and (2) are correct.

BCL421

31. The long arm and short arm of chromosome which appears during metaphase of cell cycle is represented by :-
(1) Long arm –s, Short arm–p
(2) Long arm–p, Short arm–q
(3) Long arm–p, Short arm–s
(4) Short arm–p, Long arm–q

BCL422

32. Pick the mis-match one :-
(1) Golgi body – Glycosylation
(2) Radial spoke–Connected to singlet
(3) Inter doublet bridge–is present between A-B (microtubule of doublet)
(4) Centrioles form basal body of cilia or flagella.

BCL423

33. Ribosomes of prokaryote is mostly 70S and ribosomes which is present in cytoplasm of eukaryote is 80S. So here S represent :-
(1) Svedberg's unit, which indirectly measure shape and size.
(2) Svedberg's unit, which indirectly measure density and size.
(3) Svedberg's unit, which indirectly measure shape and density.
(4) Svedberg's unit, which indirectly measure molecules number of ribosomes.

BCL424

34. How many statements are correct for mitochondria :-
(a) Matrix of mitochondria is heterogenous.
(b) Matrix of mitochondria is homogenous.
(c) Matrix of mitochondria has high H^+ concentration.
(d) The space between outer and inner membrane of mitochondria is called intra-membrane space.
(e) It is power house of all living cells.
(1) Two (2) One
(3) Three (4) Four

BCL425

35. Pick the incorrect statement :-
(1) Steroids pass easily through lipid layer
(2) Non polar substance pass easily through lipid layer
(3) Small polar molecule may be passes easily by lipid layer.
(4) Both polar charged and non-polar passes easily by lipid layer.

BCL426

AIM : FREE EDUCATION TO ALL APUL

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	4	4	4	1	1	1	1	1	3	1	4	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	2	4	1	2	1	2	4	3	3	4	1	2	4
Question	31	32	33	34	35										
Answer	4	2	2	2	4										

Exercise - I
Cell Cycle and Cell Division

1. Chromosomes with variations in progeny cells are found in case of _____.
(1) Mitosis (2) Meiosis
(3) Amitosis (4) Fission
BCD213

2. Bouquet stage and synapsis is absent in case of:
(1) Mitosis
(2) Heterotypic division
(3) Meiosis-I
(4) Reductional division
BCD214

3. In which type of division nuclear division takes place twice but division of chromosome occurs only once ?
(1) Meiosis (2) Fission
(3) Mitosis (4) Amitosis
BCD215

4. If the cell had 4n number of chromosome at G_1, after S phase the number of chromosome in the cell will be :-
(1) 2n (2) n (3) 4n (4) 8n
BCD216

5. Nuclear membrane, ER, Golgibody, Nucleolus reappears in which stage of karyokinesis :-
(1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase
BCD217

6. Which phase of prophase-I represents transition of metaphase I?
(1) Zygotene (2) Pachytene
(3) Diakinesis (4) Diplotene
BCD218

7. Bivalents arrange on equator (congression) of cell to form :
(1) Pole (2) Metaphase plate
(3) Telophase plate (4) Near to pole
BCD219

8. Interkinesis is present between.....
(1) Meiosis-I and Meiosis-II
(2) Leptotene and Zygotene
(3) Pachytene and diplotene
(4) Diplotene and diakinesis
BCD220

9. The term meiosis was given by :-
(1) Farmer and Moore
(2) Watson and Crick
(3) Alexander Fleming
(4) Flaming and Skoog
BCD221

10. Nuclear membrane and Nucleolus will disappear at the end of :-
(1) Leptotene (2) Zygotene
(3) Diplotene (4) Diakinesis
BCD222

11. Which enzyme is responsible for recombination process in meiosis-I ?
(1) Recombinase
(2) Protease
(3) DNA ligase only
(4) DNA Polymerase
BCD223

12. *Yeast* for example can progress through the cell cycle in only about.....
(1) 40 minutes (2) 90 minutes
(3) 1440 minutes (4) 60 minutes
BCD224

13. First check point occurred during cell cycle is:
(1) $G_1 \longrightarrow$ S-phase
(2) $G_2 \longrightarrow$ Prophase
(3) S $\longrightarrow$ G_2 phase
(4) Metaphase $\longrightarrow$ Anaphase
BCD225

14. During which phase(s) of cell cycle, amount of DNA in a cell remains at 8C level. If the initial amount is denoted as 4C ?
(1) G_0 and G_1 (2) G_1 and S
(3) Only G_2 (4) G_2 and Telophase
BCD226

15. Which of the following is wrong about S-phase in animal cell?
(1) DNA duplication occur in S-phase
(2) Cell synthesized Histone protein in S-phase
(3) Centriole duplication occur in S-phase
(4) Synthesis of tubulin protein in S-phase
BCD227

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

16. During cell division, centriole duplication takes place in :-
(1) G_2-phase (2) M-phase
(3) S-phase (4) G_1-phase
BCD228

17. Which of the following is smallest phase of the cell cycle ?
(1) Prophase (2) Interphase
(3) Anaphase (4) Metaphase
BCD229

18. Best material for the study of mitosis in laboratory is :-
(1) Anther (2) Onion root tip
(3) Leaf tip (4) Ovary
BCD230

19. Prophase is marked by :
(1) The initiation of condensation of chromosomal material
(2) DNA Duplication
(3) Transcription and translation
(4) Duplication of centriole
BCD231

20. Which of the following is not related to mitosis?
(1) It is an equational division.
(2) It involves recombination and crossing over.
(3) Sister chromatids are separated during Anaphase.
(4) Nuclear membrane disappears during late prophase.
BCD232

21. By which method, cytokinesis occurs in Animals:
(1) Centripetal
(2) Centrifugal
(3) Oblique
(4) Equational
BCD233

22. Meiosis process does not occur in_____.
(1) Somatic cell
(2) Germ cell
(3) Microspore mother cell
(4) Megaspore mother cell
BCD234

23. A cell has 48 chromosomes at each pole in mitotic telophase. In this division the number of chromatids at the metaphase was :
(1) 24 (2) 48 (3) 69 (4) 46
BCD235

24. Refer to the following statements describing Prophase I of meiosis and answer the question.
(i) Thin thread like chromosomes
(ii) Appearance of recombination nodules
(iii) Formation of bivalents/tetrads
(iv) Terminalisation of chiasmata
(v) Appearance of chiasmata
Arrange the given statements in the correct sequence their occurrence during prophase I
(1) (i) → (iii) → (ii) → (iv) → (v)
(2) (i) → (ii) → (iii) → (iv) → (v)
(3) (i) → (iv) → (v) → (ii) → (iii)
(4) (i) → (iii) → (ii) → (v) → (iv)
BCD236

25. If the female gamete of an organism has 10 pg of DNA in its nucleus. How much DNA would a diploid cell of same organism have in zygotene (A) diplotene (B) and anaphase-I (C)? (pg = Picogram)
(1) A = 40 pg, B = 20 pg, C = 10 pg
(2) A = 20 pg, B = 40 pg, C = 10 pg
(3) A = 40 pg, B = 40 pg, C = 40 pg
(4) A = 40 pg, B = 40 pg, C = 20 pg
BCD237

26. If in metaphase I, n = 20. Then which of the following is true?
(1) 20 tetrads, 40 chromatids
(2) 80 chromatids, 80 kinetochores
(3) 10 bivalents, 80 chromatids
(4) 10 bivalents, 40 kinetochores
BCD238

27. Which statement is correct w.r.t. anaphase I chromosomes at each pole?
(1) Twice number of chromosomes as compared to gametes
(2) One chromatid in each chromosome.
(3) Two chromatids in each chromosome.
(4) Same number of chromosomes as that of parent cell.
BCD239

28. Given below are two statements :
Statement-I : During G_0-phase of cell cycle, the cell is metabolically inactive.
Statement-II : Heart cells enter G_0-phase and divide on when required.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BCD240

29. In which phase (stage) synthesis of proteins for formation of spindle fibres takes place :-
(1) G_1-phase (2) S-phase
(3) M-phase (4) G_2-phase

BCD241

30. Given below are two statements :-
Statement-I : Chromosomes decondense in telophase I to form chromatin to extended state of interphase nucleus.
Statement-II : Chromosomes recondense in prophase II.
(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Both Statement-I and Statement-II are incorrect.

BCD242

AIM : FREE EDUCATION TO ALL APUL

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	3	4	3	2	1	1	4	1	2	1	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	2	1	2	1	1	2	4	3	2	3	3	4	2

Exercise - II
Cell Cycle and Cell Division

1. In a diploid cell before S-phase quantity of DNA is 40 picogram (pg) after meiosis-I what will be the quantity of DNA is each daughter cell ?
(1) 10 pg (2) 20 pg
(3) 5 pg (4) 40 pg

BCD243

2. The specialised kind of cell division that reduces the chromosome number by half results in the production of haploid daughter cells. This kind of division is called .....
(1) Mitosis (2) Meiosis
(3) Amitosis (4) Fission

BCD244

3. Which one of the following statement is correct :-
(1) Meiosis-I equivalent to mitosis division.
(2) The major objective of meiosis-I is segregation of homologous chromosome after recombination.
(3) The major objective of meiosis-II is Segregation of Homologous chromosome after Recombination.
(4) Division of chromosomes or centromere occurs during meiosis-I.

BCD245

4. Recombination process occur between.....
(1) Non sister chromatids of Homologous chromosomes.
(2) Sister chromatid of Homologous chromosome.
(3) Sister chromatids of non homologous chromosomes.
(4) Non sister chromatids of Non homologous chromosomes.

BCD246

5. What is the significance of meiosis :-
(1) The conservation of specific chromosome number of each species is achieved across generations in sexually reproducing organisms.
(2) It increases the genetic variability in the population of organisms from one generation to the next.
(3) Meiosis decreases the genetic variability in the population of organisms from one generation to the next.
(4) 1 and 2 option is correct.

BCD247

6. Cells at the end of prophase, when viewed under the microscope do not show_____.
(1) Golgi complexes, endoplasmic reticulum, nucleolus and the nuclear envelope
(2) Only nuclear envelope
(3) Only endoplasmic reticulum
(4) Only nucleolus

BCD248

7. **Statement-I :** If the cell has diploid or '2n' number of chromosomes at G_1 then after 'S' phase the number of chromosomes will be double (4n).
Statement-II : During prophase, protein are synthesized in preparation for mitosis.
In the light of the above statements, choose the most appropriate answer from the options given below:-
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II are correct.
(4) Both Statement-I and II are correct.

BCD249

8. **Statement-I :** Centriole begin to move towards opposite poles of the cell in anaphase.
Statement-II: Condensation of chromosomes is completed in metaphase.
In the light of the above statements, choose the most appropriate answer from the options given below:-
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

(3) Statement-I is incorrect but Statement-II are correct.
(4) Both Statement-I and statement-II are correct.

BCD250

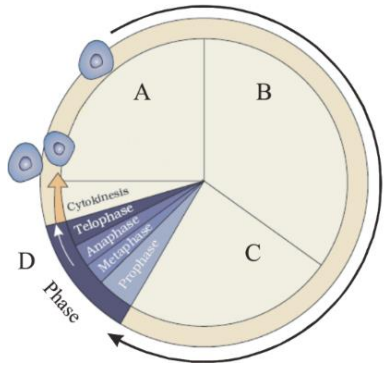
9. **Assertion :** Amount of DNA get doubled in 'S' phase of interphase.
Reason : DNA replication occur in 'S' phase of interphase.
In the light of the above statements, choose the most appropriate answer from the options given below:-
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

BCD251

10. **Assertion :** The individual chromosomes can no longer be seen and chromatin material tends to collect in a mass in the poles in telophase.
Reason : Centromeres split and chromatids separates in telophase.
In the light of the above statements, choose the most appropriate answer from the options given below:-
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

BCD252

11. A diagrammatic view of cell cycle indicating formation of two cells from one cell, is given below. Identify the different phases (A, B, C & D) of cell cycle with their corresponding events:-



I. DNA replication
II. Organelles duplication
III. Protein synthesis
IV. Actual phase of cell division

	I	II	III	IV
(1)	B	A	A, B, C	D
(2)	B	A, C	C	D
(3)	A, C	B	C	D
(4)	B	A	C	D

BCD253

12. **Assertion :** Meiosis is essential for maintaining specific number of chromosome in a species across generations.
Reason : Meiosis increase genetic variability in sexual reproducing organisms.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

BCD254

13. **Assertion (A) :** In coconut, liquid endosperm represents syncytium.
Reason (R) : In coconut liquid karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises in endosperm.
(1) Both (A) and (R) are correct but (R) is the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

BCD255

AIM : FREE EDUCATION TO ALL APUL

14. **Statement-1 :** In meiosis, DNA replication occurs once whereas nuclear division occurs twice.

Statement-2 : Crossing over occurs between non-sister chromatids of non-homologous chromosomes.

(1) Statement-1 is true and statement-2 is correct explanation for statement-1

(2) Statement-1 is true and statement-2 is also true but statement-2 is not correct explanation of statement-1

(3) Statement-1 is true but statement-2 is false

(4) Statement-1 and statement-2 both are false

BCD256

15. Which of the following option correctly describes the cell as shown in figure ?



(1) 8 chromosomes ; 8 homologous pair

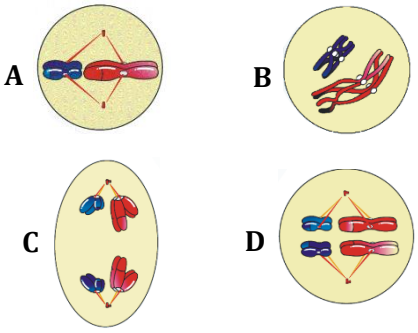
(2) 16 chromosomes ; 4 homologous pair

(3) 8 chromosomes ; 4 homologous pair

(4) 16 chromosomes ; 8 homologous pair

BCD257

16. Which of the following is a correct sequences of meiotic cell cycle w.r.t given diagram ?



(1) A → C → D → B

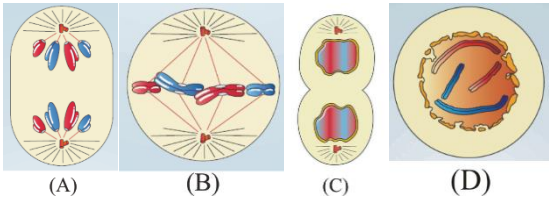
(2) B → D → C → A

(3) B → A → C → D

(4) D → B → A → C

BCD258

17. The four figures (A, B, C and D) given below represent four different stages of mitotic cell division. Which one of these is correctly identified in the options given, along with its correct description and key events ?



		Stage	Description	Key event
(1)	D	Prophase	Followed by G_2 Phase	Bivalent Chromosomes align on the equatorial plate
(2)	B	Metaphase	Condensation of chromosomes completed	Daughter chromosomes move to opposite poles
(3)	C	Telophase	Longer and highly complex	Chromosomes undergo some dispersion
(4)	A	Anaphase	Centromeres are towards poles and arms trailing behind	Centromere split and chromatids separate

BCD259

18. Which one of the following pairs is **correctly** matched :-

(1)	Anaphase I	Homologous chromosomes are separated
(2)	Metaphase I	The microtubules from opposite poles of the spindle get attached to the kinetochores of sister chromatids.
(3)	Meiotic Interphase	A nuclear envelope encloses each haploid set of chromosomes.
(4)	Prophase I	Non-homologous chromosomes undergo crossing over.

BCD260

AIM : FREE EDUCATION TO ALL APUL

19. How many of the above are correctly matched?

	Phase	Characteristic feature
(i)	Metaphase	Spindle fibres attach to kinetochores
(ii)	G_2 phase	Centriole duplication in cytoplasm
(iii)	G_1 phase	Interval between mitosis phase and initiation of DNA replication in nucleus
(iv)	Telophase I	Chromosomes do not reach extremely extended state of interphase nucleus
(v)	Prophase	Movement of duplicated centrioles towards equator of the cell

(1) Four (2) Three
(3) One (4) Two

BCD261

20. Match column I and column II.

	Column-I		Column-II
(I)	Chromosomes are moved to spindle equator	(A)	Pachytene
(II)	Pair of synapsed homologous chromosomes	(B)	Zygotene
(III)	Pairing between homologous chromosomes	(C)	Bivalent
(IV)	Crossing over between homologous chromosomes	(D)	Metaphase

(1) I→A, II→B, III→C, IV→D
(2) I→B, II→C, III→D, IV→A
(3) I→D, II→C, III→B, IV→A
(4) I→C, II→A, III→D, IV→A

BCD262

21. **Assertion:** Mitosis maintains the genetic similarity of somatic cells.
Reason: Chromosomes do not undergo crossing over during mitotic division.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BCD263

22. In meiosis I, the centromere undergoes:
(1) No division
(2) Division between anaphase and interphase
(3) Division between prophase and metaphase
(4) Division but the daughter chromosomes do not separate

BCD264

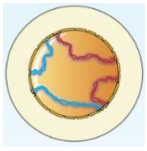
23. If the number of chromosomes in Metaphase I is n = 15; which of the following options is true?
(1) 15 bivalents, 30 tetrads
(2) 30 chromatids, 60 kinetochores
(3) 15 tetrads, 60 chromatids
(4) 60 kinetochores, 20 tetrads

BCD265

24. In which of the following phases, nuclear envelope, nucleolus, ER, GB is not seen?
Early Prophase, Anaphase, Telophase, Metaphase I, Metaphase II, Anaphase II, Late Prophase
(1) Four (2) Five
(3) Three (4) Seven

BCD266

25. Find the incorrect statement w.r.t the given diagram.



(1) First stage of karyokinesis of mitosis after the G_2 phase of interphase.
(2) Chromosomal material becomes entangled during the process of chromatin condensation.
(3) The centrosome begins to move towards opposite poles of the cell.
(4) The two asters together with spindle fibres form mitotic apparatus.

BCD267

AIM : FREE EDUCATION TO ALL APUL

26. Which of the following is an incorrect statement?
(1) Once the chromatids reach the two poles, the chromosomal elongation starts, nucleolus and the nuclear membrane reappear. This stage is called the telophase.
(2) Chromosome condensation occurs during prophase.
(3) During metaphase II the bivalents arrange on the equatorial plate.
(4) The stages through which a cell passes from one division to the next is called the cell cycle.

BCD268

27. Mitotic anaphase differs from metaphase per cell in possessing:
(1) Double number of chromosomes and half number of chromatids.
(2) Half the number of chromosomes and same number of chromatids.
(3) Half the number of chromosomes and double number of chromatids.
(4) Double number of chromosomes and same number of chromatids.

BCD269

28. A bivalent of meiosis-I consists of:
(1) Two chromatids and one centromere
(2) Two chromatids and two centromeres
(3) Four chromatids and two centromeres
(4) Four chromatids and four centromeres.

BCD270

29. **Assertion:** Morphology of chromosomes is studied in metaphase stage.
Reason: Condensation of chromosomes is completed in this stage and can be easily visualized under microscope.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BCD271

30. A cell has 50 chromosomes and 50 pg DNA in G_1 phase. What would be the number of chromosomes and DNA content in anaphase?
(1) 50 and 50 respectively
(2) 100 and 50 respectively
(3) 50 and 100 respectively
(4) 100 and 100 respectively

BCD272

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	1	4	1	1	3	4	2	1	1	1	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	1	2	3	1	1	3	2	2	3	4	3	1	4

Exercise - I
Plant Growth and Development

1. Arithmetic growth can be observed in :
(1) BGA colony (2) Yeast colony
(3) Bacteria colony (4) Root elongation
BPD225

2. What is the basic reason behind more than one type of leaves in a plant?
(1) Due to different growth rates
(2) Due to lack of nutrients
(3) Due to nutrients toxicity
(4) Due to plasticity
BPD226

3. Which of the following PGRs shows dual role, growth promotion and growth inhibition?
(1) Cytokinin (2) Gibberellin
(3) Ethylene (4) Abscisic acid
BPD227

4. What is the basic reason behind leaves remaining above the water in rice plant ?
(1) Bolting due to gibberellin
(2) Apical dominance due to auxin
(3) Internodal elongation due to ethylene
(4) Due to photoperiodism
BPD228

5. Foolish seedling disease or Bakanae disease in rice caused by a fungus helped in the discovery of :-
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Ethylene
BPD229

6. Triple response hormone on stem is :-
(1) Abscisic acid (2) Ethylene
(3) Cytokinin (4) Auxin
BPD230

7. Hormone related with α–amylase induction is :
(1) Cytokinin (2) ABA
(3) Ethylene (4) GA
BPD231

8. Which of the following is a synergistic function of auxin and cytokinin ?
(1) Promotion of flowering in pineapple
(2) Morphogenesis in plants
(3) Increase in absorption surface in plants
(4) Induction of dormancy in seeds
BPD232

9. Which of the following is/are biennial and monocarpic plant(s) ?
(1) Sugar beet (2) Wheat
(3) Barley (4) All of the above
BPD233

10. Which one of the following is incorrect ?
(1) Auxin was first isolated from human urine.
(2) Differentiation in plant is open.
(3) Ethylene promote root growth and root hair formation.
(4) Auxin help to overcome apical dominance.
BPD234

11. Secondary growth does occur in-
(1) All angiosperms and no gymnosperms
(2) Some angiosperms and gymnosperms
(3) No angiosperms and only gymnosperms
(4) All angiosperms and all gymnosperms
BPD235

12. The three phases of growth in correct order is
(1) Meristematic, maturation, elongation
(2) Elongation, meristematic, maturation
(3) Meristematic, elongation, maturation
(4) Elongation, maturation, meristematic
BPD236

AIM : FREE EDUCATION TO ALL APUL

13. PGR Ethylene can fit into-

(1) Group of plant growth promoters

(2) Group of plant growth inhibitors

(3) Both the groups of promoter & inhibitors but mostly inhibitor

(4) Both the groups of promoter & inhibitors but mostly promoter

BPD237

14. Ethylene in fruits causes-

(1) Rise in rate of respiration called respiratory anti-climactic

(2) Rise in rate of respiration called respiratory climactic

(3) Fall in rate of respiration called respirastory anti-climactic

(4) Fall in rate of respiration called respiratory climactic

BPD238

15. Photoperiods are perceived by-

(1) Shoot apices

(2) Flowering apices

(3) Nodal buds

(4) Leaves

BPD239

AIM : FREE EDUCATION TO ALL APUL

Exercise – I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	3	2	2	4	2	1	4	2	3	3	2	4

Exercise - II
Plant Growth and Development

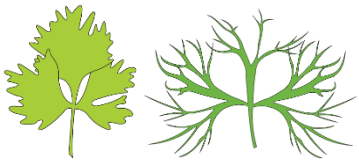
1. Number of female flowers increase in cucumber, due to :
(1) Feminising effect of auxin
(2) Feminising effect of cytokinin
(3) Feminising effect of ethylene
(4) Feminising effect of gibberellins
BPD240

2. Development in plants :- (choose the correct one)
(1) Is only under control of intrinsic factors
(2) Includes plasmatic growth followed by cell division, elongation and differentiation
(3) Is governed by intercellular factors like genetic factors
(4) Is not influenced by environmental factors
BPD241

3. Open form of growth involves.
(1) Continuous division in all meristematic cells forming new set of cells which divide further necessarily.
(2) Division in some cells forming new set of cells which do not divide further necessarily
(3) No division at all
(4) Division in some cells at the time of injury which forms new set of cells to heal the injury
BPD242

4. Choose the correct set of option for size or weight of organ against time-
(i) $W_1 = W_0 e^{rt}$
(ii) $L_t = L_0 rt$
(iii) Linear growth curve
(iv) Sigmoid growth curve
(v) Arithmetic growth
(vi) Geometric growth
(1) (i), (iii), (iv)
(2) (ii), (iii), (vi)
(3) (i), (iv), (vi)
(4) (ii), (iv), (v)
BPD243

5. The given figure show-



(1) Developmental heterophylly in buttercups
(2) Environmental heterophylly in larkspur
(3) Environmental heterophylly in buttercups
(4) Developmental heterophylly in larkspur
BPD244

6. Match the PGR with the plants which played role in their discovery-

	Column-I		Column-II
(a)	Auxin	(i)	Tobacco
(b)	Gibberellin	(ii)	Rice
(c)	Cytokinin	(iii)	Orange
(d)	Ethylene	(iv)	*Avena*

(1) a-iii, b-i, c-ii, d-iv (2) a-iv, b-ii, c-i, d-iii
(3) a-i, b-iii, c-iv, d-ii (4) a-ii, b-iv, c-iii, d-i
BPD245

7. **Assertion :** Auxin is used in plant propagation widely.
Reason : Auxin initiate rooting.
(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion
(3) Assertion is correct but Reason is incorrect
(4) Both Assertion & Reason are incorrect
BPD246

8. **Assertion :** Spraying gibberellins on fruits extend its market period.
Reason : Gibberellins delays senescence.
Choose the best option-
(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion.
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect
(4) Both Assertion & Reason are incorrect
BPD247

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

9. **Assertion :** Juvenile conifers are sprayed with GA.
Reason : GA delays senescence and malting period.
Which of the given options are correct?
(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion
(3) Assertion is correct but Reason is incorrect
(4) Both Assertion & Reason are incorrect
BPD248

10. **Statement-I :** Ethylene promotes internode/ petiole elongation in deep water rice plants.
Statement-II : Ethylene helps upper part of shoot to remain above water.
Choose the appropriate option-
(1) Statement-I is correct and Statement-II are incorrect.
(2) Statement-I is incorrect and statement-II is correct.
(3) Statement-I and Statement-II are correct.
(4) Statement-I and Statement-II are incorrect.
BPD249

11. Read the given statements –
Statement-I : Some plants require a periodic exposure to light to induce flowering.
Statement-II : Some plants are able to measure the duration of exposure to light.
Choose the correct option-
(1) Both statements are correct and are about photoperiodism
(2) One of the statements is correct about photoperiodism
(3) Both statements are correct and is about vernalization
(4) One of the statements is correct about vernalization
BPD250

12. Identify the correct and incorrect statements from the following.
(i) More than 17,500 new cells are produced per hour by a single maize root apical meristem.
(ii) With the help of length, growth of pollen tube is measured.
(iii) The growth of the leaf is measured in terms of volume.
(iv) Cells in a watermelon may increase in size by upto 3,50,000 times.
(1) (i),(ii),(iii)are correct and (iv) is incorrect.
(2) (i),(ii),(iv)are correct and (iii) is incorrect.
(3) (ii), (iii) are correct and (i), (iv) are incorrect.
(4) (i), (iv) are correct and (ii), (iii) are incorrect.
BPD251

13. Which of the following is an incorrect statement?
(1) Long day plants flower if the night length is shorter than a critical period.
(2) Short-day plants flower when night length exceeds a critical dark period.
(3) Day-neutral plants are insensitive to day length.
(4) The process of flowering does not occur in day neutral plants
BPD252

14. **Assertion :-** Development is a term that includes all changes that an organism goes through during its life cycle.
Reason :- Plants follows different pathways in response to environment or phases of life to form different kinds of structures.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.
BPD253

AIM : FREE EDUCATION TO ALL APUL

15. **Assertion :** Cytokinins helps in the delay of leaf senescence.
Reason : Cytokinins promote nutrient mobilisation.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BPD254

16. **Assertion :** A sigmoid curve is a characteristic of living organism growing in a natural environment.
Reason : Sigmoid curve is a straight line curve.
(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

BPD255

17. Match the phytohormones given in column-I with their functions given in column-II. Choose the answer with correct combination of alphabets of the two column :-

Column-I Phytohormones)		**Column-II (Functions)**	
(A)	Auxins	(p)	Elongation of genetic dwarf plants
(B)	Gibberellins	(q)	Inducing fruit ripening
(C)	Cytokinins	(r)	Formation of abscission layer
(D)	Ethylene	(s)	Root initiation
		(t)	Richmond lang effect

(1) A-r, B-s, C-p, D-t (2) A-p, B-r, C-q, D-s
(3) A-s, B-t, C-r, D-q (4) A-s, B-p, C-t, D-q

BPD256

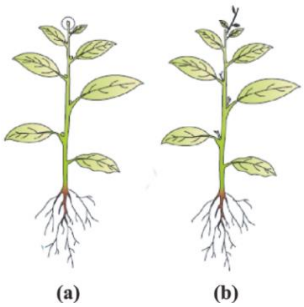
18. Match column-I with column-II.

Column-I		**Column-II**	
(I)	IAA	(A)	Terpenes
(II)	GA	(B)	Indole compounds
(III)	ABA	(C)	Adenine derivatives
(IV)	Kinetin	(D)	Carotenoid derivatives

(1) I - B, II - D, III - A, IV - A
(2) I - B, II - A, III - D, IV - C
(3) I - C, II - A, III - D, IV - B
(4) I - D, II - B, III - A, IV - C

BPD257

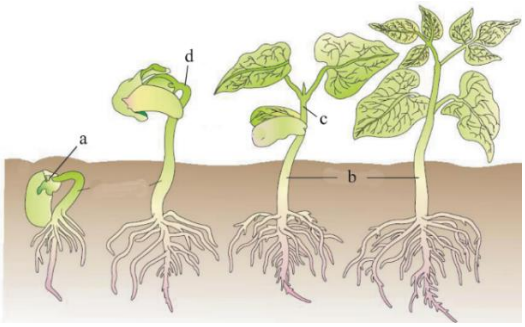
19. Represented below are the two phenomena in plants. Among the following two phenomena (a) and (b) which phenomena is widely applied in tea plantation?



(1) Phenomena (a)
(2) Phenomena (b)
(3) Phenomena (a) and (b) both
(4) Neither phenomena (a) nor (b)

BPD258

20. Find out correct labelling sequence in given figure:



(1) a = Cotyledon, b = Hypocotyle,
c = Epicotyle, d = Epicotyle hook

AIM : FREE EDUCATION TO ALL APUL

(2) a = Hypocotyle, b = Cotyledon, c = Epicotyle, d = Epicotyle hook
(3) a = Epicotyle, b = Epicotyle hook, c = Cotyledon, d = Hypocotyle
(4) a = Cotyledon, b = Hypocotyle, c = Epicotyle hook, d = Epicotyle

BPD259

21. **Statement-I :** Growth is measured by a variety of paraments some of which are increase in fresh weight, dry weight, length area, volume and cell number.
Statement-II : A single maize root apical meristem can give rise to more than 17500 new cells per hour.
(1) Both Statement-I & II are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & II is incorrect.

BPD260

22. **Statement-I :** In plant growth, the role of PGR is of only one kind of extrinsic control.
Statement-II : Along with genomic control and extrinsic factors, PGR play an important role in plant growth and development.
(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect.
(3) Only Statement-I is correct.
(4) Only Statement-II is correct.

BPD261

23. **Statement-I :** Auxins was first isolated from human urine.
Statement-II : Cytokinin was discovered as kinetin from the autoclaved herring sperm DNA.
(1) Both Statement-I & II are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & II is incorrect.

BPD262

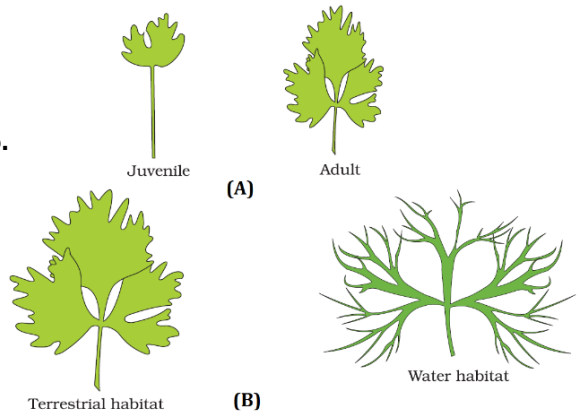
24. Auxin helps in -
1) Promotes flowering in pineapple
(2) Preparation of weed free lawns
(3) Cell division
(4) All of the above

BPD263

25. Which hormone helps the plants to increase their absorption surface ?
(1) Abscisic acid (2) Cytokinin's
(3) Ethylene (4) Auxin

BPD264

26.



(A) & (B) represent -
(1) (A)-Heterophilly in tomato, (B)-Heterophilly in buttercup
(2) (A)-Heterophilly in larkspur, (B)-Heterophilly in buttercup
(3) (A)-Heterophilly in pea, (B)-Heterophilly in buttercup
(4) All of the above

BPD265

27. Match the following.

(A)	Ethylene	(i)	Root hair formation
(B)	Kinetin	(ii)	Corn kernels
(C)	IAA	(iii)	Dormin
(D)	Zeatin	(iv)	Human urine
(E)	ABA	(v)	Herring sperm DNA

(1) A-(ii), B-(iii), C-(v), D-(ii), E-(i)
(2) A-(i), B-(v), C-(iv), D-(ii), E-(iii)
(3) A-(i), B-(iv), C-(iii), D-(ii), E-(v)
(4) A-(ii), B-(iv), C-(iii), D-(iii), E-(v)

BPD266

28. Which hormone is present in extract of vascular tissues ?
(1) Gibberellins (2) Ethylene
(3) Cytokinins (4) Indole acetic acid
BPD267

29. **Assertion :** Plant retain the capacity for unlimited growth throughout their life.
Reason : This capacity of plant is due to the meristem presence at certain locations.
(1) Both (A) and (R) is correct and (R) is the correct explanation of (A).
(2) (A) is incorrect but (R) is correct.
(3) Both (A) and (R) is correct and (R) is not the correct explanation of (A).
(4) Both (A) and (R) are incorrect.
BPD268

30. **Assertion :** Growth at a cellular level is a consequence of increase in the amount of protoplasm.
Reason : Increase in protoplasm is difficult to measure directly.
(1) Both (A) and (R) is correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) is correct and (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) (A) is incorrect but (R) is correct.
BPD269

AIM : FREE EDUCATION TO ALL APUL

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	1	3	3	2	1	1	3	3	1	2	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	2	2	1	1	4	1	4	3	2	2	3	1	2

Exercise - I
Respiration in Plants

1. First CO_2 release in aerobic respiration occurs during :-
(1) 3-PGAL formation
(2) Pyruvic acid formation
(3) Acetyl CoA formation
(4) α-Ketoglutaric acid formation
BRP203

2. At how many steps substrate level phosphorylation occurs during glycolysis?
(1) One (2) Two
(3) Three (4) Four
BRP204

3. All enzymes are present in matrix of mitochondria for TCA cycle except :-
(1) α-Ketoglutaric acid dehydrogenase
(2) Citrate synthase
(3) Succinate dehydrogenase
(4) Decarboxylase
BRP205

4. ETS is inhibited at complex-IV by :-
(1) Cyanide (2) Rotenone
(3) Antimycin (4) Oligomycin
BRP206

5. Select the correct match for one glucose molecule:-
(1) EMP pathway – 12 ATP
(2) Link reaction – 8 ATP
(3) Krebs cycle – 24 ATP
(4) Anaerobic respiration – 6 ATP
BRP207

6. Select the correct statement about anaerobic respiration :-
(1) More than 7% of total energy of glucose is generated
(2) Yeast poison themselves to death when concentration of alcohol reaches about 30%
(3) Net gain is of 2ATP.
(4) CO_2 is produced in all types of fermentations
BRP208

7. Match the column-I with column-II and select the correct option :-

Column-I		Column-II	
(A)	Alcoholic fermentation	(i)	3-carbon compound
(B)	Lactic acid fermentation	(ii)	CO_2 production
(C)	Pyruvic acid	(iii)	Muscles during vigorous exercise in animals

(1) A-iii, B-ii, C-i (2) A-iii, B-i, C-ii
(3) A-ii, B-iii, C-i (4) A-i, B-ii, C-iii
BRP209

8. RQ is more than one if respiratory substrate is:
(1) Tripalmitin (2) Glucose
(3) Protein (4) Malic acid
BRP210

9. Select the correct statement regarding R.Q. ?
(1) RQ depends upon type of respiratory substrate
(2) R.Q. is less than one in case of organic acids
(3) R.Q. is always more than one in case of carbohydrate
(4) R.Q. is 0.7 in succulent plants
BRP211

10. If 15 Acetyl CoA enter into Krebs cycle then how many CO_2 molecules are produced after complete decarboxylation?
(1) 60 CO_2 molecule (2) 15 CO_2 molecule
(3) 30 CO_2 molecule (4) 75 CO_2 molecule
BRP212

11. Which of the following enzymes is called as mitochondrial marker enzyme?
(1) Carboxylase
(2) Decarboxylase
(3) Alcohol dehydrogenase
(4) Succinate dehydrogenase
BRP213

12. The C–C bond of complex compound in broken by which process in respiration?
(1) Oxidation
(2) Reduction
(3) Hydrogenation
(4) None of the above
BRP214

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

13. Choose the correct equation :-
(1) $C_6H_{12}O_6 + 12O_2 \rightarrow 6H_2O + 6H_2O + Energy$
(2) $C_6H_{12}O_6 + 3O_2 \rightarrow 2CO_2 + 3H_2O + Energy$
(3) $C_6H_{12}O_6 + 6CO_2 \rightarrow 6H_2O + Energy$
(4) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy$
BRP215

14. Glycolysis is a how many steps of process?
(1) Ten (2) Eight
(3) Eleven (4) Five
BRP216

15. One molecule of glucose is converted into how many molecules of pyruvic acid?
(1) 1 (2) 2
(3) 3 (4) 4
BRP217

16. How many molecules of ATP are directly produced in one complete glycolysis of one glucose molecule?
(1) 1 (2) 2 (3) 3 (4) 4
BRP218

17. In alcoholic fermentation, pyruvate is converted to which among the following?
(1) Ethanol, CO_2, NADH
(2) CO_2 and Methanol
(3) CO_2 and Ethanol only
(4) CO_2 and Carboxylic acid
BRP219

18. The range beyond which yeasts poison themselves to death in alcohol fermentation when the concentration of alcohol reaches to?
(1) 13% (2) 15% (3) 12% (4) 17%
BRP220

19. Pyruvate enters to the mitochondrial matrix and undergoes.
(1) Reductive decarboxylation
(2) Oxidative carboxylation
(3) Reductive carboxylation
(4) Oxidative decarboxylation
BRP221

20. One molecule of glucose synthesizes how many molecules of NADH + H^+ in TCA cycle?
(1) 6 (2) 7 (3) 8 (4) 10
BRP222

21. For each ATP produced, _____ passes through F_0 from intermembrane space to the matrix down the electrochemical proton gradient.
(1) H^+ (2) $2H^+$
(3) $3H^+$ (4) $4H^+$
BRP223

22. In the balance sheet of fermentation, net gain is :-
(1) 12 ATP molecules (2) 38 ATP molecules
(3) 2 ATP molecules (4) 8 ATP molecules
BRP224

23. What is RQ if carbohydrates are used as a respiratory substrate?
(1) 1 (2) 0.8 (3) 0.7 (4) 0.9
BRP225

24. Choose the correct.
(1) RQ = volume of CO_2 evolved/volume of O_2 consumed
(2) RQ = volume of O_2 consumed /volume of CO_2 evolved
(3) RQ = volume of O_2 evolved/volume of CO_2 consumed
(4) None of these
BRP226

25. What is the final end product of TCA cycles?
(1) 3 NADH + H^+ (2) 1 ATP
(3) 1 $FADH_2$ (4) All of the above
BRP227

26. Oxidation of NADH to NAD^+ is :-
(1) Slower in fermentation than aerobic respiration
(2) Faster in fermentation than aerobic respiration
(3) Equal in fermentation than aerobic respiration
(4) Cannot be compared
BRP228

27. In ETS, O_2 accept the electrons and get reduced to which of the following?
(1) Water
(2) Carbon dioxide
(3) Palmitic acid
(4) None of the above
BRP229

28. How many statements are correct about fermentation?

(i) Very low amount of energy is released, < 7% of energy in glucose is released in fermentation

(ii) In animal cells, when oxygen is inadequate acetic acid is formed during respiration

(iii) It is dangerous process as it leads to acid and alcohol formation.

(1) 0 (2) 1

(3) 2 (4) 4

BRP230

AIM : FREE EDUCATION TO ALL APUL

Exercise - I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	3	1	3	3	3	4	1	3	4	1	4	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	4	3	1	4	1	4	3	1	1	4	1	1	3		

Exercise - II
Respiration in Plants

1. Select the incorrect statement regarding amphibolic pathway :-
(1) All carbohydrates are usually converted into glucose before they used for respiration
(2) It includes both anabolism and catabolism
(3) Acetyl CoA is the connecting link between fat, carbohydrate and protein metabolism
(4) Proteins can enter into the respiratory pathway only at one point i.e. as α-ketoglutaric acid.

BRP231

2. Which one of the following is incorrect about cellular respiration?
(1) Glycolysis takes place inside cytoplasm of all living cells.
(2) Value of RQ for fat is 0.9
(3) Yeast poison themselves to death when the concentration of alcohol reaches about 13%.
(4) During complete oxidation of glucose, oxidation take place in both cytoplasm and mitochondria.

BRP232

3. **Statement-I :** Cytochrome c is a small protein remain attached to outer surface of inner mitochondrial membrane.
Statement-II : Cytochrome c is a mobile e^- carrier between complex-III and IV.
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BRP233

4. **Statement-I :** ATP act as energy currency of cell
Statement-II : Energy released through respiration is trapped as bio-chemical energy in the from of ATP.
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BRP234

5. Which among the following step in glycolysis yields energy?
(i) Conversion of BPGA to PGA
(ii) Conversion of fructose-6-phosphate to fructose 1, 6-bisphosphate
(iii) Conversion of PEP to pyruvate acid
(iv) Conversion of glucose-6-phosphate to fructose-6-phosphate
(1) Both ii and iv (2) i, ii, iii
(3) Both i and iii (4) All of the above

BRP235

6. Which among the following is wrong?
(i) Glycolysis occurs in all living organism.
(ii) TCA cycle and ETS only occurs in aerobes
(iii) Complete oxidation of pyruvate occurs by removal of all hydrogen atom in TCA cycle.
(1) (i) (2) (ii)
(3) (iii) (4) None of the above

BRP236

7. Which among the following is the role of O_2 in whole respiration process?
(i) Act as hydrogen removal from the system
(ii) Act as final hydrogen acceptor
(iii) It bond with C atom and released CO_2, one of the by product of respiration.
(1) Both ii and iii
(2) iii only
(3) Both i and ii
(4) All of the above

BRP237

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL ♥ APUL

8. The respiratory balance sheet is calculated on some assumptions.
Which of the following assumption is correct?
(1) The pathway is sequential, with series of glycolysis, ETS and TCA cycle in the same order for a given molecules.
(2) NADH is transferred to chloroplast where oxidative phosphorylation occurs, leading to formation of 3 ATP.
(3) Only glucose is the substrate and none other substrate or intermediate enters or leaves the pathway.
(4) None of these.

BRP238

9. Choose the correct option according to the correct sequence (from substrate to end product) :-
(i) Glucose 6-phosphate
(ii) Pyruvic acid
(iii) Carbohydrate
(iv) Fructose-1, 6-bisphosphate
(v) Glucose
(vi) Dihydroxy Acetone Phosphate $\rightleftharpoons$ Glyceraldehyde 3-phosphate
(1) i, iii, iv, v, vi, ii
(2) iii, iv, v, ii, i, vi
(3) iii, v, i, ii, iv, vi
(4) iii, v, i, iv, vi, ii

BRP239

10. Match the following :-

1. $NADH + H^+$	i. 1 ATP
2. $FADH_2$	ii. 2 ATP
3. GTP	iii. 3 ATP

(1) 1-i, 2-ii, 3-iii (2) 1-i, 2-iii, 3-ii
(3) 1-iii, 2-ii, 3-i (4) 1-ii, 2-i, 3-iii

BRP240

11. **Assertion :-** Two redox-equivalents are removed from PGAL during glycolysis.
Reason :- Two molecules of pyruvic acids are formed at the end of glycolysis.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BRP241

12. Which among the following is wrong about ATP synthase?
(i) It is also called complex V.
(ii) This is used to synthesis ATP by utilising the energy released during ETS.
(iii) It works on the basis of proton and F_0.
(iv) It consist of two major components, F_1 and F_0.
(1) Only ii (2) Both i and iii
(3) i and iv (4) None of the above

BRP242

13. Match the following :-

1. Amino acids	i. Pyruvic acid
2. Fatty acid	ii. Dihyroxy Acetone Phosphate
3. Glycerol	iii. Acetyl CoA

(1) 1-i, 2-iii, 3-ii (2) 1-ii, 2-i, 3-iii
(3) 1-iii, 2-i, 3-ii (4) 1-ii, 2-iii, 3-i

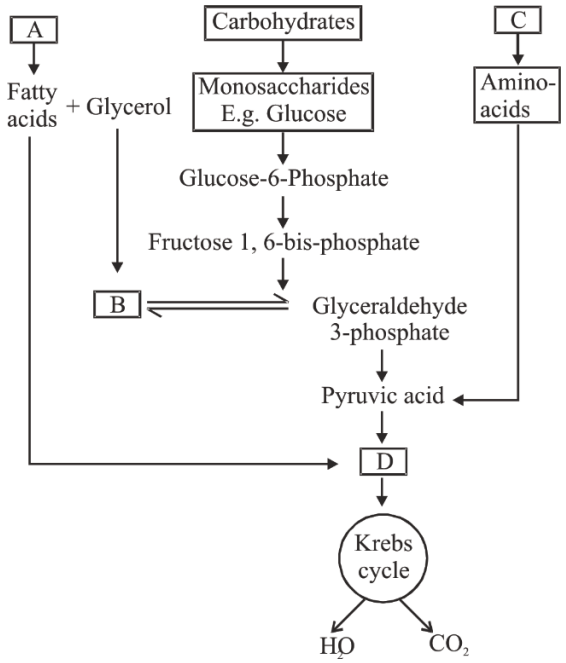
BRP243

14. **Assertion:** The respiratory pathway comes into the picture both during breakdown and synthesis of fatty acids.
Reason : fatty acids would be broken down to acetyl CoA before entering the respiratory pathway. But when the organism needs to synthesise fatty acids, acetyl CoA would be withdrawn from the respiratory pathway.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BRP244

AIM : FREE EDUCATION TO ALL APUL

15.

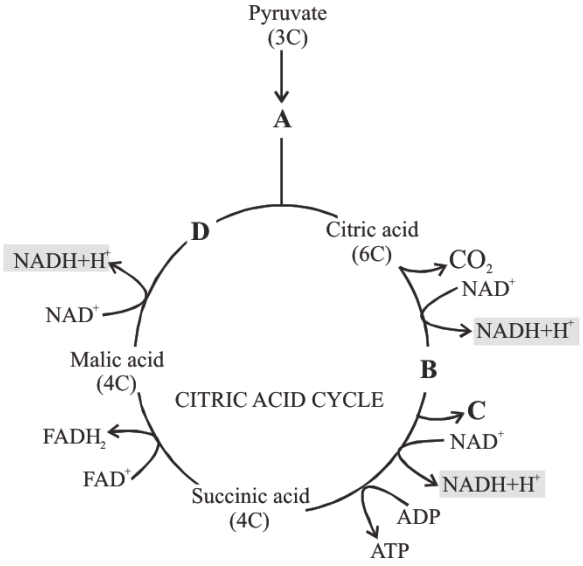


The above figure indicates the interrelationship among metabolic pathways. Now identify A to D.

	A	B	C	D
(1)	Protein	Acetyl CoA	Fats	DHAP
(2)	Fats	DHAP	Protein	Acetyl CoA
(3)	Acetyl CoA	Fats	DHAP	Protein
(4)	Fats	DHAP	Acetyl CoA	Protein

BRP245

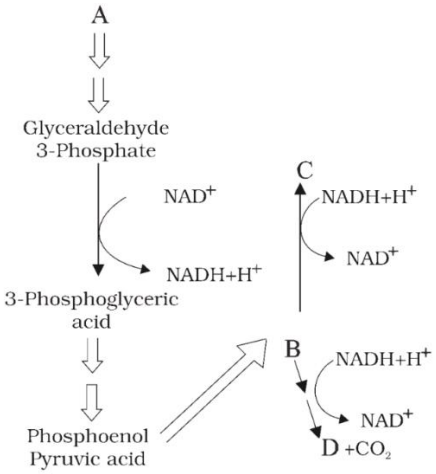
16. In the given representation of Krebs cycle, Identify A, B, C and D.



	A	B	C	D
(1)	OAA	Acetyl CoA	CO_2	α-ketoglutaric acid
(2)	OAA	Acetyl CoA	CoA	α-ketoglutaric acid
(3)	Acetyl CoA	OOA	CoA	CO_2
(4)	Acetyl CoA	α-ketoglutaric acid	CO_2	OAA

BRP246

17. Choose the correct option for A to D.



(1) A-Fat, B-key product of glycolysis, C-Lactic acid, D-Ethanol

(2) A-Carbohydrate, B-Pyruvic acid, C-Ethanol, D-Lactic acid

(3) A-Respiratory substrate, B-Key product of glycolysis, C-lactic acid, D-Ethanol

(4) A-Proteins, B-Pyruvic acid, C-Lactic acid, D-Ethanol.

BRP247

18. Match the column :-

Column-I		Column-II	
(A)	NADH dehydrogenase	(1)	Complex IV
(B)	Cytochrome bc_1	(2)	Complex III
(C)	Cytochrome a-a_3	(3)	Complex I
(D)	ATP synthase	(4)	Complex V

(1) A–1, B–2, C–3, D–4

(2) A–3, B–2, C–1, D–4

(3) A–4, B–1, C–3, D–2

(4) A–1, B–4, C–2, D–3

BRP248

AIM : FREE EDUCATION TO ALL APUL

19. Which one of the following options give the correct number of ATP molecules released during different processes (A, B, C) of aerobic respiration of one glucose?

	A Glycolysis	B One Link Reaction	C One Krebs cycle
(1)	6	3	12
(2)	8	2	11
(3)	8	3	11
(4)	8	2	12

BRP249

20. Select the wrong match with respect to respiratory quotient (R.Q.) :-

(1) Proteins = 0.9 (2) Carbohydrate = 1

(3) Tripalmitin = 4 (4) Organic acid > 1

BRP250

21. Select correct option with respect to number of carbon present in respiratory intermediate:

	Pyruvic acid	Acetyl CoA	Citric acid	OAA
(1)	3C	2C	5C	4C
(2)	2C	2C	6C	4C
(3)	3C	2C	6C	4C
(4)	3C	2C	6C	5C

BRP251

22. Match the Column-I with Column-II and choose the correct combination from the option given below:-

	Column-I		Column-II
(A)	Inner mitochondrial membrane	I	Krebs cycle
(B)	Pyruvic acid is converted into CO_2 and ethanol	II	ETC
(C)	Cytoplasm	III	Fermentation
(D)	Mitochondrial matrix	IV	Glycolysis

(1) A-IV, B-III, C-II, D-I

(2) A-I, B-II, C-IV, D-III

(3) A-II, B-I, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

BRP252

23. **Statement-I :** During glycolysis two redox-equivalents are removed from PGAL and transferred to a molecule of NAD+.

Statement-II : PGAL is reduced to 1,3-Bis PGA during glycolysis.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

BRP253

24. **Statement-I :** Fermentation accounts for only a partial breakdown of glucose whereas in aerobic respiration it is completely breakdown to CO_2 and H_2O.

Statement-II : In fermentation there is a net gain of only two molecules of ATP for each molecules glucose breakdown to alcohol and CO_2.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

BRP254

25. **Statement-I :** The electron, as they move through the electron transport system release enough energy that are trapped to synthesize ATP, called oxidative phosphorylation.

Statement-II : In oxidative phosphorylation O_2 is the ultimate acceptor of electron and it gets reduced to water.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

BRP255

26. Chemiosmotic hypothesis of ATP synthesis in respiration based on-

(1) Proton gradient between lumen and stroma.

(2) Proton gradient between matrix and inner membrane.

(3) Proton gradient between matrix and inner membrane space.

(4) Proton gradient between matrix and inter membrane space.

BRP256

AIM : FREE EDUCATION TO ALL APUL

27. Which is not true for glycolysis -
(1) End product is pyruvic acid
(2) Expenditure of ATP
(3) Formation of $NADPH_2$
(4) Consumption of ATP

BRP257

28. RQ value 0.7 represent that -
(1) Glucose is used as respiratory substrate
(2) Organic acid is used as respiratory substrate
(3) Protein is used as respiratory substrate
(4) Fat is used as respiratory substrate

BRP258

29. Which of the following process use oxygen as the final hydrogen acceptor-
(1) Fermentation
(2) Glycolysis
(3) Oxidative phosphorylation
(4) Krebs cycle

BRP259

30. End product of fermentation are-
(1) Ethanol + CO_2 (2) Lactic acid
(3) Pyruvic acid + CO_2 (4) Both (1) and (2)

BRP260

31. In which step of glycolysis $NADH_2$ formation occurs :-
(1) Triose phosphate $\rightarrow$ Triose bisphosphate
(2) DHAP $\rightarrow$ G3P
(3) 3PGA $\rightarrow$ 2PGA
(4) 2 Phosphoenol pyruvic acid $\rightarrow$ pyruvic acid

BRP261

32. If 5 molecules of water are produced in ETS than find the number of $NADH_2$ oxidised and electron required respectively, in mitochondria :-
(1) 10, 10 (2) 5, 10
(3) 6, 15 (4) 8, 20

BRP262

33. Select incorrect statement :-
(1) For each ATP, $4H^+$ passes through F_0 by active transport
(2) Oxygen acts as a final hydrogen accepter.
(3) F_0 is an integral membrane protein complex
(4) Proton passes through intermembrane space to the matrix down the electrochemical proton gradient.

BRP263

34. Select the incorrect assumption for calculations on respiratory balance sheet-
(1) There is a sequential orderly pathway.
(2) $NADH_2$ synthesised in glycolysis transferred in to the mitochondria.
(3) Intermediates in the pathway are utilised to synthesise any other compound.
(4) Only glucose is being respired.

BRP264

35. Pyruvic acid + CoA + NAD^+ $\xrightarrow[\text{Pyruvate dehyrogenase}]{(A)}$ Acetyl CoA + CO_2 + NADH + H^+
A is –
(1) Coenzyme A (2) Mg^{+2}
(3) K^+ (4) ATP

BRP265

Exercise - II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	4	4	3	4	3	3	4	3	2	4	1	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	2	1	3	3	4	3	1	1	4	3	4	3	4
Question	31	32	33	34	35										
Answer	1	2	1	3	2										

Exercise - I
Photosynthesis in Higher Plants

1. O_2 evolved during light reaction of photosynthesis comes from H_2O was proved by:
(1) T.W. Engelmann
(2) Joseph Priestley
(3) Ruben and Kamen
(4) Julius Von Sachs
BPP323

2. Which of the following pigments is an universal photosynthetic pigment ?
(1) Chlorophyll-a (2) Chlorophyll-b
(3) Chlorophyll-c (4) Chlorophyll-d
BPP324

3. Experiment for Emerson enhancement and red drop was conducted on :
(1) *Chlamydomonas*
(2) *Polysiphonia*
(3) *Chlorella*
(4) *Dictyota*
BPP325

4. How many additional ATPs are required in C_4 plants as compared to C_3 plants to synthesize 2 molecules of hexose sugar ?
(1) 12 ATP (2) 24 ATP
(3) 36 ATP (4) 48 ATP
BPP326

5. Select the incorrect statement :-
(1) Primary CO_2 acceptor in C_3 plants is RuBP
(2) Primary CO_2 acceptor in CAM plants is PEP
(3) High temperature sensitive enzyme PEP case is present in bundle sheath cells of C_4 plants
(4) CO_2 is stored in CAM plants as malic acid inside vacuole during night.
BPP327

6. Red drop occurs due to :
(1) Low CO_2 in atmosphere
(2) High CO_2 in atmosphere
(3) Supply of wavelengths of light below 680 nm
(4) Supply of monochromatic light of 700 nm
BPP328

7. In an experiment where a part of leaf is enclosed in test tube containing KOH soaked cotton & exposed to light will-
(1) Test positive for starch.
(2) Test negative for starch due to inability to absorb light inside test tube.
(3) Test negative for starch due to inability to absorb CO_2.
(4) Test negative for starch due to absence of water.
BPP329

8. _____ from a suitable oxidizable compound reduces CO_2 to Carbohydrates.
(1) Oxygen
(2) Hydrogen
(3) Carbon
(4) Both (1) and (2)
BPP330

9. Sugar is synthesized -
(1) Non-enzymatically in grana
(2) Non-enzymatically in stroma
(3) Enzymatically in grana
(4) Enzymatically in stroma
BPP331

10. Leaf pigments are separated by -
(1) Crystallization
(2) Gel electrophoresis
(3) Blotting
(4) Paper chromatography
BPP332

11. Maximum absorption by chlorophyll a occurs in:
(1) Blue & green region
(2) Red & green region
(3) Blue & red region
(4) Yellow & red region
BPP333

12. Reaction centre is formed by-
(1) Only one chlorophyll 'a' molecule
(2) A few chlorophyll 'a' molecule
(3) One chlorophyll 'a' and a few accessory
(4) A few chlorophyll 'a' and a few accessory pigments.
BPP334

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

13. The movement of excited electrons in Noncyclic Photophosphorylation:
(1) Uphill in terms of reduction potential scale
(2) Downhill in terms of reduction potential scale
(3) Uphill and downhill in terms of oxidation-reduction potential scale
(4) Both (1) and (3)
BPP335

14. The Z scheme is named so because-
(1) It was discovered by a scientist with 'Z' as initial letter of name
(2) The carriers of ETS present in thylakoid membrane are in 'Z' shape.
(3) It forms 'Z' shape when the carriers of ETS are arranged in sequence of redox potential scale.
(4) Both (1) and (3)
BPP336

15. The O_2 is released in -
(1) Lumen of thylakoid
(2) Outer side of thylakoid
(3) Stroma
(4) Cytoplasm
BPP337

16. Non-cyclic photo phosphorylation involves-
(1) PS I
(2) PS II
(3) Both PS I & PS II
(4) None of the these, only enzymes in stroma
BPP338

17. A possible location of cyclic photo phosphorylation under full light is :-
(1) Stroma
(2) Stroma lamellae
(3) Cristate
(4) Outer membrane of chloroplast
BPP339

18. The proton gradient may be formed in photosynthesis due to-
(1) Splitting of water
(2) Reduction of $NADP^+$
(3) Both (1) and (2)
(4) None of these
BPP340

19. Which of these is a H carrier?
(1) PS II (2) Ferredoxin
(3) Plastocyanin (4) Plastoquinone
BPP341

20. ATP formation occurs upon-
(1) Build up of potential gradient
(2) Break down of potential gradient
(3) Both of these
(4) None of these
BPP342

21. The transmembrane channel allows movement across membrane for ATP synthesis -
(1) Osmosis of protons
(2) Simple diffusion of proton
(3) Facilitated diffusion of electron
(4) Facilitated diffusion of proton
BPP343

22. The first product of CO_2 fixation was identified to be in the Calvin cycle.
(1) PGA (2) RUBP
(3) Citric acid (4) OAA
BPP344

23. The Calvin cycle starts with _____ ends with-
(1) RuBisCO, regeneration of RuBisCO
(2) RUBP, regeneration of RuBisCO
(3) RUBP, regeneration of RUBP
(4) PGA, regeneration of PGA
BPP345

24. Reduction involves use of _____ molecules of NADPH to reduce fixed CO_2.
(1) 1 (2) 2 (3) 3 (4) 4
BPP346

25. For formation of 1 glucose, how many molecules of ATP are required by C_3 cycle?
(1) 12 (2) 16 (3) 18 (4) 10
BPP347

26. Which of the following is true?
(1) C_4 plants lack RuBisCO
(2) Mesophyll cells of C_4 plants lack RuBisCO
(3) Bundle sheath cells of C_4 plants lack RuBisCO
(4) C_3 plants lack RuBisCO
BPP348

27. PEP case enzyme in C_4 plants is
(1) Absent in mesophyll cells
(2) Present in bundle sheath cells
(3) Both (1) and (2)
(4) None of these

BPP349

28. In photorespiration-
(1) Sugar and ATP are formed
(2) Sugar is formed but not ATP
(3) ATP is formed but not sugar
(4) Sugar and ATP are not formed

BPP350

29. C_3 plants show saturation for CO_2 concentration at-
(1) 450 μlL^{-1} (2) 390 μlL^{-1}
(3) 540 μlL^{-1} (4) 240 μlL^{-1}

BPP351

30. Water stress causes-
(1) Reduced CO_2 availability
(2) Leaf wilting
(3) Reduced surface area of leaf
(4) All of these

BPP352

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	3	2	3	4	3	2	4	4	3	1	3	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	3	4	2	4	1	3	2	3	2	4	4	1	4

AIM : FREE EDUCATION TO ALL APUL

Exercise - II

Photosynthesis in Higher Plants

1. Select the incorrect statement regarding photosynthesis:
(1) It is an endothermic (endergonic) process
(2) It is a physico-chemical process
(3) It is a redox process
(4) It is a non enzymatic process

BPP353

2. Select the correct statement about photophosphorylation :
(1) Cyclic photophosphorylation involves photolysis of water
(2) Cyclic photophosphorylation always occurs in granal thylakoid
(3) External electron donor is required in cyclic photophosphorylation
(4) Only PS-I is involved in cyclic photophosphorylation

BPP354

3. Which of the following is most strong natural oxidant involved in photophosphorylation?
(1) Pheophytin
(2) Plastoquinone
(3) $NADP^+$
(4) Cytochrome b_6 & f complex

BPP355

4. Select the incorrect statement about photorespiration :-
(1) Photorespiration occurs in high light intensities
(2) Photorespiration occurs in high temperature
(3) Photorespiration occurs under high O_2 concentration
(4) Photorespiration occurs under high CO_2 concentration

BPP356

5. Select the correct statement about CO_2 in photosynthesis :
(1) Rate of photosynthesis decreases with increase in CO_2 concentration upto 0.05%
(2) Green house crops (C_3 plants) show CO_2 fertilization effect
(3) Atmospheric CO_2 is rarely a limiting factor for C_3 plants
(4) High concentration of CO_2 in C_3 plants causes photorespiration

BPP357

6. Select the incorrect statement about photosystems :-
(1) PS-I is located at non-appressed parts of grana and stroma thylakoid
(2) PS-II is involved in photolysis of water
(3) Both PS-II and PS-I participate in cyclic photophosphorylation
(4) PS-II requires external electrons in non-cyclic photophosphorylation

BPP358

7. Which one of the following statements is incorrect?
(1) The first stable product of calvin cycle is 3PGA
(2) Sugarcane and maize are C_4 plants
(3) Light is a major limiting factor for photosynthesis.
(4) Chl-a is the most abundant pigment in the world.

BPP359

8. **Statement-1 :** Kranz anatomy is characteristic feature of C_4 plants only.
Statement-2 : There is a linear relationship between incident light and CO_2 fixation rate under low light intensity.
Options :
(1) Only statement-1 is correct
(2) Only statement-2 is correct
(3) Both the statements are correct
(4) Both the statements are incorrect

BPP360

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

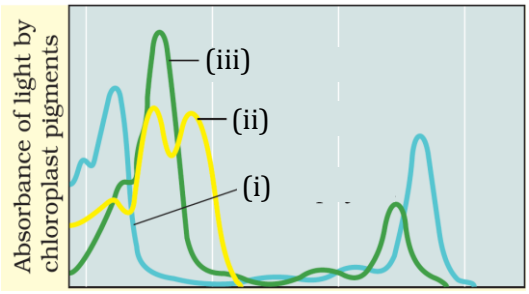
9. Match the experiment objective with the scientist who performed it-

(i)	Priestly	(I)	Production of glucose in photosynthesis
(ii)	Julius von Sachs	(II)	Role of sunlight in photosynthesis
(iii)	Jan Ingenhousez	(III)	Role of air in photosynthesis

(1) (i)–I, (ii)–III, (iii)–II
(2) (i)–II, (ii)–I, (iii)–III
(3) (i)–II, (ii)–III, (iii)–I
(4) (i)–III, (ii)–I, (iii)–II

BPP361

10.



Identity correct-
(1) (i) Chl a, (ii) Chl b, (iii) Carotenoids
(2) (i) Carotenoids, (ii) Chl a, (iii) Chl b
(3) (i) Chl b, (ii) Chl a, (iii) Carotenoids
(4) (i) Chl a, (ii) Carotenoids, (iii) Chl b

BPP362

11. **Assertion :** Chlorophyll 'a' is the chief pigment associated with photosynthesis.
Reason : Chlorophyll 'a' maximum absorption coincides with maximum photosynthesis.
(1) Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Assertion & Reason are correct but Reason is not correct explanation of Assertion
(3) Assertion is correct & Reason is incorrect
(4) Assertion & Reason are incorrect

BPP363

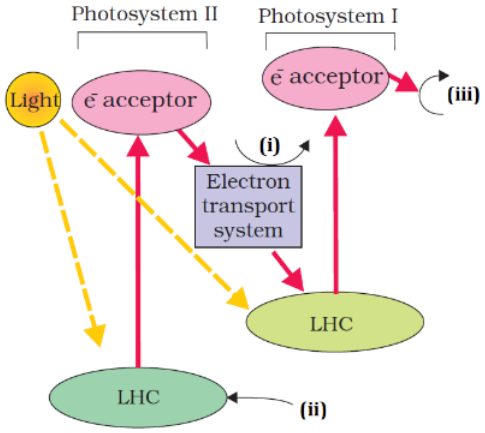
12. Advantage of accessory pigments include-
(1) They help by photolyzing the water
(2) They protect chl 'a' from photooxidation
(3) They enable narrower range of wavelength of incoming light to be used for photosynthesis
(4) Both (1) & (2)

BPP364

13. Choose correct order of events in light reaction-
(i) ATP & NADPH formation
(ii) Water Splitting
(iii) Oxygen release
(iv) Light absorption
(1) III → IV → II → I (2) IV → II → I → III
(3) IV → II → III → I (4) III → III → IV → I

BPP365

14. Identify the correct site for I-water splitting, II-$NADP^+$ reduction, III-ATP synthesis.



(1) (i)–I, (ii)–II, (iii)–III
(2) (i)–II, (ii)–I, (iii)–III
(3) (i)–III, (ii)–II, (iii)–I
(4) (i)–III, (ii)–I, (iii)–II

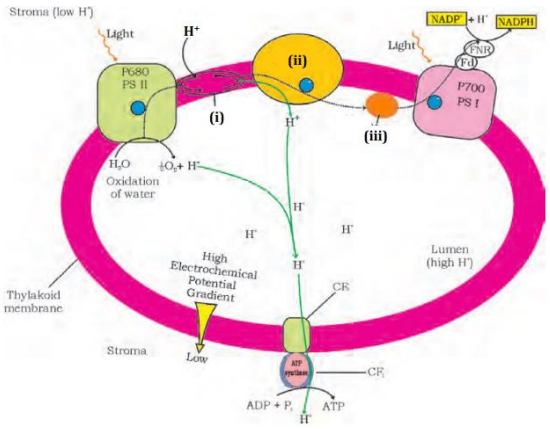
BPP366

15. **Assertion :** Cyclic photophosphorylation occurs in stroma lamellae.
Reason : Stroma lamellae lacks PS II and NADP reductase.
(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion
(3) Assertion is correct but Reason is incorrect
(4) Both Assertion & Reason are incorrect

BPP367

AIM : FREE EDUCATION TO ALL APUL

16.



Identify correct labels-

(1) (i) PQ, (ii) PC, (iii) Cyt B_6f

(2) (i) Cyt B_6f, (ii) PQ, (iii) PC

(3) (i) PC, (ii) Cyt B_6f, (iii) PC

(4) (i) PQ, (ii) Cyt B_6f, (iii) PC

BPP368

17. **Statement A :** Biosynthetic reaction is independent of direct presence of light

Statement B : Biosynthetic process continues for some time after the light becomes unavailable and then stops.

(1) Both A & B are correct

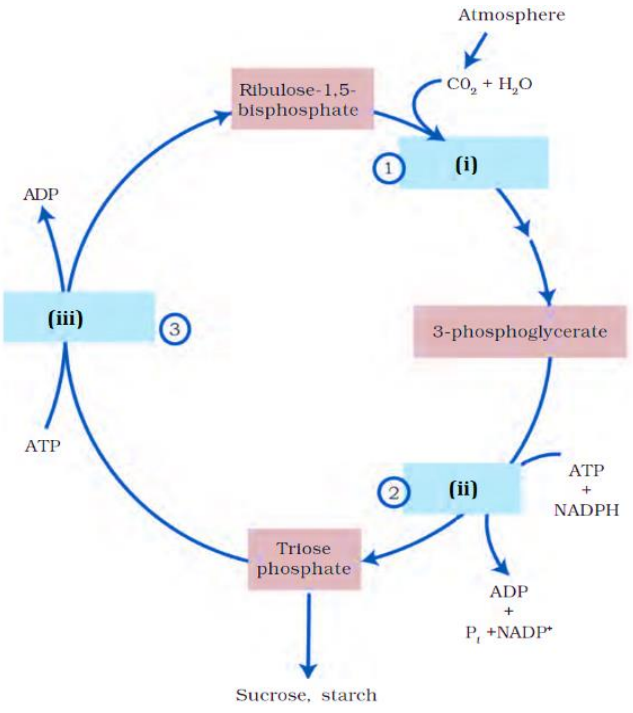
(2) A is correct and B incorrect

(3) A is incorrect and B is correct

(4) A & B are incorrect

BPP369

18. The three steps in Calvin cycle are :



	(i)	(ii)	(iii)
(1)	Carboxylation	Oxidation	Regeneration
(2)	Reduction	Carboxylation	Regeneration
(3)	Carboxylation	Reduction	Regeneration
(4)	Carbonation	Reduction	Regeneration

BPP370

19. **Statement A :** In CO_2 fixation cycle, the molecules of ATP used is more than NADPH used.

Statement B : To meet the difference in number of ATP & NADPH used in dark reaction, cyclic phosphorylation take place.

(1) A is correct but B is wrong

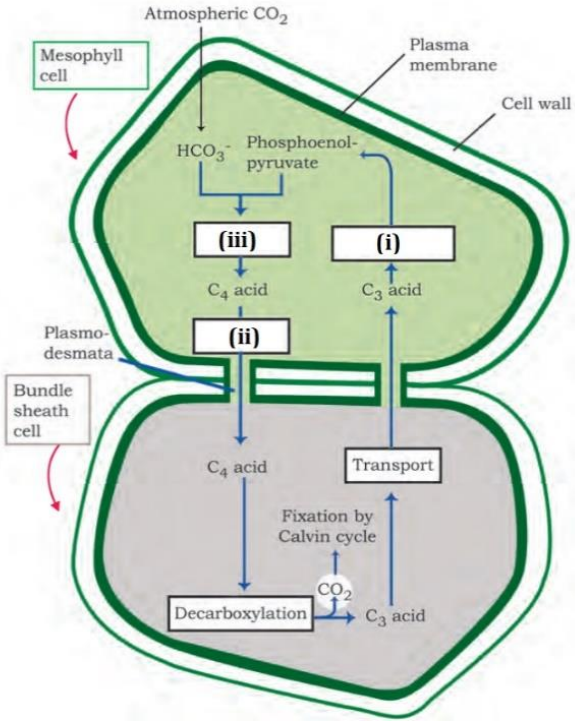
(2) B is correct but A is wrong

(3) A & B are correct and A explains B

(4) A & B are incorrect

BPP371

20. Identify correct lables-



	(i)	(ii)	(iii)
(1)	Fixation	Decarboxylation	Regeneration
(2)	Transport	Fixation	Regeneration
(3)	Regeneration	Transport	Fixation
(4)	Regeneration	Decarboxylation	Fixation

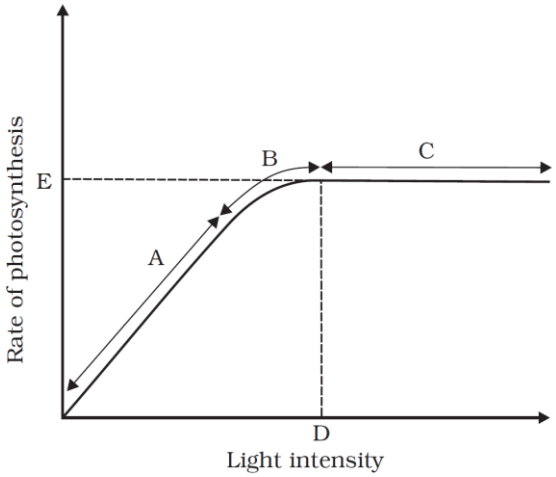
BPP372

AIM : FREE EDUCATION TO ALL APUL

21. **Assertion :** Photorespiration doesn't occur is C_4 plants.
Reason : CO_2 concentration at enzyme site in high in C_4 plants.
(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Assertion is correct and Reason is wrong
(3) Assertion is wrong and Reason is correct
(4) Both Assertion & Reason are wrong
BPP373

22. **Assertion :** Greenhouse crops like tomatoes and bell pepper are grown in CO_2 enriched atmosphere for higher yields.
Reason : C_4 plants respond to higher CO_2 concentration by showing decreased rate of photosynthesis.
(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion
(3) Assertion is correct but Reason is incorrect
(4) Both Assertion & Reason are incorrect
BPP374

23. In the given graph, in which region will plant not respond to increase in light intensity?



(1) A (2) B
(3) C (4) None
BPP375

24. According to Blackman's Law, the rate of chemical process is determined by the factor which.
(1) Is nearest to its maximum value
(2) Is nearest to its minimum value
(3) Both (1) & (2)
(4) None of these
BPP376

25. Which of the following is true?
(1) C_4 plants lack RuBisCO
(2) Mesophyll cells of C_4 plants lack RuBisCO
(3) Bundle sheath cells of C_4 plants lack RuBisCO
(4) C_3 plants lack RuBisCO
BPP377

26. **Statement A :** The accumulation of protons in stroma cause decrease in pH of stroma.
Statement B : The accumulation of proton in lumen cause formation of potential gradient cross thylakoid membrane.
(1) A is correct but B is incorrect
(2) B is correct but A is incorrect
(3) A & B are correct
(4) A & B are incorrect
BPP378

27. When O_2 binds to RuBisCO, RUBP is converted to _____ and _____.
(1) Phosphoglycerate, phosphoenolpyruvate
(2) Phosphoglycerate, phosphoglycerate
(3) Phosphoenolpyruvate, phosphoglycerate
(4) Phosphoglycolate, phosphoglycerate
BPP379

28. **Statement-I :** C_3 plants respond to higher CO_2 concentration by showing increased rates of photosynthesis leading to higher productivity.
Statement-II : C_4 plants respond to lower temperature and show higher rate of photosynthesis.

AIM : FREE EDUCATION TO ALL APUL

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

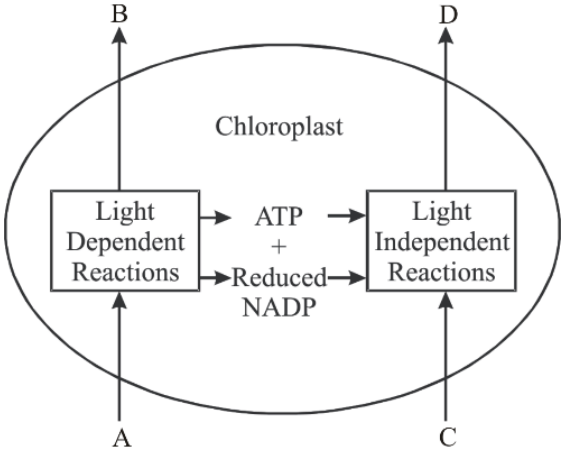
BPP380

29. Regeneration of __(i)__ takes place at expense of __(ii)__ ATP & __(iii)__ NADPH.

	(i)	(ii)	(iii)
(1)	RUBP	0	1
(2)	RUBP	1	0
(3)	RuBisCO	0	1
(4)	RuBisCO	1	0

BPP381

30. The diagram shows the movement of substances in and out of a chloroplast :-

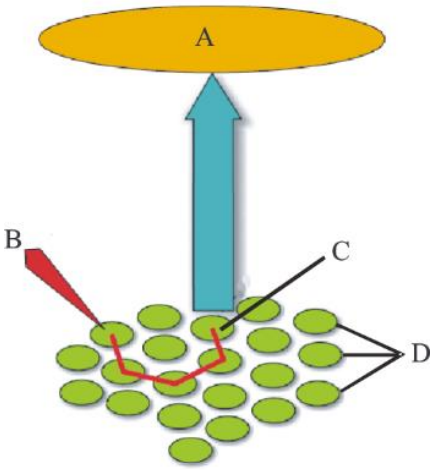


What do labels A, B, C, D represent?

	A	B	C	D
(1)	CO_2	ATP	H_2O	Starch
(2)	CO_2	H_2O	Sugar	O_2
(3)	H_2O	O_2	CO_2	Sugar
(4)	Sugar	H_2O	ATP	O_2

BPP382

31. Choose the correct option for A to D :-



(1) A-Photon B-Reaction centre, C-Pigment molecules, D-Primary e^- acceptor

(2) A-Pigment molecules, B-Primary e^- acceptor, C-Photon, D-Reaction centre

(3) A-Primary e^- acceptor, B-Photon, C-Reaction centre, D-Pigment molecules

(4) A-Reaction centre, B-Pigment molecules, C-Primary e^- acceptor, D-Photon

BPP383

32. To synthesise one molecule of glucose the requirements of CO_2 : ATP : NADPH + H^+ in C_3 plants are C_4 plants are :-

	C_3 Plants CO_2 : ATP : NADPH + H^+	**C_4 Plant CO_2 : ATP : NADPH + H^+**
(1)	6 : 30 : 12	6 : 18 : 12
(2)	1 : 5 : 2	1 : 3 : 2
(3)	6 : 18 : 12	6 : 30 : 12
(4)	1 : 3 : 2	1 : 5 : 2

BPP384

33. Match the following.

1	Non cyclic ETS	a	Stroma thylakoid
2	Cyclic ETS	b	Only grana thylakoid
3	Chemiosmotic theory	c	Stroma
4	Dark reaction	d	Formation of ATP

AIM : FREE EDUCATION TO ALL APUL

(1) 1-a, 2-b, 3-d, 4-c
(2) 1-b, 2-d, 3-c, 4-a
(3) 1-b, 2-a, 3-d, 4-c
(4) 1-a, 2-b, 3-c, 4-d

BPP385

34. **Assertion:** Cyclic flow, results only in the synthesis of ATP, but not of NADPH + H^+.
Reason : The excited electron does not pass on to NADP+ but is cycled back to the PS I complex through the electron transport chain.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BPP386

35. **Statement-I :** Except for plants in shade or in dense forests, light is rarely a limiting factor in nature.
Statement-II : light saturation occurs at 10 per cent of the full sunlight.
(1) Both Statement-I & II are correct.
(2) Both Statement-I & II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPP387

36. **Statement-I :** The NADP reductase enzyme is located on the lumen side of thylakoid membrane.
Statement-II : The ATP synthase enzyme consist of two parts : one called the CF_0 and the other portion is called CF_1.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BPP388

37. Given below are two statements:
Statement-I : The particularly large cells around the vascular bundles of the C_4 plants are called bundle sheath cells.
Statement-II : C_4 plants lack a process called photorespiration.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BPP389

38. **Assertion :** Yellow green colour chlorophyll is the chief pigment associated with photosynthesis.
Reason : Yellow green chlorophyll shows maximum absorption of blue and red regions of wavelengths.
(1) Both (A) and (R) is correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) is correct and (R) is not the correct explanation of (A).
(3) Both (A) and (R) are incorrect.
(4) (A) is incorrect but (R) is correct.

BPP390

AIM : FREE EDUCATION TO ALL APUL

39. **Assertion :** In photosystem-II, the electron become exited and jump into an orbit farther from the atomic nucleus.

Reason : In photosystem-II, the reaction centre chlorophyll a absorbs 700 nm wavelength of red light.

(1) Both (A) and (R) is correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are incorrect.

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

BPP391

40. Primary electron acceptor (A) transfers its electron on H carrier (B) in photosynthesis.

How many statement about (A) and (B) is correct.

(i) B removes proton from stroma and transfer it to the lumen.

(ii) A located towards the inner side the membrane.

(iii) A has lower redox potential compare to B

(iv) B passes its electron to the Fe-S protein.

(1) (i), (ii)

(2) (i), (ii), (iii)

(3) (i), (ii), (iv)

(4) All the above

BPP392

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	4	2	3	3	3	4	4	1	2	3	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	3	3	3	1	3	3	2	2	2	4	2	2	3
Question	31	32	33	34	35	36	37	38	39	40					
Answer	3	3	3	1	1	3	4	4	3	2					

AIM : FREE EDUCATION TO ALL APUL

Exercise - I Sexual Reproduction in Flowering Plants

1. Occurrence of more than one embryo in a seed is referred as _____.
(1) Apomixis (2) Polyembryony
(3) Parthenocarpy (4) Parthenogenesis
BSR405

2. The plants in which some of the nucellar cells surrounding the embryo sac start dividing protrude into the embryo sac and develops into the embryos are :-
(1) *Citrus* and mango (2) Papaya and *Citrus*
(3) *Citrus* and litchi (4) Mango and guava
BSR406

3. In flowering plants, after fertilisation, ovary develops into _____ and ovule matures into _____ respectively.
(1) Seed, fruit
(2) Fruit, seed
(3) Seed, embryo
(4) Embryo, endosperm
BSR407

4. Which one of the following structure is known as the progenitor of the next generation ?
(1) Ovule (2) Embryo
(3) Endosperm (4) Fruit
BSR408

5. Flowers are morphological and embryological marvels and the site of :-
(1) Apomixis
(2) Parthenocarpy
(3) Sexual reproduction
(4) Fragmentation
BSR409

6. The integumented megasporangium is :-
(1) Locule (2) Ovule
(3) Pollen sac (4) Embryo sac
BSR410

7. In a typical angiosperm anther is :-
(1) Monothecous and bisporangiate
(2) Dithecous and tetrasporangiate
(3) Monothecous and monosporangiate
(4) Dithecous and bisporangiate
BSR411

8. The innermost wall layer of micro-sporangium is called.
(1) Epidermis (2) Endothecium
(3) Middle layer (4) Tapetum
BSR412

9. The exine of pollen grain is made up of _____ and has germ pores :-
(1) Sporopollenin (2) Pectin
(3) Cellulose (4) 2 & 3 both
BSR413

10. The mature embryo sac of a typical angiosperm is _____ celled and _____ nucleated :-
(1) 8, 7 (2) 7, 8 (3) 6, 8 (4) 8, 6
BSR414

11. The central cell after triple fusion becomes the:
(1) Primary endosperm cell (PEC)
(2) Primary zygote nucleus
(3) Ovule
(4) Embryo
BSR415

12. Number of nuclei participating in the process of double fertilization is :-
(1) 2 (2) 3 (3) 5 (4) 6
BSR416

13. Seeds are endospermic in :-
(1) Pea & castor
(2) Pea & coconut
(3) Castor and coconut
(4) Coconut & pea
BSR417

14. The portion of embryonal axis above the level of cotyledons is the ..... and portion below the level of cotyledons is ..... in dicotyledonous embryo :-
(1) Epicotyl, hypocotyl
(2) Hypocotyl, epicotyl
(3) Epicotyl, plumule
(4) Hypocotyl, plumule
BSR418

AIM : FREE EDUCATION TO ALL APUL

15. ..... is a form of asexual reproduction that mimics sexual reproduction :-
(1) Apomixis
(2) Parthenocarpy
(3) Polyembryony
(4) Amphimixis

BSR419

16. If the number of chromosomes in stem cells is 14, what will be the number of chromosomes in nucellus cells of ovule of that plant ?
(1) 7
(2) 14
(3) 21
(4) Incomplete information

BSR420

17. Megasporophyll is equivalent to which structure?
(1) Ovule (2) Nucellus
(3) Carpel (4) Anther

BSR421

18. A functional megaspore develops after three successive mitotic divisions into :
(1) Embryo (2) Ovule
(3) Embryo sac (4) Zygote

BSR422

19. Archesporium of ovule is :
(1) Single celled derived from integuments
(2) Single celled derived from nucellar hypodermis
(3) Multicellular derived from nucellar epidermis
(4) Multicellular derived from nucellar hypodermis

BSR423

20. One of the most resistant known biological material is :-
(1) Lignin (2) Hemicellulose
(3) Sporopollenin (4) Lignocellulose

BSR424

21. Mature pollen grain contains :-
(1) One Cell (2) Two Cells
(3) Three Cells (4) Four Cells

BSR425

22. In a mature dehisced anther, the anther wall consists of :-
(1) Epidermis and middle layer
(2) Epidermis and endothecium
(3) Endothecium and tapetum
(4) Epidermis and tapetum

BSR426

23. 'Flower is a modified shoot, for which objective it is modified :-
(1) formation of leaf
(2) Reproduction
(3) Respiration
(4) Photosynthesis

BSR427

24. An oily layer found on the outside of the mature pollen grains of many insect pollinated species is known as :-
(1) Pollen sac (2) Pollinia
(3) Pollen Kitt (4) Pollen cap

BSR428

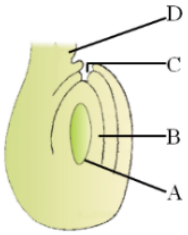
25. Consider the following statements regarding angiosperms and choose the correct option.
(a) Microspore represent the mature male gametophyte.
(b) Pollen tablets are used as food supplement
(c) Polar nuclei are situated below the egg apparatus.
(1) a and b are true but c is false
(2) b and c are true but a is false
(3) a and c are true but b is false
(4) a, b and c are false

BSR429

26. Which of the following statement regarding sporopollenin is true
(A) Exine is made up of sporopollenin
(B) Sporopollenin is most resistant inorganic material in nature
(C) No enzyme can degrade sporopollenin
(D) Sporopollenin form intine of pollen grain
(1) A, B & C (2) A & C
(3) A & B (4) Only C

BSR430

27. Find out the structure A, B, C and D in given figure :



	A	B	C	D
(1)	Embryo sac	Chalaza	Funicle	Hilum
(2)	Embryo sac	Hilum	Chalaza	Micropyle
(3)	Embryo sac	Nucellus	Funicle	Hilum
(4)	Embryo sac	Nucellus	Micropyle	Funicle

BSR431

28. (A) Polar nuclei are situated below the egg apparatus in the large central cell.
(B) Filiform apparatus plays important role in guiding the entry of pollen tube into embryo sac.
(1) Both (A) and (B) are correct
(2) Both (A) and (B) are incorrect
(3) Only (A) is correct
(4) Only (B) is correct

BSR432

29. **Statement-I:** The hard inner layer of pollen grain intine, has prominent apertures called germ pores.
Statement-II: Cytoplasm of pollen grain is surrounded by Tapetum.
(1) Statement-I & II are correct
(2) Statement I is correct & Statement-II is Incorrect
(3) Both Statements are Incorrect
(4) Statement II is only correct

BSR433

30. **Statement-I:** The process of true fertilization was discovered by Strasburger in *Monotropa*.
Statement-II: In most of angiosperms, chalazal megaspore remains functional.
(1) Both Statement -I and II are correct.
(2) Statement -I is correct but statement II is incorrect.
(3) Statement -I is incorrect and statement -II is correct.
(4) Both Statement -I and statement -II are incorrect.

BSR434

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	2	2	3	2	2	4	1	2	1	3	3	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	3	2	3	2	2	2	3	2	2	4	1	3	1

AIM : FREE EDUCATION TO ALL APUL

Exercise - II
Sexual Reproduction in Flowering Plants

AIM : FREE EDUCATION TO ALL APUL

1. How many given structures are haploid ?
Ovary, anther, egg, pollen grain, male gamete, zygote
(1) 3 (2) 4 (3) 5 (4) 6
BSR435

2. A typical angiospermic microsporangium appears near circular in outline. It is generally surrounded by four wall layers. Find correct option with correct sequence from outside to inner side :-
(1) Epidermis, hypodermis, middle layers and tapetum
(2) Epidermis, endothecium, middle layers and tapetum
(3) Endothecium, epidermis, middle layers and tapetum
(4) Endothecium, hypodermis, middle layers and tapetum
BSR436

3. Zygote develops into the ..... and the primary endosperm cell forms the ..... tissue in angiosperm :-
(1) Embryo, endosperm
(2) Endosperm, embryo
(3) Seed, embryo
(4) Endosperm, seed
BSR437

4. In Angiosperm, formation of endosperm always precedes development of the ..... :-
(1) Embryo
(2) Embryosac
(3) Male gametophyte
(4) Female gametophyte
BSR438

5. The coconut water form tender coconut respectively is ..... and the white kernel is the ..... :-
(1) Nuclear endosperm, cellular endosperm
(2) Cellular endosperm, nuclear endosperm
(3) Cellular endosperm, cellular endosperm
(4) Nuclear endosperm, nuclear endosperm
BSR439

6. **Statement-I :** Transfer of pollen grains from anther to the stigma of different plant is known as xenogamy.
Statement-II : Plants use two biotic (wind and water) and one abiotic (animals) agents to achieve pollination.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II are correct
(4) Both Statement-I and statement-II are correct
BSR440

7. **Statement-I :** Most of zygotes divide only after certain amount of endosperm is formed.
Statement-II : In the pea family the cotyledon is called scutellum that situated towards one side (lateral) of the embryonal axis.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II are correct
(4) Both Statement-I and statement-II are correct
BSR441

8. Read the following statements and choose the set of correct statements :-
(a) Pollen grains represents the male gametophytic generation.
(b) Some angiosperms produce more than one embryo in their seed this phenomenon is called polyembryony.

AIM : FREE EDUCATION TO ALL APUL

(c) In angiosperm the fruit is the only final product of sexual reproduction.
(d) The portion of embryonal axis above the level of cotyledons is hypocotyl which terminates with the plumule or stem tip.
(e) Embryo development precedes endosperm development.

Choose the correct answer from the options given below.
(1) (a) and (b) only
(2) (a), (b) and (e) only
(3) (a), (b) and (d) only
(4) (c), (d) and (e) only

BSR442

9. Read the following statement and choose the set of correct statements :-
(a) Majority of insect pollinated flowers are large.
(b) Pollination does not guarantee the transfer of right type of pollen.
(c) Pollination by wind is quite rare in flowering plants and is limited to about 30 genera mostly monocotyledons.
(d) Both wind and water pollinated flowers are very colourful and produce nector.
(e) Opposite the micropylar end, is the chalaza, representing the apical part of the ovule.

Choose the correct answer from the options given below.
(1) (a), (b) and (c) only
(2) (a), (b) and (e) only
(3) (a) and (b) only
(4) (c), (d) and (e) only

BSR443

10. Given below are two statements : one is labelled as Assertion(A) and the other is labelled as Reason (R).
Assertion : Pollen grains can withstand high temperatures and strong acid and alkali.
Reason : Their hard outer layer exine is made up of sporopollenin.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A).

BSR444

11. Given below are two statements : one is labelled as Assertion(A) and the other is labelled as Reason (R).
Assertion : Tapetum is innermost layer of anther which is single layered.
Reason : Tapetum is polyploid and multinucleated.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A).

BSR445

12. Match List-I with List-II

List-I		List-II	
(a)	Autogamy	(i)	Transfer of pollen grains from anther to stigma of different plant
(b)	Xenogamy	(ii)	Transfer of pollen grain from anther to stigma of same plant but different flower
(c)	Cleistogamous	(iii)	Transfer of pollen grain from anther to stigma of same flower
(d)	Geitonogamy	(iv)	Flower do not open and never are exposed stigma & anther

Choose the correct answer from the options given below:
(1) a-i, b-iii, c-iv, d-ii (2) a-iii, b-iv, c-i, d-ii
(3) a-iii, b-i, c-iv, d-ii (4) a-iii, b-iv, c-ii, d-i

BSR446

AIM : FREE EDUCATION TO ALL APUL

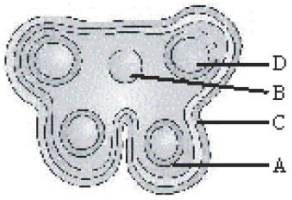
13. Match the column-I with column-II and select the right option :-

	Column-I		Column-II
(A)	Male gametophyte	(i)	Embryosac
(B)	Megasporangium	(ii)	Pollen grain
(C)	Megasporophyll	(iii)	Ovule
(D)	Female gametophyte	(iv)	Pistil/carpel

	A	B	C	D
(1)	i	ii	iii	iv
(2)	ii	iv	i	iii
(3)	ii	iii	iv	i
(4)	ii	iii	i	iv

BSR447

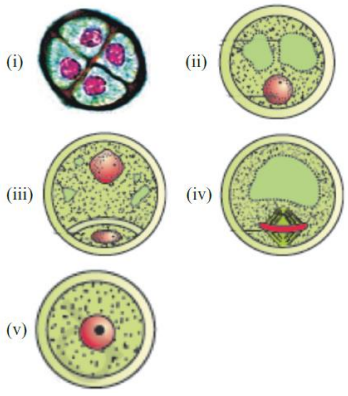
14. Given below is the diagrammatic sketch of T.S. of mature anther. Identify the labelled parts and select the right option-



	Part A	Part B	Part C	Part D
(1)	Endothecium	Sporogenous Tissue	Tapetum	Connective
(2)	Tapetum	Sporogenous Tissue	Endothecium	Connective
(3)	Endothecium	Connective	Tapetum	Sporogenous Tissue
(4)	Tapetum	Connective	Endothecium	Sporogenous Tissue

BSR448

15. Few events of male gametophyte development in angiosperm are shown in figure given below. Arrange the diagram in progressive order of their occurrence and select the correct order from options given below :



Options :

(1) i, ii, iii, iv, v (2) i, v, iv, iii, ii

(3) i, ii, iv, iii, v (4) i, v, ii, iv, iii

BSR449

16. Identify the correct match from the columns I, II and III :-

	Column-I	Column-II	Column-III
(1)	Microsporangium	(a) Megasporangium	(i) Embryosac
(2)	Ovule	(b) Pollen sac	(ii) Microsporogenesis
(3)	Aril	(c) Endothecium	(iii) Litchi
(4)	Stomium	(d) Integument	(iv) Dehiscence

(1) 1–b–ii, 2–a–i, 3–d–iii, 4–c–iv

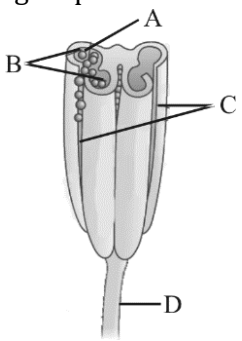
(2) 1–a–ii, 2–b–i, 3–d–iii, 4–c–iv

(3) 1–c–iii, 2–d–ii, 3–b–iv, 4–a–i

(4) 1–d–ii, 2–c–iii, 3–a–i, 4–b–iv

BSR450

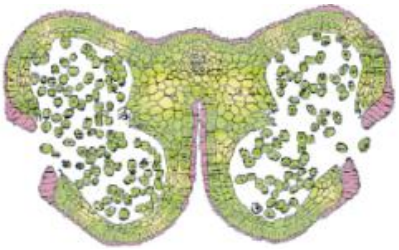
17. Identify the structure A, B, C, D in given fig and select the right option :-



	A	B	C	D
(1)	Pollen grains	Pollen chamber	Line of dehiscence	Filament
(2)	Pollen sac	Pollen grain	Line of dehiscence	Filament
(3)	Pollen grains	Pollen sac	Line of dehiscence	Filament
(4)	Pollen grains	Micro sporangia	Line of dehiscence	Filament

BSR451

18. How many layers are found in angiospermic anther at this stage :-

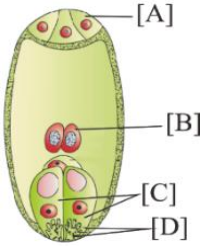


(1) 1 (2) 2 (3) 3 (4) 4

BSR452

AIM : FREE EDUCATION TO ALL APUL

19. Given below is the diagrammatic view of an embryo sac. Identify the parts labelled A, B, C and D and select the right option about them :-



	Part A	Part B	Part C	Part D
(1)	Antipodals	Polar nuclei	Synergids	Filiform apparatus
(2)	Synergids	Polar nuclei	Antipodals	Filiform apparatus
(3)	Antipodals	Filiform apparatus	Synergids	Polar nuclei
(4)	Synergids	Filiform apparatus	Antipodals	Polar nuclei

BSR453

20. How many given structures are diploid?
Pollen grain, embryo sac, nucellus, sporogenous tissue, anther, pistil
(1) 4 (2) 2 (3) 3 (4) 1

BSR454

21. Microsporophyll is equivalent to:
(1) Pistil (2) Stamen
(3) Anther (4) Ovule

BSR455

22. What is the function of tapetum ?
(1) It helps in the dehiscence of anther.
(2) It provides nutrition to the developing spores.
(3) It provides protection to anther.
(4) Both (1) and (2)

BSR456

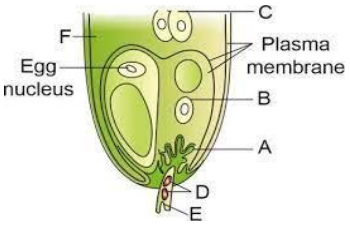
23. Choose the correct statement :
Assertion: Female gametophyte or embryo sac is embedded in nucellus.
Reason: Nucellus provides nutrition to the developing embryo sac.
(1) Both A and R are correct but R is not the correct explanation of A.
(2) A is correct but R is incorrect.
(3) A is not correct but R is correct.
(4) A and R both are correct, and R is the correct explanation of A.

BSR457

24. Read the following statements and choose the set of correct statements:
(A) Persistent nucellus is found in beet and black pepper.
(B) Wheat, maize, barley and castor are non-albuminous seed.
(C) Parthenocarpic fruits are always seedless.
(D) Apomixis is seen in some species of Asteraceae and grasses.
(1) A, C and D (2) B, C and D
(3) C and D (4) A and D

BSR458

25. Identify the structure A, B, C, D, E and F in the given figure and select the correct option :



	A	B	C	D	E	F
(1)	Synergid	Egg cell	Secondary nucleus	Male gamete	Male gamete	Central cell
(2)	Filiform apparatus	Synergid	Polar nuclei	Male gametes	Vegetative nucleus	Central cell
(3)	Egg cell	Egg apparatus	Central cell	Male gamete	Male gamete	Egg cell
(4)	Synergid	Synergid	Polar nuclei	Male gamete	Vegetative nucleus	Central cell

BSR459

26. **Statement-I:** Typical anther is bilobed and dithecous.
Statement-II : In Malvaceae , anthers are monothecous and bisporangiate.
(1) Both Statement -I and II are correct.
(2) Statement -I is correct but statement II is incorrect.
(3) Statement -I is incorrect and statement -II is correct.
(4) Both Statement -I and statement -II are incorrect.

BSR460

27. Pollen grains of rice and wheat loose viability within :
(1) 1 hour (2) 40 minutes
(3) 30 minutes (4) 10 minutes.

BSR461

28. Arrange them in sequential order and choose the correct option :

(A) Ploidy of megaspore is haploid

(B) Out of 4 megaspores, 3 degenerate.

(C) Megaspore mother cell divides meiotically to form megaspores.

(D) Megaspore develops into female gametophyte.

(1) B, C, D and A (2) C, B, A and D

(3) A, B, C and D (4) D, C, B and A

BSR462

29. **Statement-I:** Endothecium helps in the dehiscence of anther.

Statement-II: Tapetum acts as a nutritive layer.

(1) Both Statement -I and II are correct.

(2) Statement -I is correct but statement II is incorrect.

(3) Statement -I is incorrect and statement -II is correct.

(4) Both Statement -I and statement -II are incorrect.

BSR463

30. Pollen grains of which family retain viability for months?

(1) Rosaceae, Cucurbitaceae and Solanaceae.

(2) Solanaceae, Asteraceae and Cucurbitaceae.

(3) Solanaceae, Rosaceae and Solanaceae.

(4) Rosaceae, Leguminosae and Solanaceae.

BSR464

Exercise – II **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	1	1	1	2	2	1	3	4	1	3	3	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	1	1	2	2	4	1	2	1	3	2	1	4

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Exercise - I
Principles of Inheritance and Variation

1. (i) Segregation of alleles is a ___(a)___ process
(ii) Dwarf character shows itself in ___(b)___ condition.
Fill the correct option respectively :
(1) a-Sequenced, b-heterozygous
(2) a-Random, b-homozygous
(3) a-Random, b-heterozygous
(4) a-Sequenced, b-homozygous

BPV638

2. In monohybrid cross, What is the possibility of obtaining genotype TT, tt and Tt respectively in F_2 generation :
(1) 3/4, 1/4, 1/2 (2) 1/4, 1/4, 1/2
(3) 1/4, 1/2, 1/4 (4) 1/2, 1/2, 1/4

BPV639

3. Which statement is correct about polygenic inheritance :-
(1) One character is controlled by many genes
(2) Characters controlled by cytogenes
(3) Two pair of allelic genes are essential in dominant form to produce a particular character
(4) Gene which inhibit the expression of another non allelic gene

BPV640

4. 'Mendel work remain unrecognised till 1900'. Which statement is not correct regarding above statement :-
(1) Communication was not easy
(2) Mendel did not provide any physical proof of factor
(3) Factors which did not blend with each other, was not accepted by his contemporaries
(4) Mendel's approach of using mathematics was totally new and unacceptable

BPV641

5. Physical association of two genes is known as:
(1) Recombination (2) Linkage
(3) Crossing over (4) Sex determination

BPV642

6. Analysis of traits in a several generation of a family is called :-
(1) Mendelian disorder
(2) Chromosomal disorder
(3) Pedigree
(4) Aneuploidy

BPV643

7. Myotonic dystrophy is an example of :-
(1) Autosomal recessive trait
(2) Autosomal dominant trait
(3) Sex linked recessive trait
(4) Sex linked dominant trait

BPV644

8. Following symbol shows :-



(1) Affected individual
(2) Mating between non relative
(3) Mating between relative
(4) Parents with male child

BPV645

9. Presence of one additional copy of a chromosome known as :-
(1) Monosomy
(2) Tetrasomy
(3) Trisomy
(4) Double monosomy

BPV646

10. Which disease is sex linked recessive disease?
(1) Haemophilia
(2) Sickle cell anaemia
(3) Phenylketonuria
(4) Thalassemia

BPV647

11. Which symbol used for unspecified sex :-
(1) ▱ (2) ○ (3) ◇ (4) ■

BPV648

12. Which type of genetic disorder is related to chromosomal disorder :-
(1) Haemophilia (2) Cystic fibrosis
(3) Phenylketonuria (4) Down's syndrome

BPV649

13. Which of the following type of gamete will not be produce by a genotype AAbbCcDd :-
(1) AbCd (2) AbcD
(3) AbCD (4) AbCcd

BPV650

14. Which part is removed at the time of emasculation in unisexual flower :-
(1) Anther (2) Carpel
(3) Stamen (4) None

BPV651

AIM : FREE EDUCATION TO ALL ♥ APUL

15. Given below are two statements:

Statement-I : Heterozygous female for haemophilia may transmit the disease to sons.

Statement-II : Sickle-cell anaemia is caused by the substitution of valine (val) by glutamic acid (Glu) of β-globin chain.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BPV652

16. Given below are two statements:

Statement-I : Dominant disorder in human cause due to formation of any undesired chemical/protein.

Statement-II : Recessive disorder occurs due to lack of protein in animals.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BPV653

17. Given below are two statements:

Statement-I : One gene affect many characters called pleiotropic gene.

Statement-II : Dominance is an autonomous feature of a gene or the gene product.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BPV654

18. Given below are two statements:

Statement-I : Behaviour of chromosome is parallel to the behaviour of genes.

Statement-II : Sutton and Boveri used chromosomal movement to explain Mendel's laws.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BPV655

19. Read the all statements carefully :

(i) Mendel conducted artificial pollination /cross pollination experiments using several true breeding pea line

(ii) True breeding having undergone continuous self pollination, show stable trait inheritance

(iii) Mendel selected 14 true breeding pea plant varieties.

(1) Only i and iii are correct

(2) i, ii and iii are correct

(3) Only iii is incorrect

(4) i and ii are incorrect

BPV656

AIM : FREE EDUCATION TO ALL APUL

20. Read the following statement and choose incorrect statement : (According to Mendel experiment)
(1) Mendel crossed tall and dwarf pea plant to study the inheritance of one gene known as Monohybrid cross.
(2) In monohybrid cross, F_1 progeny plants were similar to one of its parents
(3) In F_2 generation of monohybrid cross, only one of the parental trait was expressed
(4) In F_2 generation of monohybrid cross both the traits were expressed in the proportion 3 : 1.

BPV657

21. Match column-I to column-II and then arrange all correctly :

Column-I		Column-II	
(i)	Factor	(a)	Different forms of the same gene
(ii)	Allele	(b)	Morphological appearance of a particular character
(iii)	Phenotype	(c)	Genetic constitution of a character
(iv)	Genotype	(d)	Pass down from parent to offspring stably in Mendel experiments

(1) i-a, ii-b, iii-c, iv-d
(2) i-b, ii-a, iii-d, iv-c
(3) i-d, ii-a, iii-b, iv-c
(4) i-d, ii-a, iii-c, iv-b

BPV658

22. Which are related to incomplete dominance in following :-
(i) Flower colour in *Mirabilis Jalapa*
(ii) Coat colour in cattle
(iii) ABO blood group
(iv) Flower colour in snapdragon
(v) Coat colour in mice
(1) i, ii (2) i, iii (3) i, iv (4) ii, iii

BPV659

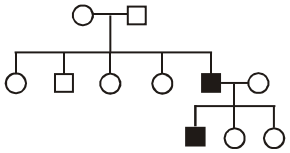
23. Which genetic disorder presented in Queen Victoria's family :-
(1) Colour blindness (2) Haemophilia
(3) Cystic fibrosis (4) Thalassemia

BPV660

24. Which statement is not correct regarding the sickle cell anaemia :-
(1) This is autosomal linked recessive trait
(2) Heterozygous (Hb^AHb^S) are resistant to malarial disease
(3) Heterozygous (Hb^AHb^S) individuals are diseased phenotype
(4) Defect is caused by the substitution of glutamic acid (Glu) by valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule

BPV661

25. In the following pedigree chart, the mutant trait is shaded black. The gene responsible for the trait is :-



(1) Dominant and sex linked
(2) Dominant but autosomal
(3) Y-linked
(4) Recessive and autosomal

BPV662

26. Which is not a correct match :-
(1) Deletion - Loss of a segment of DNA result in alteration in chromosome
(2) Insertion - Gain of a segment of DNA, result in alteration in chromosome
(3) Aberration - Result of chromosomal alteration
(4) Sickle cell anaemia - A classical example of frame shift mutation

BPV663

27. Which match is not correct :-
(1) Crossing over - Recombination of genes
(2) Turner's syndrome - Additional copy of X-chromosome
(3) Test cross - Tt × tt
(4) Polyploidy - more than two sets of chromosomes

BPV664

28. Which statement is not correct :-
(1) ABO blood group is an example of multiple allelism.
(2) Law of independent assortment is based on monohybrid cross.

AIM : FREE EDUCATION TO ALL APUL

(3) A set of haploid chromosome known as genome.
(4) X-body name given by Henking.

BPV665

29. Which of the following is not a test cross?
(1) Aabb x aaBb (2) AaBb x aabb
(3) AAbb x aaBb (3) AabbCc x aaBbcc

BPV666

30. The possibility of a human female being colorblind is :
(1) Very rare
(2) Very high
(3) Equal to male
(4) No specific pattern as such

BPV667

31. Regarding the following cross
AaBbCcDd × AABbCcDD
Which statement is correct?
(1) The possibility of one specific genotype cannot be calculated.
(2) The probability of genotype AaBBCCDd is 1/4.
(3) The probability of having an offspring with all dominant alleles is zero.
(4) The probability of genotype aabbccdd is zero.

BPV668

32. Among the following disorders
Turner's syndrome, Cri-du-chat syndrome, Down's syndrome, and Thalassemia
How many of them are caused due to trisomy?
(1) One (2) Two
(3) Three (4) Four

BPV669

33. **Assertion:** Translocation is an illegal cross over.
Reason: Translocation is the exchange of chromosomal segments between nonhomologous chromosomes.
(1) Both Assertion and Reason are correct, but reason is not the correct explanation of Assertion.
(2) Assertion is incorrect but reason is correct.
(3) Both Assertion and Reason are correct, and Reason is the correct explanation of Assertion.
(4) Assertion is correct but Reason is incorrect.

BPV670

34. A person having genotype I^A and I^B will show AB blood group. This is because of-
(1) Segregation (2) Dominance
(3) Co-dominance
(4) Independent assortment

BPV671

35. Select the correct statement regarding dihybrid cross-
(1) Genes of the same chromosomes that are loosely linked, show similar recombination as the tightly linked genes do.
(2) Tightly linked genes like body colour and eye colour genes in *Drosophila* show very less recombination.
(3) Tightly linked genes of the same chromosomes show high recombination.
(4) Genes that are far apart and present on the same chromosomes show very little recombination.

BPV672

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	1	2	2	3	2	3	3	1	3	4	4	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	4	2	3	3	3	2	3	4	4	2	2	3	1
Question	31	32	33	34	35										
Answer	4	1	3	3	2										

Exercise - II
Principles of Inheritance and Variation

1. Which statement is incorrect :-
(1) Pleotropic gene has multiple phenotype
(2) 9 : 7 ratio is produced due to co-dominance
(3) Two identical alleles known as homozygous
(4) Incomplete dominance is present in *Antirrhinum majus*

BPV673

2. Which statement is not true :-
(1) Genetics deals with the inheritance, as well as the variation of characters from parents to offspring
(2) Characters are controlled by discrete units called factor
(3) Co-dominance present in 'AB' blood group
(4) Linkage $\propto$ distance between two gene present in one chromosome

BPV674

3. For human ABO blood group there are three different _____, there are six different_____ and there are four different_____.
(1) Alleles, phenotypes, genotypes
(2) Phenotypes, genotypes, alleles
(3) Genotypes, Alleles, phenotypes
(4) Alleles, genotypes, phenotypes

BPV675

4. How many dominant traits are present in pea plant from the given list ?
Yellow seed colour, Green pod colour, Wrinkled seed shape, Violet flower colour, Terminal flower position, Full pod shape
(1) 3 (2) 4 (3) 2 (4) 5

BPV676

5. Which of the following is correctly matched :-
(1) Stem height – chromosome number 1
(2) Seed colour – chromosome number 2
(3) Pod colour – chromosome number 5
(4) Pod shape – chromosome number 7

BPV677

6. The Punnett square shown below represents the pattern of inheritance in dihybrid cross when yellow (Y) is dominant over white (y) and round (R) is dominant over wrinkled (r) seeds:-

	YR	Yr	yR	ry
YR	F	J	N	R
Yr	G	K	O	S
yR	H	L	P	T
ry	I	M	Q	U

A plant of 'H' type will produce seeds with the genotype identical to seeds produced by the plants of:-
(1) Type M (2) Type J
(3) Type P (4) Type N

BPV678

7. Given below are two statements :
Statement-I : Every gene, as you know by now, contains the information to express a particular trait.
Statement-II : In a diploid organism, there are two copies of each gene i.e. as a pair of alleles. Now these two alleles always be identical, as in heterozygote.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and II are incorrect
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and II are correct.

BPV679

8. Given below are two statements :
Statement-I : According to Sutton and Boveri the important things to remember are that chromosomes as well as genes occur in pairs.
Statement-II : The two alleles of a gene pair are located on homologous sites on homologous chromosomes.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and II are incorrect
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and II are correct.

BPV680

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

9. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome, called aneuploidy.
Reason (R) : Down's syndrome and Turner's syndrome are example of aneuploidy.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BPV681

10. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Female heterogamety is found in birds.
Reason (R) : Two different types of gametes in terms of the sex chromosomes, are produced by female birds.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BPV682

11. Identify the set of correct statement :
(a) Mendel investigated inheritance with usage of statistical analysis and mathematical logics.
(b) Mendel's experiments had a small sampling size.
(c) Mendel conducted artificial/cross pollination for the hybridisation experiment.
(d) Mendel selected 14 true-breeding pea plant varieties.
Choose the correct answer from the option given below :
(1) (a) and (c) only
(2) (b), (c) and (d) only
(3) (a), (c) and (d) only
(4) (a), (b), (c) and (d) all

BPV683

12. Identify the set of correct statement :
(a) Mendel conducted test cross for the first time between a tall plant from F_2 with a dwarf plant.
(b) Test cross can easily predict the genotype of the test organism.
(c) Law of dominance explains the proportion of 3 : 1 obtained at the F_2.
(d) Alleles show blending and both the characters are not recovered as such in F_2 in pea.
(e) Factor occurs in pairs.
Choose the correct answer from the option given below :
(1) (a) and (e) only
(2) (b), (c) and (d) only
(3) (a), (b) and (c) only
(4) (a), (b), (c) and (e) only

BPV684

13. Identify the set of incorrect statement/s :
(a) Gene which code for a pair of contrasting traits are known as Allele.
(b) Mendelian factors are discrete units.
(c) A large sample size is required to make any quantitative conclusion.
(d) Graphical representation to calculate the probability of all possible genotype of offsprings genetic cross, developed by Morgan.
Choose the correct answer from the option given below :
(1) (a) and (c) only
(2) (d) only
(3) (a), (b) and (d) only
(4) (b) and (d) only

BPV685

14. Identify the set of incorrect statement/s.
(a) Sometime, F_1 do not resemble either of two parents for flower colour in some plants.

AIM : FREE EDUCATION TO ALL APUL

(b) Occasionally a single gene product may produce more than one phenotype.
(c) After maturation of seeds, BB seeds are round and Bb seeds are wrinkled.
(d) Dominance is an autonomous feature of a gene.

Choose the correct answer from the option given below :
(1) (c) only
(2) (a), (c) and (d) only
(3) (c) and (d) only
(4) (b) and (c) only

BPV686

15. Identify the set of incorrect statement/s.
(a) In 1902, De Vries, Correns and Tschermark independently rediscovered Mendel's result on inheritance.
(b) By 1902, the chromosome movement during meiosis had been worked out by Sutton and Boveri.
(c) Two Alleles of a gene pair are located on different loci of homologous chromosomes.
(d) Chromosome segregate at the time of gamete formation which occurs in pairs in an individual.

Choose the correct answer from the options given below :
(1) (a) and (c) only
(2) (a) only
(3) (c) only
(4) (a), (c) and (d) only

BPV687

16. Identify the set of incorrect statement/s.
(a) *Drosophila* complete their life span in about two weeks.
(b) *Drosophila* has heteromorphic sex chromosomes in male.
(c) Morgan carried out test cross in *Drosophila*.
(d) Morgan attributed linkage as physical association of two genes.
(e) Morgan et al. found in *Drosophila* that the genes white and yellow were loosely linked than white and miniature.

Choose the correct answer from the options given below :
(1) (e) only
(2) (a), (b) and (c) only
(3) (a), (c) and (e) only
(4) (a), (b), (c) and (e) only

BPV688

17. Identify the set of correct statement/s.
(a) Today genetic maps are extensively used as a starting point in genome sequencing in HGP.
(b) In Morgan's experiment's, wild type allele and mutant type allele are represented with (+) and (–) sign in superscript, respectively.
(c) Strength of linkage between Y and W is higher than W and M.
(d) Pleiotropy in most cases is the effect of a gene on metabolism pathways.

Choose the correct answer from the option given below :
(1) (a), (b) and (d) only
(2) (b), (c), (d) only
(3) (a) and (d) only
(4) (a), (b), (c) and (d) only

BPV689

18. Identify the set of correct statement/s.
(a) Grasshopper is an example of female heterogamety.
(b) In birds, total number of chromosome is same in both male and female.
(c) In honey bee, fertilized egg develops queen and unfertilized egg develops worker.
(d) Haplo-diploid sex-determination system found in Honey bee.

Choose the correct answer from the option given below :
(1) (a) and (d) only
(2) (b), (c) and (d) only
(3) (a), (b), (c) and (d)
(4) (a), (b) and (c) only

BPV690

19. Identify the set of correct statement :
(a) Mutation is a phenomenon which results in change in genotype and phenotype of an organism.

AIM : FREE EDUCATION TO ALL APUL

(b) Chromosomal aberrations are commonly observed in cancer cells.
(c) There are many chemical and physical factors referred as mutagens.
(d) Analysis of traits in a several of generations of a family is called the pedigree analysis.
(e) Deletions and insertions of base pairs of DNA causes point mutation.

Choose the correct answer from the options given below :
(1) (a), (b), (c), (d) only
(2) (a), (b), (c) only
(3) All are incorrect
(4) All are correct

BPV691

20. Identify the set of incorrect statement/s.
(a) Colour blindness is a sex-linked dominant disorder.
(b) Colour blindness occurs in about **0.4** per cent of males and **8** per cent of females.
(c) Male are in heterozygous condition for haemophilia and colour blindness.
(d) Colour blindness is due to absence of some specific proteins which forms cones.

Choose the correct answer from the option given below :
(1) only (a)
(2) (a), (b) and (c) only
(3) (a) and (b) only
(4) only (d)

BPV692

21. Given below are two statements:
Statement-I : Mendel's study found that the factors independently assort and combine in all permutations and combinations.
Statement-II : Later, it was found that Mendel's law of independent assortment does not hold true for the genes located on the same chromosomes.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BPV693

22. Given below are two statements:
Statement-I : ABO blood group system provide a good example of multiple alleles.
Statement-II : Multiple Allele can be found in an individual of Human population.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and II are correct.

BPV694

23. Given below are two statements:
Statement-I : Modified allele could be responsible for production of normal enzyme.
Statement-II : Normal/unmodified allele is responsible for production of normal enzyme.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BPV695

24. **Assertion (A) :** Law of segregation is based on the fact that the alleles do not show any blending.
Reason (R) : Factor or alleles of a pair segregate from each other such that a gamete receives only one of the two factor.

AIM : FREE EDUCATION TO ALL APUL

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPV696

25. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Unmodified Allele, which represents the original phenotype is the dominant allele.

Reason (R) : A dominant allele could be modified into recessive allele by only crossing over.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPV697

26. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : A gene 'T' is responsible for tallness in pea plant.

Reason (R) : 'T' allele is functional, that produces normal enzyme.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPV698

27. **Assertion (A) :** Starch synthesis gene in Heterozygous (Bb) produce intermediate size starch grains but round seed.

Reason (R) : Gene for starch branching enzyme (SBE) show multiple allelism.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPV699

28. **Assertion (A) : Phenylketonuria** is an example of **pleiotropy**.

Reason (R) : Phenylketonuria manifests mental retardation and a reduction in hair and skin pigmentation besides too much phenylalanine in the body.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPV700

29. **Assertion (A) :** In human, we don't just have tall or short people as two distinct alternatives but a whole range of possible heights.

Reason (R) : Human skin colour is a classic example of pleiotropic gene.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

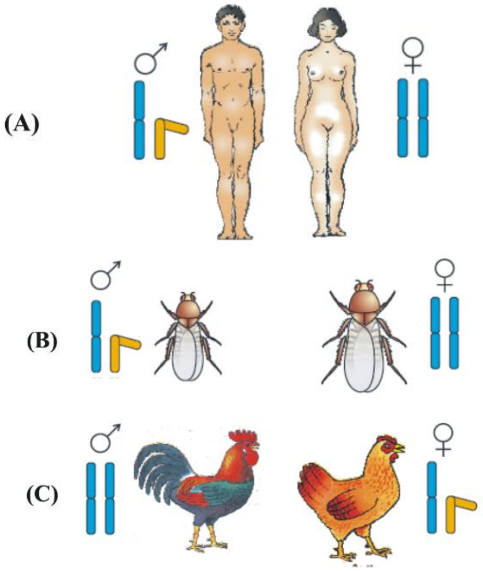
BPV701

AIM : FREE EDUCATION TO ALL APUL

30. **Assertion (A) :** Maximum 50% non-parental gene combinations are possibly produces during gamete formation.
Reason (R) : Maximum frequency of crossing over could be 100% because all four chromatids of tetrad are involves in crossing over.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPV702

31. The figure below shows three types of sex determination. Select the option giving correct identification.



(1) B - XY males, XX females
(2) A - XO Males, XX females
(3) B - XY females, XX males
(4) C - ZZ females, XX males

BPV703

32. Match the Column - I with Column - II.

	Column-I		Column-II
(I)	Phenylketonuria	(A)	8 percent of males
(II)	Colour blindness	(B)	Colour of human being
(III)	Incomplete dominance	(C)	Mental retardation
(IV)	Polygenic inheritance	(D)	Size of starch grain in pea seeds

(1) I - (D); II - (A); III - (C); IV - (B)
(2) I - (D); II - (B); III - (C); IV - (A)
(3) I - (C); II - (A); III - (D); IV - (B)
(4) I - (A); II - (C); III - (B); IV - (D)

BPV704

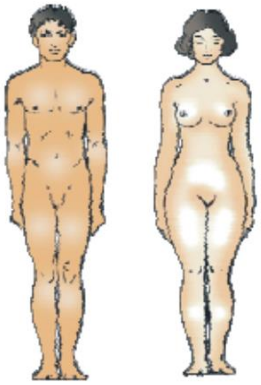
33. Match the column-I with column-II and select the correct answer using codes given below:-

	Column-I		Column-II
A.	Turner syndrome	i.	Trisomy of 21
B.	Klinefelter's syndrome	ii.	AA + XO
C.	Y-chromosome	iii.	Karyotype of 47
D.	Down's syndrome	iv.	Testis determining fa ctor

(1) A-ii, B-i, C-iv, D-iii
(2) A-ii, B-i, C-iii, D-iv
(3) A-iv, B-ii, C-i, D-iii
(4) A-ii, B-iii, C-iv, D-i

BPV705

34. Identify A and B, in the given diagrammatic representation of genetic disorder.

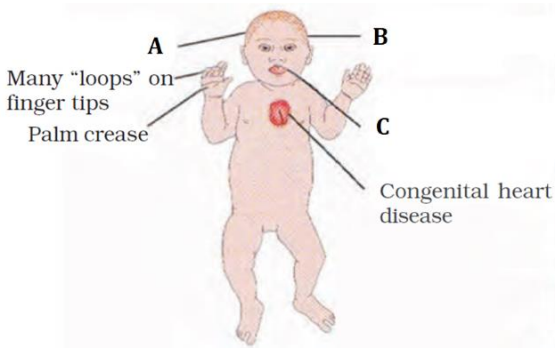


(1) A = Tall stature with feminised character
B = Turner's Syndrome
(2) A = Turner's Syndrome
B = Klinefelter Syndrome
(3) A = Klinefelter Syndrome
B = Turner's Syndrome
(4) Both (1) and (3)

BPV706

AIM : FREE EDUCATION TO ALL APUL

35. Identify A, B and C in the given representative figure showing an individual inflicted with down syndrome.



	A	B	C
(1)	Flat back of head	Broad flat face	Small and wrinkled tongue
(2)	Flat back of head	Broad flat face	Big and wrinkled tongue
(3)	Oval back head	Broad flat face	Big and wrinkled tongue
(4)	Oval back head	flat face	Wrinkled tongue

BPV707

36. Match column I with column II and choose the correct option

Column-I	Column-II
A. Linkage theory	I. Sturtevant
B. Chromosomal theory of inheritance	II. Henking
C. Discovery of X chromosome	III. Sutton and Boveri
D. Genetic mapping	IV.Morgan

(1) A-I, B-IV, C-III, D-II (2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II (4) A-II, B-I, C-III, D-IV

BPV708

37. Which of the following is not matched correctly?
(1) Incomplete dominance — Exception of Mendel's first law
(2) Co-dominance — Exception of law of purity of gametes
(3) Multiple alleles — Studied in population
(4) Linkage — Exception of independent assortment

BPV709

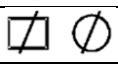
38. **Assertion:** Unlike humans, eggs are responsible for sex determination in birds.
Reason: Female birds show heterogamy.
(1) Both Assertion and Reason are correct, and Reason is the correct explanation of Assertion.
(2) Assertion is correct but the reason is incorrect.
(3) Assertion is incorrect but the reason is correct.
(4) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.

BPV710

39. **Assertion:** The possibility of a female being colorblind is extremely rare.
Reason: For a female to be colorblind, her mother must be colorblind.
(1) Both Assertion and Reason are correct, and Reason is the correct explanation of Assertion.
(2) Assertion is correct but the reason is incorrect.
(3) Assertion is incorrect but the reason is correct.
(4) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.

BPV711

40. Match the following and select the correct option-

	Column I		Column II
A.		P.	Death
B.		Q.	Five unaffected offspring
C.		R.	Sex unspecified
D.		S.	Female
E.		T.	Male

(1) A→P; B→Q; C→T; D→S; E→R
(2) A→P; B→Q; C→S; D→R; E→T
(3) A→P; B→Q; C→R; D→S; E→T
(4) A→T; B→S; C→R; D→Q; E→P

BPV712

AIM : FREE EDUCATION TO ALL APUL

41. Which of the following is correctly matched

	A Mendel	B Morgan	C Bateson
(1)	Segregation	Sex linkage	Dominance
(2)	Segregation	Linkage	Modern Genetics
(3)	Dominance	Assortment	Linkage
(4)	Linkage	Segregation	Allele

BPV713

42. **Assertion:** Sickle cell anaemia is an autosomal disorder that shows co-dominance in heterozygous condition.
Reason: Round and Sickle-shaped, both types of RBCs are found in a person heterozygous for Sickle cell anaemia.
(1) Both Assertion and Reason are correct, and Reason is the correct explanation of Assertion.
(2) Assertion is correct but the reason is incorrect.
(3) Assertion is incorrect but the reason is correct.
(4) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.

BPV714

43. **Statement I:** Male honeybees do not have fathers and sons.
Statement II: Male grasshoppers have two X chromosomes besides autosomes.
(1) Statement I is incorrect, but Statement II is correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

BPV715

44. Which of the following statement(s) is/are correct?
(1) Crossing over occurs during mitosis in gametes.
(2) Crossing over leads to the exchange of chromosomal segments.
(3) Crossing over is done between non-sister chromatids of homologous chromosomes.
(4) Both (1) and (3)

BPV716

45. Read the following statements and select the correct option.
Statement I: Segregation of alleles during cell division takes place due to disjunction.
Statement II : Segregation of alleles takes place during Anaphase-I.
(1) Both Statement I and Statement II are correct.
(2) Statement I is incorrect, but Statement II is correct.
(3) Statement I is correct, but Statement II is incorrect.
(4) Both Statement I and Statement II are incorrect.

BPV717

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	4	2	3	4	2	4	1	4	3	4	2	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	2	1	2	4	2	4	4	2	4	2	4	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	4	4	3	2	2	1	2	4	2	1	2	1	1

Exercise - I — Molecular Basis of Inheritance

1. The component of a nucleotide in DNA are :-
(1) Nucleoside + phosphate group
(2) Nitrogenous base + deoxyribose sugar + phosphate group
(3) Nitrogenous base + ribose sugar + phosphate group
(4) (1) and (2) both are correct
BMI535

2. Which type of bond is not present in DNA :-
(1) Phosphodiester bond
(2) Glycosidic bond
(3) Hydrogen bond
(4) Disulphide bond
BMI536

3. In RNA, which type of nitrogen base is not present as a structural component :-
(1) Uracil (2) 5-methyl uracil
(3) Cytosine (4) Adenine
BMI537

4. Nature of histone protein is :-
(1) Basic and positive charged
(2) Acidic and positive charged
(3) Acidic and negative charged
(4) Basic and negative charged
BMI538

5. Replication is continuous in :-
(1) The leading strand
(2) The lagging strand
(3) Strand where okazaki fragments are present
(4) Both (2) & (3)
BMI539

6. Hershey and chase use which type of radioactive medium in experiment :-
(1) Radioactive sulphur containing medium only.
(2) Radioactive phosphorus containing medium only.
(3) One medium radioactive sulphur containing and another medium radioactive phosphorus containing.
(4) Both medium were radioactive sulphur containing.
BMI540

7. Which was the first genetic material on earth:
(1) DNA (2) Protein
(3) RNA (4) Carbohydrate
BMI541

8. Which one of the following synthesize a repressor protein :-
(1) Regulator gene (2) Promoter gene
(3) Structural gene (4) Operator gene
BMI542

9. DNA fingerprinting involve identifying differences in some specific regions in DNA sequence called as :-
(1) Euchromatin
(2) Repetitive DNA
(3) Genophore
(4) Heterochromatin
BMI543

10. Allelic sequence variation has traditionally been described as a DNA polymorphism if more than one variant (allele) at a locus occurs in human population with a frequency:
(1) Greater than 0.1 (2) Greater than 0.01
(3) Less than 0.1 (4) Less than 0.01
BMI544

11. Human genome consists how many number of genes :-
(1) 3000 (2) 30000
(3) 3×10^9 (4) 40000
BMI545

12. Which major approach of HGP used for identifying all the genes that are expressed as RNA :-
(1) Bio informatics (2) ESTs
(3) Sequence annotation (4) BAC
BMI546

13. Which is wrongly match with its specificity :-
(1) YAC – A type of vector
(2) SNPs – Single nucleotide polymorphisms
(3) Human largest gene – Dystrophin
(4) Chromosome No. 1 – 231 genes
BMI547

14. A sequence of three Nitrogen base of mRNA can function as :-
(1) Gene (2) Nucleoside
(3) Codon (4) Amino acid
BMI548

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

15. Find out incorrect match :-
(1) Genome of ϕ×174 bacteriophage - 5386 nucleotides
(2) Catalytic RNA - Ribozyme
(3) Sudden chemical change in DNA - Mutation
(4) Purine - Adenine and cytosine
BMI549

16. Find out incorrect match :-
(1) Inducer - Lactose
(2) Euchromatin - Inactive chromatin
(3) DNA packaging - Nucleosome
(4) Prokaryotic DNA - Nucleoid
BMI550

17. Which statement is not correct :-
(1) Heterochromatic region is more densely packed
(2) DNA replication takes place in 3' → 5' direction
(3) Intron is non coding sequence
(4) DNA have acidic nature
BMI551

18. Which statement is not correct :-
(1) DNA polymerase is the main enzyme for RNA synthesis
(2) Primer is formed by primase enzyme
(3) In human, less than 2% of genome codes for protein
(4) AUG codes for methionine amino acid
BMI552

19. Give the correct sequence of Hershey-Chase experiment :-
(1) Infection, centrifugation, blending
(2) Blending, infection, centrifugation
(3) Infection, blending, centrifugation
(4) Centrifugation, infection, blending
BMI553

20. The process of copying genetic information from one strand of the DNA into RNA is known is :-
(1) Replication (2) Transcription
(3) Translation (4) Termination
BMI554

21. Which of the following region is not present in transcriptional unit in DNA :-
(1) Promoter (2) Structural gene
(3) Terminator (4) N-terminal
BMI555

22. Process of polymerisation of amino acids to form a polypeptide known as :-
(1) Transcription (2) Translation
(3) Replication (4) Duplication
BMI556

23. Which type of rRNA act as ribozyme in bacteria :-
(1) 23s-rRNA (2) 16s-rRNA
(3) 28s-rRNA (4) 18s-rRNA
BMI557

24. Given below are two statements:
Statement-I : Splicing refers to joining of exons in a defined order.
Statement-II : In capping an unusual nucleotide is added to 3'-end of hn-RNA.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.
BMI558

25. Given below are two statements:
Statement-I : Split-gene arrangements represent probably an ancient feature of the genome.
Statement-II : Process of splicing represent the dominance of RNA-world.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.
BMI559

AIM : FREE EDUCATION TO ALL APUL

26. **Statement-I :** Process of translation is easy to conceptualise on the basis of complementarity.

Statement-II : Nirenberg's cell-free system for protein synthesis finally helped the code to be deciphered.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BMI560

27. **Statement-I :** Genetic code is nearly universal.

Statement-II : AUG codes of methionine (met), and it also acts as 'stop' codon.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BMI561

28. **Statement-I :** 23s rRNA also acts as ribozyme for formation of peptide bond and known as peptidyl transferase enzyme.

Statement-II : In prokaryotes, control of the rate of transcriptional initiation is the predominant site for control of gene expression.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and II are correct.

BMI562

29. Which one of the following is correctly matched.

	m-RNA	**t-RNA**	**r-RNA**
(1)	Highly stable	Soluble RNA	Act as an enzyme
(2)	Adapter RNA	Largest RNA	Single stranded
(3)	Have message for synthesis protein	Adapter RNA	Play catalytic role
(4)	Soluble RNA	Highly stable	Double stranded

BMI563

30. Match column I with column II and select the correct option.

	Column-I		**Column-II**
(I)	Termination	(A)	Aminoacyl tRNA synthetase
(II)	Charging of tRNA	(B)	Okazaki fragments
(III)	Transcription	(C)	GTP dependent release factor
(IV)	DNA replication	(D)	RNA polymerase

(1) I-B, II-A, III-C, IV-D
(2) I-D, II-C, III-A, IV-B
(3) I-C, II-A, III-D, IV-B
(4) I-B, II-C, III-A, IV-D

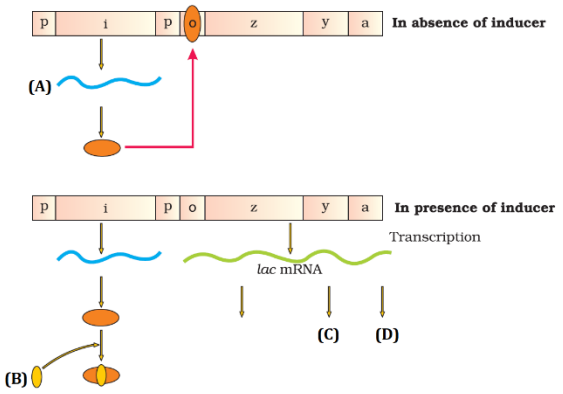
BMI564

31. Which of the following is not a part of the DNA fingerprinting process?

(1) Transfer of DNA fragments on nitrocellulose membrane.
(2) Denaturation of DNA using alkaline solution.
(3) Gel electrophoresis.
(4) Formation of complementary DNA using reverse transcription process.

BMI565

32. Carefully read the diagram and select the correct option.



Identify A, B, C, D :-

(1) A = inducer, B = repressor,
C = inhibitor, D = mRNA

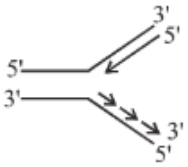
(2) A = repressor mRNA, B = inducer,
C = permease, D = transacetylase,

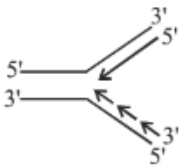
(3) A = repressor, B = operator,
C = β-galactosidase, D = transacetylase,

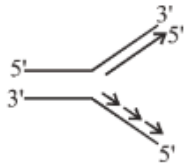
(4) A = operator, B = inducer, C = permease,
D = RNA polymerase

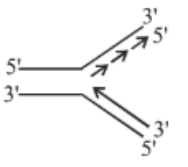
BMI566

33. Which of the following correctly represent DNA replication?

(1) 

(2) 

(3) 

(4) 

BMI567

34. Select the set of correct statements only

(A) Eukaryotic mRNA is polycistronic while prokaryotic mRNA is monocistronic.

(B) Capping and tailing are done for eukaryotic mRNA only.

(C) The precursor for eukaryotic mRNA is called hnRNA.

(D) During replication only one strand of DNA is copied into RNA.

(E) In prokaryotes, transcription and translation are coupled.

(1) B, C and E only

(2) A, B, and D only

(3) A, B, C and E only

(4) A and D only

BMI568

35. Choose the correct set of statements :-

(A) RNA is the genetic material in QB bacteriophage.

(B) DNA is chemically less reactive and more stable than RNA.

(C) Viruses with RNA genomes have shorter life spans and evolve slowly.

(1) A and B only

(2) A only

(3) B and C only

(4) A, B and C

BMI569

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	1	1	3	3	1	2	2	2	2	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	1	3	2	4	2	1	2	4	3	2	4	3	3
Question	31	32	33	34	35										
Answer	4	2	1	1	1										

AIM : FREE EDUCATION TO ALL APUL

Exercise - II
Molecular Basis of Inheritance

1. If in double stranded DNA molecule, cytosine is 40%, then the percentage of adenine would be :-
(1) 60% (2) 40% (3) 20% (4) 10%
BMI570

2. Match the column-I to column-II :-

Column-I		Column-II	
(i)	Griffith	(a)	Transduction
(ii)	Hershey and Chase	(b)	Nucleoid
(iii)	Euchromatin	(c)	Transformation
(iv)	Prokaryotic DNA	(d)	Active chromatin

(1) i - c, ii - a, iii - b, iv - d
(2) i - c, ii - a, iii - d, iv - b
(3) i - c, ii - b, iii - d, iv - a
(4) i - a, ii - b, iii - c, iv - d
BMI571

3. The correct match is :-

Column-I		Column-II	
(i)	Helicase	(a)	Polymerisation of nucleotides
(ii)	Topoisomerase	(b)	Ligate okazaki fragments
(iii)	Ligase	(c)	Remove stress at the time of unwinding
(iv)	DNA polymerase-II	(d)	Opening of DNA

(1) i - c, ii - d, iii - a, iv - b
(2) i - d, ii - c, iii - a, iv - b
(3) i - d, ii - c, iii - b, iv - a
(4) i - b, ii - a, iii - c, iv - d
BMI572

4. The base sequence in one strand of DNA is AAGCTACCGA, then what would be the sequence of bases in another strand is :-
(1) AAGCTACCGA (2) TTCGATGGCT
(3) TTCCATGGCT (4) TTCGATGCCT
BMI573

5. RNA is more reactive or unstable to DNA due to presence of :-
(1) –OH group on 2nd position of ribose sugar
(2) –OH group on 3rd position of deoxy ribose sugar
(3) –Phosphate group on 5th position of ribose sugar
(4) –OH group on 1st position of ribose sugar
BMI574

6. Which statement is not correct :-
(1) DNA and RNA are able to mutate
(2) RNA being unstable, mutate at a faster rate compare to DNA
(3) The presence of thymine at the place of uracil also confers additional stability to DNA
(4) DNA is not dependent on RNA for synthesis of protein.
BMI575

7. (i) Transcriptional level
(ii) Processing level
(iii) Transport of m-RNA from nucleus to the cytoplasm
(iv) Translational level
Which of the above level includes in regulation of gene expression in eukaryotes :-
(1) (i) and (ii) only
(2) (iii) only
(3) (i), (ii), (iii) and (iv)
(4) (i) and (iv) only
BMI576

8. Match the column-I to column-II :-

Column-I		Column-II	
(i)	i-gene	(a)	Codes for transacetylase
(ii)	z-gene	(b)	Codes for permease
(iii)	y-gene	(c)	Codes for the repressor of lac operon
(iv)	a-gene	(d)	Codes for beta-galactosidase

(1) i - c, ii - d, iii - b, iv - a
(2) i - a, ii - b, iii - c, iv - d
(3) i - b, ii - c, iii - d, iv - a
(4) i - a, ii - c, iii - b, iv - d
BMI577

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

9. Which statement is not correct :-
(1) Amino acid attach with acceptor end of t-RNA
(2) Release factor binds to the stop codon for terminating translation
(3) All 64 codons code for amino acids
(4) In splicing, introns are removed and exons are joined

BMI578

10. What is the basic concept of transformation experiment of Griffith :-
(1) R-II strains convert in new R-II strains
(2) R-II strains transform in S-III strains
(3) S-III strains transform in R-II strains
(4) S-III strains transform in new S-III strains

BMI579

11. Match the column-I to column-II :-

Column-I		Column-II	
(i)	Griffith experiment	(a)	Use of protein, RNA and DNA digestive enzymes
(ii)	Avery, Macleod and McCarty experiment	(b)	S-III and R-II strain of Pneumococcus
(iii)	Hershey and Chase experiment	(c)	Discovery of nuclein
(iv)	Miescher	(d)	Use if P^{82} and S^{35} (radioactive P and S)

(1) i - a, ii - c, iii - b, iv - d
(2) i - b, ii - d, iii - c, iv - a
(3) i - c, ii - d, iii - a, iv - b
(4) i - b, ii - a, iii - d, iv - c

BMI580

12. Match the column-I to column-II :-

Column-I		Column-II	
(i)	m-RNA	(a)	Attach on promoter
(ii)	t-RNA	(b)	Provide the template during translation
(iii)	r-RNA	(c)	Brings amino acids and read genetic code
(iv)	RNA polymerase	(d)	Play structural and catalytic role during translation

(1) i - c, ii - d, iii - b, iv - a
(2) i - d, ii - c, iii - b, iv - a
(3) i - a, ii - b, iii - c, iv - d
(4) i - b, ii - c, iii - d, iv - a

BMI581

13. **Statement-I :** Deoxy ribonucleic acid (DNA) and ribonucleic acid (RNA) are the two types of nucleic acids found in living system.
Statement-II : DNA acts as the genetic material in most of the organisms. RNA though it also acts as a genetic material in some viruses, mostly functions as a messenger.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and statement II are incorrect
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

BMI582

14. **Statement-I :** The genetic material should be stable enough not to change with different stages of life cycle, age or with change in physiology of the organism.
Statement-II : Stability as one of the properties of genetic material was not very evident in Griffith's 'transforming principle'.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and statement II are incorrect
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

BMI583

15. **Assertion (A) :** RNA is also now known to be catalytic, hence reactive.
Reason (R) : DNA is chemically less reactive and structurally more stable when compared to RNA.
In the light of the above statements, choose the correct answer from the options given below:

AIM : FREE EDUCATION TO ALL APUL

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BMI584

16. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The DNA replication scheme was termed as semi-conservative.
Reason (R) : After the completion of replication, each DNA molecule would have one parental and one newly synthesised strand.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BMI585

17. Identify the **correct** set of statement :
(a) The length of DNA is usually defined as number of nucleotides.
(b) Number of pair of nucleotides is characteristic of an organism or species.
(c) A bacteriophage $\phi \times 174$ has 5386 base pairs (bp).
(d) Diploid content of human DNA is 3.3×10^9 bp.
(e) Human genome contains 3164.7 million bp.
Choose the correct answer from the option given below :
(1) (a), (b), (e) only
(2) (a), (b), (c) and (d) only
(3) (a) and (e) only
(4) All are correct

BMI586

18. Identify the correct set of statement :
(a) A nucleotide has three components – a nitrogenous base, a pentose sugar and a phosphate group.
(b) Uracil is present in RNA at the place of thymine.
(c) In adenosine, a Nitrogenous base linked to pentose sugar through a N-glycosidic bond.
(d) DNA polymer has a free phosphate moiety at 3'- end of sugar at one end.
(e) Backbone of a polynucleotide chain is formed due to sugar and phosphates.
Choose the **correct** answer from the option given below :
(1) (a), (b), (c), (e) only
(2) (a), (d) and (e) only
(3) (a), (b), (c) only
(4) All are correct

BMI587

19. Identify the **incorrect** set of statement :
(a) 5-methyl uracil is another chemical name of **thymine**.
(b) Elucidation of structure of DNA was proposed by **Watson** and **crick**.
(c) Each strand from a parental DNA act as a template for synthesis of new strand.
(d) Two chain of DNA have parallel polarity.
(e) Two chain of DNA are coiled in a left-handed fashion only.
Choose the **correct** answer from the option given below :
(1) (a), (b) and (c) only
(2) (b), (d) and (e) only
(3) (d) and (e) only
(4) All are incorrect

BMI588

20. Identify the correct set of statement :
(a) Base pairs are said to be complementary to each other.
(b) Guanine is bonded with cytosine with three H-bonds.
(c) Pitch of the DNA helix is 3.4×10^{-6}m.
(d) Distance between two base pair in a helix is approximately 3.4 nm.
(e) DNA is scattered throughout the cell in E.coli.
Choose the correct answer from the option given below :
(1) (a) and (b) only
(2) (a), (b), (c) and (d) only
(3) (a), (b) and (c) only
(4) All are correct

BMI589

AIM : FREE EDUCATION TO ALL APUL

21. Identify the correct set of statement/s.
(a) Histones are rich in the basic amino acid residues lysine and arginine.
(b) Lysine and arginine carry positive charges in their side chain.
(c) Nucleosomes constitute the repeating unit of a structure in nucleus called chromatin.
(d) Packaging of chromatin at higher level requires additional set of proteins that collectively referred as non-histone chromosomal protein.
(e) *Heterochromatin* is said to be transcriptionally active chromatin.
Choose the correct answer from the option given below :
(1) (a) and (c) only
(2) (a), (b), (c) and (d) only
(3) (a), (c), (d) only
(4) All are correct

BMI590

22. Identify the correct set of statement/s.
(a) Some "transforming principle" transferred from heat-killed S-strain to R-strain in Hershey and Chase experiment.
(b) Proteases and RNase do not affect transformation.
(c) S-strain bacteria have genes for synthesis of mucous coat.
(d) Genetic material should be stable chemically and do not provide the scope for any changes.
Choose the correct answer from the option given below :
(1) (c) and (d) only
(2) (a), (c) and (d) only
(3) (b) and (c) only
(4) All are incorrect

BMI591

23. Identify the **incorrect** set of statement/s.
(a) RNA chemically is less reactive and structurally more stable as compared to DNA.
(b) The presence of uracil at the place of 5-methyl uracil also confer additional stability to RNA.
(c) Both DNA and RNA are able to mutate.
(d) For transmission of genetic information, RNA is better.
Choose the correct answer from the option given below :
(1) (a) and (b) only
(2) (c) and (d) only
(3) (a) only
(4) (a), (b) and (c) only

BMI592

24. Identify the correct set of statement/s.
(a) Cscl (cesium chloride) density gradient require for distinguish normal DNA from heavy DNA.
(b) ^{15}N is radioactive isotope.
(c) In Meselson and Stahl experiment DNA extracted after 2 generation have equal amount of hybrid and heavy DNA.
(d) DNA in chromosomes also replicate semi conservatively.
Choose the correct answer from the option given below :
(1) (a) and (b) only (2) (a) only
(3) (a), (b), (c) only (4) (a) and (d) only

BMI593

25. Identify the correct set of statement :
(a) DNA polymerase uses a DNA template to catalyse the polymerisation of deoxynucleotides.
(b) Diploid content of DNA in human genome, completes the process of replication within 18 minutes.
(c) dNTPs serve dual purpose in DNA replication.
(d) Discontinuously synthesised fragments are later joined by DNA polymerase enzyme.
Choose the correct answer from the option given below :
(1) (a) and (c) only (2) (a) and (d) only
(3) (d) only (4) All are correct

BMI594

26. Identify the incorrect set of statement :
(a) Principle of complementarity governs the process of transcription.
(b) If both strands of DNA transcribed, they would code for RNA with same sequences.
(c) Structural gene in transcription unit could be said as monocistronic in eukaryotes.

AIM : FREE EDUCATION TO ALL APUL

(d) Transcription and translation can be coupled in bacteria.
(e) There is a clear cut division of labour in RNA polymerase of eukaryotes.
Choose the correct answer from the option given below :
(1) (b) only
(2) (b), (d) and (e) only
(3) (a) and (b) only
(4) (b), (c), (d) and (e) only

BMI595

27. **Statement-I :** Mutations may result in loss or gain of a gene.
Statement-II : Amino acids have no structural specialities to read the code uniquely.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BMI596

28. Given below are two statements:
Statement-I : Secondary structure of t-RNA has been depicted that looks like a clover-leaf but in actual structure, it is a compact molecule which looks like inverted L.
Statement-II : Charging of t-RNA is the first phase in which amino acids are activated without using ATPs.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BMI597

29. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : If the bacteria do not have lactose around them they would no longer require the synthesis of the enzyme beta-galactosidase.
Reason (R) : β-galactosidase is used to catalyse the hydrolysis of lactose into galactose and glucose.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI598

30. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.
Assertion (A) : A very low level of expression of lac operon has to be present in the cell all the time.
Reason (R) : Lactose regulates switching on and off of the operon and termed as "*inducer*".
In the light of the above statements, choose the **correct** answer from the options given below :
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI599

31. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.
Assertion (A) : Regulation of an operon by repressor is referred to as negative regulation.
Reason (R) : Operon under control of positive regulation are regulated by activators.

AIM : FREE EDUCATION TO ALL APUL

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI600

32. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : In HGP genes which remains unexpressed as RNA referred as expressed sequence tags (ESTs) in HGP.

Reason (R) : One major approach focused on identifying all the genes that are expressed as RNA in methodologies in HGP.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI601

33. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : In HGP, DNA fragments were sequenced using automated DNA sequencers that worked on the principle of sanger's method.

Reason (R) : HGP was a mega project.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI602

34. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : In repetitive sequences a small stretch of DNA is repeated many time.

Reason (R) : The bulk DNA forms a major peak and the other small peak are referred as satellite *DNA* during centrifugation.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI603

35. **Assertion (A) :** VNTRs (Variable Number of Tandem repeats) is the basis of "DNA fingerprinting".

Reason (R) : VNTRs are small DNA sequences whose copy numbers varies from chromosome to chromosome in an individual.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

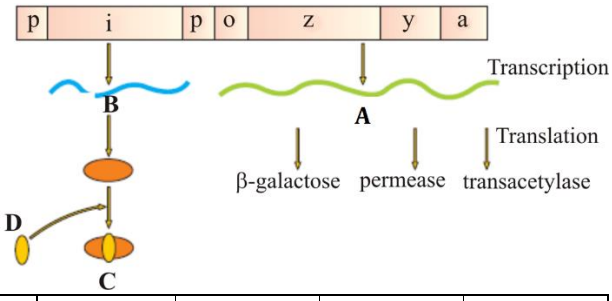
(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BMI604

AIM : FREE EDUCATION TO ALL APUL

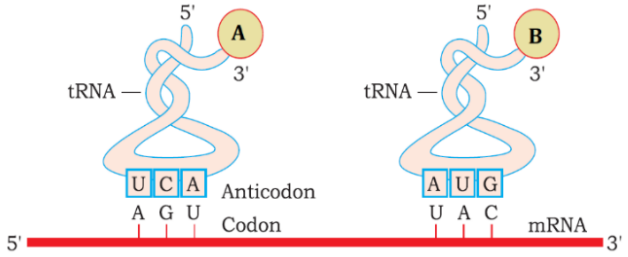
36. Examine the diagram below and select the right option given all the correct match w.r.t A, B, C & D.



	A	B	C	D
(1)	Lac mRNA	Repressor m-RNA	Inactive repressor	Inducer
(2)	Repressor m-RNA	Inactive repressor	Inducer	In presence of inducer
(3)	Inactive repressor	Inducer	In presence of inducer	Repressor m-RNA
(4)	Inducer	In presence of inducer	Repressor m-RNA	Inactive repressor

BMI605

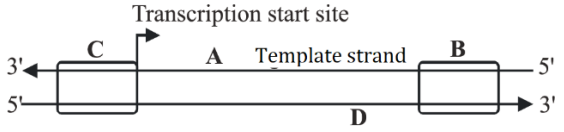
37. Identify A and B in the given diagram :



(1) A = Tyrosine, B = Serine
(2) A = Valine, B = Leucine
(3) A = Serine, B = Tyrosine
(4) A = A larine, B = Uracil

BMI606

38. In the given schematic structure of a transcription unit, identify the correct labelling for A to D.



	A	B	C	D
(1)	Promotor	Template strand	Terminator	Promotor
(2)	Structural gene	Terminator	Promotor	Coding strand
(3)	Coding strand	Terminator	Promotor	Template strand
(4)	Template strand	Promotor	Terminator	Coding strand

BMI607

39. Which of the salient features of double helical structure of DNA are correct?
(i) It is made up of two polynucleotide chain.
(ii) Two chains are coiled in right handed fashion .
(iii) Bases in two strands are joined by phosphoester bonds.
(iv) Pitch of the helix is 3.4 nm.
(1) i, ii are correct
(2) i, ii and iv are correct
(3) iii, iv are correct
(4) i, iii, iv are correct

BMI608

40. Select the incorrect match regarding lac operon.
(1) Structural genes — Transcribe mRNA for polypeptide synthesis
(2) Repressor protein — Functional in the presence of lactose
(3) Operator gene — Interacts with a regulator molecule
(4) Promoter — Site for RNA polymerase binding

BMI609

41. **Assertion :** DNA fingerprinting involves identifying difference in some specific regions in DNA called as repetitive DNA sequences.
Reason : These sequences show high degree of polymorphism & form the basis of DNA fingerprinting.
(1) Both Assertion & Reason are correct and the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are correct but Reason is not a correct explanation of the Assertion.
(3) Assertion is correct but the Reason is incorrect.
(4) Both Assertion & Reason are incorrect.

BMI610

42. If a DNA molecule with 200 bp has 40 Adenine bases, then find out the number of Guanine bases.
(1) 40 (2) 30
(3) 80 (4) 60

BMI611

43. Select the two correct statements out of the four (i–iv) given below about lac operon.
(i) Glucose or galactose may bind with the repressor and inactivate it.
(ii) In the absence of lactose, the repressor binds with the operator region.
(iii) The z-gene codes for permease.
(iv) lac-operon was elucidated by Francois Jacob and Jacques Monod.
(1) (ii) and (iii) (2) (i) and (iii)
(3) (ii) and (iv) (4) (i) and (ii)

BMI612

44. **Assertion :** N-base is heterocyclic compound.
Reason : Adenine and Guanine are found in both RNA & DNA
(1) Both Assertion & Reason are correct and the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are correct but Reason is not a correct explanation of the Assertion.
(3) Assertion is correct but the Reason is incorrect.
(4) Both Assertion & Reason are incorrect.

BMI613

45. Consider the following four statements (a-d) and select the option which includes all the correct ones :-
(a) The codon is read in mRNA in a contiguous fashion.
(b) AUG is a stop codon in bacteria.
(c) In actual structure, the tRNA is a compact molecule which looks like inverted 'L'
(d) 28S rRNA in bacteria also functions as ribozyme.
(1) b, c and d
(2) a, b and d
(3) a, c and d
(4) a and c

BMI614

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	2	1	4	3	1	3	2	4	4	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	1	3	1	2	3	1	4	1	1	4	2	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	1	1	4	1	3	2	2	2	1	4	3	2	4

AIM : FREE EDUCATION TO ALL APUL

Exercise - I
Microbes in Human Welfare

1. Find the correct match :-
(1) *Aspergillus niger* - butyric acid
(2) *Clostridium butylicum* - ethanol
(3) *Acetobacter aceti* - Acetic acid
(4) *Saccharomyces cerevisiae* - citric acid
BMH173

2. The micro-organism used to prepare alcoholic beverages is :-
(1) *Streptococcus*
(2) *Staphylococcus*
(3) *Lactobacillus*
(4) *Saccharomyces cerevisiae*
BMH174

3. **Statement-1 :** Baculoviruses are pathogens that attack insects and other arthropods.
Statement-2 : These viruses are excellent candidates for species-specific and broad spectrum insecticidal application.
In light of above statement, choose the correct answer from the option given below.
(1) Both statement I and II are incorrect
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and II are correct.
BMH175

4. **Statement-I :** Cyclosporin-A is a bioactive molecule used as an immune suppressive drug during organ transplantation.
Statement-II : Statins produced from yeast, is used as blood cholesterol lowering agents.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II are incorrect.
BMH176

5. **Statement-I :** A bacterium *Propionibacterium sharmanii* is used to produce more CO_2 in Swiss cheese.
Statement-II : *Saccharomyces cerevisiae* is used to make breads spongy due to much production of CO_2.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II are incorrect.
BMH177

6. Why some fermented beverages are made by distillation of fermented broth ?
(1) To reduce toxicity
(2) To enable fermentation
(3) To prevent spoilage
(4) To increase the alcohol percentage
BMH178

7. How many of the following beverages are produced by distillation of the fermented broth?
Whisky, wine, rum, brandy, beer
(1) 5 (2) 4 (3) 3 (4) 1
BMH179

8. Which of the following is correctly matched for the product produced by them?
(1) *Acetobacter aceti* : Antibiotics
(2) *Methanobacterium* : Lactic acid
(3) *Penicillium notatum* : Acetic acid
(4) *Saccharomyces cerevisiae* : Ethanol
BMH180

9. Methanogenic bacteria are not present in-
(1) Gobar gas plant
(2) Stomach of ruminants
(3) Bottom of water-logged paddy fields
(4) Activated sludge
BMH181

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

10. Which of the following is incorrect with respect to production of biogas?
(1) The type of the gas produced depends upon the microbes and the organic substrates they utilise.
(2) Certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO_2 and N_2.
(3) Methanogens are commonly found in the anaerobic sludge during sewage treatment, the common example is Methanobacterium.
(4) Methanogens are also present in the rumen of cattle where they help in the breakdown of cellulose and play an important role in the nutrition of cattle.

BMH182

11. Which of the following statements about methanogens is not correct?
(1) They produce methane gas.
(2) They can be used to produce biogas.
(3) They are found in the rumen of cattle and their excreta.
(4) They grow aerobically and breakdown cellulose-rich food.

BMH183

12. Which of the following statements is incorrect?
(1) Eradication of the pests is undesirable as without them the beneficial predatory and parasitic insects which depend upon them as food or hosts would not be able to survive.
(2) An important part of the biological farming approach is to become familiar with the various life form that inhabit the field, predators as well as pests, and also their life cycles, patterns of feeding and the habitats that they prefer.
(3) The very familiar beetle with red and black markings is the Ladybird.
(4) Bt spores are mixed with water and sprayed onto valuable plants such as brassicas and fruit trees, where these are eaten by the insect adults.

BMH184

13. The bacteria which can fix atmospheric nitrogen in it's free-living form in the soil are-
(1) *Azospirillum* and *Azorhizobium*
(2) *Azospirillum* and *Azotobacter*
(3) *Azotobacter* and *Rhizobium*
(4) *Frankia* and *Azospirillum*

BMH185

14. Which one of the following helps in the absorption of phosphorus from the soil by plants?
(1) *Anabaena* (2) *Glomus*
(3) *Rhizobium* (4) *Frankia*

BMH186

15. BOD of wastewater is represented as -
(1) Total inorganic matter
(2) Biodegradable matter
(3) Carbon dioxide evolution
(4) oxygen consumption

BMH187

16. **Assertion:** Integrated pest management (IPM) programme at the same time deals with conservation of insects and destruction of insects.
Reason: IPM programmes are specially used in dealing with ecologically sensitive areas.
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false

BMH188

17. **Assertion:** Disadvantages of synthetic pesticides can be overcome by the use of natural biopesticides.
Reason: Biopesticides are harmless agents which are used to control weeds and pest without causing any damage.

(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false

BMH189

18. **Assertion:** Chemical pesticides are more hazardous as compared to biopesticides.
Reason: Chemical pesticides are mostly nonspecific, expensive, hazardous and pollute the atmosphere.
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false

BMH190

19. **Assertion:** Leguminous plants are best preferred in rotation of crops.
Reason: They have root nodules, which have nitrogen-fixing bacteria *Clostridium*.
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false

BMH191

20. Match column I with column II and select the correct answer from the given codes.

Column-I	Column-II
A. Methanogens	(i) BOD
B. Fermentors	(ii) Methane rich fuel gas
C. Organic waste in water	(iii) Production of methane
D. Biogas	(iv) Large vessels for growing microbes

(1) A-(ii), B-(iv), C-(iii), D-(i)
(2) A-(ii), B-(iii), C-(iv), D-(i)
(3) A-(ii), B-(i), C-(iv), D-(iii)
(4) A-(iii), B-(iv), C-(i), D-(ii)

BMH192

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	2	1	1	4	3	4	4	2	4	4	2	2	4
Question	16	17	18	19	20										
Answer	2	2	1	3	4										

AIM : FREE EDUCATION TO ALL APUL

Exercise - II
Microbes in Human Welfare

1. How many of the dipteran insects given in below are killed by the proteins produced form Bt bacterium
Budworm, Flies, armyworm, mosquitoes
(1) One (2) Two (3) Three (4) Four

BMH193

2. Match the column-I and column-II :-

Column-I (Bioactive molecule)		Column-II (Source)	
(A)	Clot buster	(i)	*Monascus*
(B)	Statins	(ii)	*Trichoderma*
(C)	Butyric acid	(iii)	*Streptococcus*
(D)	Cyclosporin-A	(iv)	*Clostridium*

(1) a - i, b - iii, c - iv, d - ii
(2) a - iii, b - i, c - iv, d - ii
(3) a - iii, b - i, c - ii, d - iv
(4) a - iii, b - iv, c - i, d - ii

BMH194

3. Which of the following statement is incorrect:
(1) Green revolution was dependent to a large extent on plant breeding technique
(2) The greater the BOD of waste water, more is its polluting potential
(3) *Trichoderma* species are effective biocontrol agents of several plant pathogens
(4) Baculoviruses are pathogens that attack several Plant pathogen

BMH195

4. Choose the correct options :-

(A)	Dough used	Baker's yeast	.....A.....
(B)	Cyclosporin-A	.....B.....	Transgenic pigs
(C)	Anticoagulant	*Brassica napus*	.....C.....
(D)	RNA interference	.....D.....	*Meloidegyne incognitia*

Options are :-
(1) A-*Saccharomyces cerevisiae*, B-Immuno suppressive agent, C-Heparin, D-Gene silencing
(2) A-*Schizosaccharomyces*, B-Immuno suppressive agent, C-Hirudin, D-Gene silencing
(3) A-*Saccharomyces cerevisiae*, B-Clot buster, C-Hirudin, D-Gene silencing
(4) A-*Saccharomyces cerevisiae*, B-immuno suppressive agent, C-Hirudin, D-Gene silencing

BMH196

5. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : LAB grows in milk and converts milk into curd.
Reason (R) : LAB produces acid that coagulate and partially digest the milk protein.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is incorrect.
(3) (A) is not correct but (R) is correct.
(4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

BMH197

6. Identify the correct set of statements out of followings :
(a) Commercial production of citric acid is done in the presence of *Aspergillus niger* fungi
(b) First antibiotic was penicillin produced from a fungi
(c) *Trichoderma* is a free living fungi which is common in the root ecosystem
(d) Streptokinase is obtained from a bacterium called *Streptobacillus*
Choose the correct answer from the given options below.
(1) a, b, c and d (2) only a and b
(3) a, b and c (4) b, c and d

BMH198

7. Which of the following statements is not incorrect?
(1) LAB produces acids that coagulate and completely digest the milk proteins.

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

(2) A small amount of cheese is added to the fresh milk as inoculum or starter contain millions of LAB.
(3) LAB multiply at a suitable temperature, thus converting curd to milk, which also improves its nutritional quality by increasing vitamin B12 in our stomach.
(4) LAB plays very beneficial role in checking disease causing microbes.

BMH199

8. How many of the following statements are correct?
I. The dough which is used for making food such as dosa and idli is also fermented by yeast.
II. The puffed-up appearance of dough is due to the production of CO_2 gas.
III. The dough which is used for making bread, is fermented using brewer's yeast
IV. 'Toddy' a traditional drink of some parts of Southern India is made by fermenting sap from palms.
V. Microbes are also used to ferment fishes, soyabean and bamboo-roots to make foods.
(1) 4 (2) 3 (3) 2 (4) 1

BMH200

9. Which of the following is wrongly matched in the given table?

Microbe	Product	Application
1. *Trichoderma polysporum*	I. Cyclosporin A	1. Immunosuppressive
2. *Monascus purpureus*	II. Statins	2. Blood cholesterol lowering agent
3. *Streptococcus*	III. Streptokinase	3. Removal of clot from blood vessels
4. *Clostridium*	IV. Lipase	4. Removal of oil stains

(1) 1 (2) 2 (3) 3 (4) 4

BMH201

10. How many of the following are not incorrect with respect to wastewater treatment?
I. A major component of waste water is human excreta.
II. The municipal waste-water is called sewage which contains large amounts of inorganic matter and microbes. Many of which are pathogenic.
III. Sewage is treated in sewage treatment plants (STPs) to make it non-polluting.
IV. Treatment of waste-water is done by the aerobic microbes naturally present in the sewage.
(1) 4 (2) 3 (3) 2 (4) 1

BMH202

11. Which of the following statements is incorrect?
(1) The effluent from the secondary treatment plant is generally released into natural water bodies like rivers and streams.
(2) Microbes play a major role in treating millions of gallons of waste water everyday across the globe. This methodology has been practiced for more than a decade now, in almost all parts of the world.
(3) Till date, no manmade technology has been able to rival the microbial treatment of sewage.
(4) The Ministry of Environment and forests has initiated Ganga Action Plan and Yamuna Action Plan to save these major rivers of our country from pollution.

BMH203

12. How many of the following are correct with respect to mycorrhiza?
I. Fungi are also known to form symbiotic associations with plants (mycorrhiza).
II. Many members of the genus Glomus form mycorrhiza.
III. It provide resistance to root-borne pathogens.
IV. It provide tolerance to salinity and drought.
V. It helps in overall increase in plant growth and development.
(1) 5 (2) 4
(3) 3 (4) 1

BMH204

AIM : FREE EDUCATION TO ALL APUL

13. Match the items in Column-I and Column-II and choose the correct answer.

	Column-I		Column-II
(a)	Ladybird	(1)	*Methanobacterium*
(b)	Mycorrhiza	(2)	*Trichoderma*
(c)	Biological control	(3)	*Aphids*
(d)	Biogas	(4)	*Glomus*

Which of the following is the correct option?

(1) a-1, b-4, c-3, d-2

(2) a-3, b-4, c-2, d-1

(3) a-4, b-1, c-2, d-3

(4) a-3, b-2, c-1, d-4

BMH205

14. How many of the following are correct with respect to *Trichoderma*?

I. Developed for use in the treatment of plant diseases

II. It's species are free living fungi that are very common in root ecosystems.

III. It belongs to the division Ascomycetes

IV. They are very effective Biocontrol agents for several plant pathogens.

(1) 4 (2) 3 (3) 2 (4) 1

BMH206

15. Select the correct statements from the following:

(1) Activated sludge-sediment in settlement tanks of the sewage treatment plant is a rich source of aerobic bacteria.

(2) Methanobacterium is an aerobic bacterium found in the rumen of cattle

(3) Biogas is produced by the activity of aerobic bacteria on animal waste.

(4) Biogas, commonly called gobar gas is pure methane.

BMH207

16. Match the following list of microbes and their importance :

(A)	*Clostridium bulyticum*	(i)	Production of immunosuppressive agents
(B)	*Monascus purpureus*	(ii)	Ripening of Swiss cheese
(C)	*Trichoderma polysporum*	(iii)	Butyric acid
(D)	*Propionibacterium shermanii*	(iv)	Production of blood cholesterol lowering agents

(1) A-iii, B-i, C-iv, D-ii

(2) A-iii, B-iv, C-i, D-ii

(3) A-iv, B-iii, C-ii, D-i

(4) A-iv, B-ii, C-i, D-iii

BMH208

17. **Assertion :** *Trichoderma polysporum* is an important fungus.

Reason : Cyclosporin-A is obtained from this fungus which is used as an immunosuppressive agent in organ transplant.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BMH209

18. **Assertion :** Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering.

Reason : It is used as a "Clot buster" for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

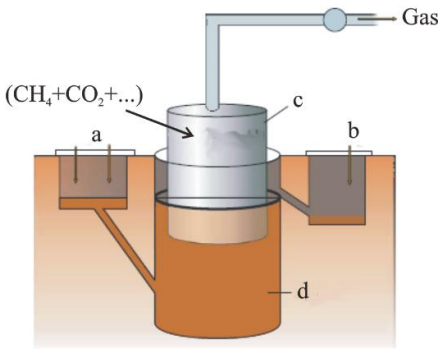
(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BMH210

AIM : FREE EDUCATION TO ALL APUL

19. Recognise the figure and find out the correct matching :-



(1) a-sludge, b-dung and water, c-digester, d-gas holder
(2) b-sludge, a-dung and water, d-digester, c-gas holder
(3) a-sludge, b-dung and water, d-digester, c-gas holder
(4) b-sludge, a-dung and water, c-digester, d-gas holder

BMH211

20. **Statement-I :** The large holes in 'Swiss-cheese' are due to production of a large amount of CO_2 by bacterium named *Propionibacterium-sharmanii*.

Statement-II : The 'Roquefort cheese' are ripened by growing a specific bacteria on them, which gives them a particular flavour.

(1) Both Statement-I and II are correct.
(2) Statement-I is correct and Statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and II is incorrect.

BMH212

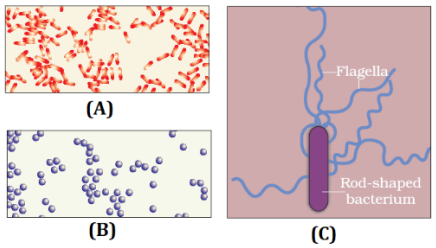
21. **Statement-I :-** Statins act by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Statement-II :- Statins are commercially used as blood cholesterol lowering agent.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect.
(3) Statement-I is correct & Statement-II is incorrect.
(4) Statement-I is incorrect & Statement-II is correct.

BMH213

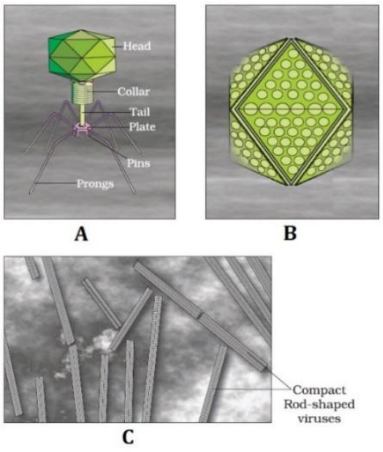
22. Identify the correct pair, in the given diagram.



(1) A = Rod shaped Bacteria – Magnified 1500X
(2) B = Spherical shaped bacteria - Magnified 50,000X
(3) C = Rod shaped Bacteria – Magnified 21,000X
(4) Both (2) and (3)

BMH214

23. Identify A, B and C in the given diagram.



	A	B	C
(1)	Bacteriophage	TMV	Adenovirus
(2)	TMV	Bacteriophage	Adenovirus
(3)	Bacteriophage	Adenovirus	TMV
(4)	Adenovirus	TMV	Bacteriophage

BMH215

24. Match the following list of bioactive substances and their roles:

Bioactive substance	Role
(i) Statin	(A) Removal of oil stains
(ii) Cyclosporin A	(B) Removal of clots from blood vessels
(iii) Streptokinase	(C) Lowering of blood cholesterol
(iv) Lipase	(D) Immunosuppressive agent

AIM : FREE EDUCATION TO ALL APUL

Choose the correct match:
(1) i-B, ii-C, iii-A, iv-D
(2) i-D, ii-B, iii-A, iv-C
(3) i-D, ii-A, iii-B, iv-C
(4) i-C, ii-D, iii-B, iv-A

BMH216

25. What would happen if oxygen availability to the activated sludge flocs is reduced?
(1) It will slow down the rate of degradation of organic matter.
(2) The centre of floes will become anoxic, which would cause the death of bacteria and eventually breakage of flocs.
(3) Flocs would increase in size as anaerobic bacteria would grow around flocs.
(4) Protozoa would grow in large numbers.

BMH217

26. **Assertion:** Greater is BOD of waste water, more is its polluting potential.
Reason: BOD is a measure of organic matter present in water.
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false

BMH218

27. Which of the following statement is correct ?
(1) Methanobacterium is an aerobic bacterium found in rumen of cattle.
(2) Biogas (commonly called gobar gas) is pure methane.
(3) Activated sludge sediment in settlement tanks of sewage treatment plants is rich source of Aerobic bacteria.
(4) Biogas is produced by the activity of aerobic bacteria on animal waste.

BMH219

28. Which one of the following is the most important role of microorganism for the well-being of humans?
(1) Sewage treatment
(2) Biological control of plant disease
(3) Conversion of milk to curd
(4) All of these

BMH220

29. Consider the following statements about organic farming.
(i) It utilizes genetically modified crops like Bt cotton.
(ii) It uses only naturally produced inputs like compost.
(iii) It does not use pesticides and urea.
(iv) It produces vegetables rich in vitamins and minerals.
Which of the above statements are correct?
(1) (ii), (iii) and (iv) (2) (iii) and (iv) only
(3) (ii) and (iii) only (4) (i) and (ii) only

BMH221

30. What are the advantages of gobar gas over conventional utilization?
(i) It is most efficient source of energy.
(ii) It is used as good fertilizer.
(iii) It reduces the chances of spreading of pathogens.
(1) (i) and (ii) only (2) (ii) and (iii) only
(3) (i) and (iii) only (4) all of these

BMH222

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	4	4	4	3	4	1	4	2	2	1	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	1	2	2	1	1	3	4	2	2	3	4	3	4

Exercise - I — Organisms & Populations

1. What is the correct order of ecological hierarchy:-
(1) Species → Genes → Family → Order → Class → Division → Kingdom
(2) Species →Community →Order → Kingdom
(3) Organism → Population → Community → Biome
(4) Organism → Community → Biosphere → Ecosystem → Biome

BOP262

2. Which of the following is an example of brood-parasitism?
(1) Orchid – Mango tree
(2) Sea anemone – Clown fish
(3) Cuckoo – Crow
(4) Fig free – Wasp

BOP263

3. The clear stratification is found in :-
(1) Tropical rain forest
(2) Temperate deciduous forest
(3) Grasslands
(4) Deserts

BOP264

4. The feeding efficiency of one species might be reduced due to the presence of the other interfering and inhibitory species, even If resources are unlimited. It is called :-
(1) Competitive release
(2) Competitive exclusion
(3) Resource partitioning
(4) Interference competition

BOP265

5. The cattle egret and grazing cattle is an example of :-
(1) Mutualism (2) Predation
(3) Parasitism (4) Commensalism

BOP266

6. A population growth becomes asymptote when population density :-
(1) is in log phase
(2) is in lag phase
(3) reaches carrying capacity
(4) crosses carrying capacity

BOP267

7. Verhulst - Pearl logistic growth pattern is :-
(1) Sigmoid (2) Parabola
(3) Hyperbola (4) J-shaped

BOP268

8. Exponential growth or geometric growth is represented by :-
(1) $N_{t+1} = N_t + [(B + I) - (D + E)]$
(2) $\frac{dN}{dt} = (b - d) \times N$
(3) $\frac{dN}{dt} = rN\left(\frac{K - N}{K}\right)$
(4) $\frac{dN}{dt} = KN$

BOP269

9. In logistic growth $\frac{dN}{dt} = rN\left(\frac{K - N}{K}\right)$ Here, $\left(\frac{K - N}{K}\right)$ represents :-
(1) Carrying capacity
(2) Intrinsic rate of natural increase
(3) Environmental resistance
(4) biotic potential and environmental resistance

BOP270

10. An 'Urn' shaped age pyramid represents :-
(1) Stable population
(2) Expanding population
(3) Declining population
(4) Threatened population

BOP271

11. Given below are two statements:
Statement-I : Immigration is the number of individuals of same species who left the habitat and gone elsewhere during the time period under consideration.
Statement-II : Natality and emigration contribute to an increase in population density.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BOP272

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

12. Given below are two statements:

Statement-I : Herbivores and plants appear to be more adversely affected by competition than carnivores.

Statement-II : Abingdon tortoise in Galapagos island became extinct within a decade after goats were introduced on island, apparently due to the greater browsing efficiency of the goat.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BOP273

13. Given below are two statements:

Statement-I : Under a particular set of selection pressures, organisms evolve towards the most efficient reproduction strategy.

Statement-II : Most of the birds and mammals breed only once in their lifetime.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BOP274

14. Given below are two statements:

Statement-I : All autotrophic plants are self-sustainable and do not need any other species for survival.

Statement-II : In nature animals, plants and microbes do not and cannot live in isolation but interact in various ways to form biological community.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

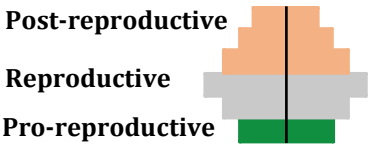
(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BOP275

15. What type of human population is represented by the following age pyramid?



(1) Stable population

(2) Declining population

(3) Expanding population

(4) Increasing population

BOP276

16. If N is the population density at time t, then its density at time t + 1 is-

(1) $N_{t+1} = N_t + (B + E) - (D + I)$

(2) $N_{t+1} = N_t + (B + D) - (E + I)$

(3) $N_{t+1} = N_t - (B + I) - (D + E)$

(4) $N_{t+1} = N_t + (B + I) - (D + E)$

BOP277

AIM : FREE EDUCATION TO ALL APUL

17. Birth rate = B, Death rate = D, Emigration = E and Immigration = I.

	Column-I		Column-II
(a)	Population is increasing	(i)	B + I = D + E
(b)	Population is decreasing	(ii)	B + I < D + E
(c)	Population is stable	(iii)	B + I > D + E

(1) a-iii, b-ii, c-i (2) a-ii, b-iii, c-i
(3) a-i, b-ii, c-iii (4) a-ii, b-i, c-iii

BOP278

18. Which of the following organism breeds only once in their life time?
(1) Pacific salmon fish (2) Oyster
(3) Bamboo (4) Both (1) and (3)

BOP279

19. (A) Predators helps in maintaining species diversity in community.
(B) It reduces the intensity of competition among competing prey species.
(1) Statement A is correct
(2) Statement B is correct
(3) Both statements are wrong
(4) Both statements are correct

BOP280

20. The feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species even if resource are abundant in known as-
(1) Interspecific predation
(2) Interfering competition
(3) Both (1) and (2)
(4) Commensalism

BOP281

21. When certain exotic species are introduced in to a geographical area they become invasive mainly because:
(1) The invaded land has unlimited resources for the introduced species.
(2) The population of the introduced species in the invaded land is very low.
(3) Introduced species do not face any competition in the introduced land.
(4) The invaded land does not have its natural predator

BOP282

22. How many statements are correct?
(a) A group of individuals resulting even from asexual reproduction is also considered a population for the purpose of ecological studies.
(b) Population ecology is an important area because it links ecology to population genetics and evolution.
(c) An individual organism has certain attributes whereas, a population does not.
(d) An individual may have births and deaths, but a population has birth rates and death rates.
(1) Two (2) Three (3) Four (4) One

BOP283

23. Select the incorrect statement from the following?
(1) The tiger census in our national parks and tiger reserves is often based on pug marks and fecal pellets.
(2) The size of a population for any species is a static parameter.
(3) Natality refers to the number of births during a given period in the population that are added to the initial population density.
(4) Emigration is the number of individuals of the population who left the habitat and gone elsewhere during the time period under consideration.

BOP284

24. **Assertion:** When resources in the habitat are unlimited, each species has the ability to realise fully its innate potential to grow in number.

Reason: When resources are limited the population grows in an exponential or geometric fashion.

(1) Both assertion and reason are true and reason is a correct explanation of the assertion.

(2) Both assertion and reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

BOP285

25. In an exponential growth equation $\frac{dN}{dt} = rN$, r represents-

(1) Intrinsic rate of natural increase

(2) (b – d)

(3) Birth rate

(4) Both (1) and (2)

BOP286

AIM : FREE EDUCATION TO ALL APUL

Exercise - I

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	1	4	4	3	1	2	3	3	1	4	2	3	2
Question	16	17	18	19	20	21	22	23	24	25					
Answer	4	1	4	4	2	4	2	2	3	4					

Exercise - II
Organisms & Populations

1. (A) If two species compete for the same resource, they could avoid competition by choosing, for instance, different times for feeding or different foraging patterns.
(B) A species whose distribution is restricted to a small geographical area due to presence of a competitively superior species is called "Competitive Exclusion Principle"
(1) Both A & B correct
(2) Only A is correct
(3) Only B is correct
(4) Both A & B incorrect

BOP287

2. The interacting organism become co-evolved in which types of interactions :-
(1) Mutualism and Competition
(2) Mutualism and Parasitism
(3) Symbiosis and Competition
(4) Parasitism and Amensalism

BOP288

3. Match list-I with list-II

List-I	**List-II**
(a) Predation	(i) Ticks on dogs
(b) Competition	(ii) Sea-anemone and Clown fish
(c) Parasitism	(iii) Fish and Flamingos
(d) Commensalism	(iv) Seeds eating sparrow

Choose the correct answer from the options given below :
(1) a-i, b-iii, c-ii, d-iv (2) a-iv, b-i, c-ii, d-iii
(3) a-iv, b-iii, c-i, d-ii (4) a-ii, b-iv, c-i, d-iii

BOP289

4. Which of the following statement is **correct**?
(1) The human liver fluke depends on one intermediate host to complete its life cycle.
(2) Parasites that feed on the external surface of the host organism are called endoparasites.
(3) Ectoparasites are those that live inside the host body at different sites.
(4) Brood parasitism in birds is a fascinating example of parasitism.

BOP290

5. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
In the light of the above statement, choose the correct answer from the options given below:
Assertion (A) : In wasp fig-association wasp larvae gets nourishment.
Assertion (R) : This is an example of mutualism.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BOP291

6. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : The interaction where one species is benefited and the other is neither benefited nor harmed, is called commensalism.
Reason (R) : An orchid growing as an epiphyte on a mango branch is an example of commensalism.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is wrong.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BOP292

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

7. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : Removal of keystone species cause serious disruption in the function of the community.

Reason (R) : Keystone species are low in abundance (or biomass) than the dominant species.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is wrong.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BOP293

8. Identify the correct set of statements :

(a) Pug marks and faecal pellets are used for measuring tiger population.

(b) Impact of biotic and abiotic factors on population growth can be assessed by studying "intrinsic rate of natural growth.

(c) When resources are not limiting, the growth plot is logistic.

(d) Logistic growth model is considered a more realistic model as compared to exponential.

(e) All population of all species in nature has its disposal unlimited resources to permit exponential growth.

Choose the correct answer from options given below :

(1) a, b and d (2) a, b and c

(3) b, d and e (4) a, b, c and d

BOP294

9. Identify the incorrect set of statements :

(a) Predators help in maintaining species diversity in a community.

(b) Predators in nature are prudent.

(c) The weed *Calotropis* produces highly poisonous cardiac glycosides.

(d) The monarch butterfly is highly distasteful to its predator because of a special chemical present in its body.

(e) A predator species is sustainable when it is highly efficient.

Choose the correct answer from options given below :

(1) a, b and e (2) a, b, c and d

(3) d and e (4) only e

BOP295

10. Identify the correct set of statements :

(a) An orchid growing as an epiphyte on a mango tree is an example of commensalism.

(b) Lichens represent an intimate mutualistic relationship between a photosynthetic fungus and saprophytic algae.

(c) Sea anemone and Clown fish show prey-predator relationship.

(d) Gause's exclusion principle justifies the co-existence of two closely related species.

Choose the correct answer from options given below :

(1) a only (2) a, b and d

(3) a, c and d (4) a, b and c

BOP296

11. Identify the correct set of statements :

(a) Primary productivity is lesser than secondary productivity.

(b) The movement of energy is always unidirectional from lower trophic level to higher trophic level in an ecosystem.

(c) Vertical distribution of different species occupying different levels is called stratification.

(d) All the organic matter produced by the plant is converted into biomass.

(e) NPP is the available biomass for the consumption to heterotrophs.

Choose the correct answer from options given below :

(1) a, c and e (2) a, b and c

(3) b, c and e (4) c, d and e

BOP297

AIM : FREE EDUCATION TO ALL APUL

12. **Assertion :-** *Cuscuta* is a parasite commonly found on hedge plants.
Reason :- *Cuscuta* has lost its chlorophyll and leaves in the course of evolution.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BOP298

13. **Assertion:** Fig tree and wasp are example of Mutualism.
Reason: Fig species can be pollinated only by its partner wasp species and wasp uses the fruit as an oviposition site and developing seeds with in the fruit for nourishing its larvae.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BOP299

14. **Assertion (A) :-** The Mediterranean orchid ophryus employs 'sexual deceit' to get pollinated by some male bees.
Reason (R) :- Plant animal interactions never involve co-evolution.
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct
(3) (A) is not correct and (R) is correct
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BOP300

15. **Assertion (A) :–** The life cycles of parasites are complex.
Reason (R) :– They involve one or two intermediate hosts or vectors to facilitate parasitism of its primary host.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BOP301

16. Which of the following are correct matching pair(s) among the options given below ?

S.N.	Species A	Species B	Name of interaction	
(i)	–	0	(A)	Mutualism
(ii)	+	–	(B)	Competition
(iii)	+	0	(C)	Parasitism
(iv)	+	+	(D)	Commensalism
(v)	–	–	(E)	Amensalism

(1) A-iii, B-v, C-i, D-iv, E-ii
(2) A-iii, B-v, C-i, D-ii, E-iv
(3) A-iv, B-v, C-ii, D-i, E-iii
(4) A-iv, B-v, C-ii, D-iii, E-i

BOP302

17. Read the following statement w.r.t. population interaction and find out correct option :
(A) Some species of insects and frogs are cryptically coloured to avoid being detected easily by the predator
(B) Lichens represent an mutualistic relationship between a fungus and photosynthesising algae.
(C) Amensalism is 0, – type of interspecific interaction
(1) Only A, C is correct (2) Only B, C is correct
(3) Only C is correct (4) All are correct

BOP303

AIM : FREE EDUCATION TO ALL APUL

18. Match the following ecological interaction with the respective examples and feature of their relationship.

(a)	Mutualism	(p)	Orchid on mango branch	(w)	–, –
(b)	Predation	(q)	Mycorrhiza	(x)	+, 0
(c)	Commensalism	(r)	Flamingos and resident fishes	(y)	+, +
(d)	Competition	(s)	Sparrow feeding on grain	(z)	+, –

(1) a → q → y ; b → r → z ; c → p → x
d → s → w

(2) a → q → z ; b → s → w ; c → p → x
d → r → y

(3) a → q → y; b → s → z ; c → p → x
d → r → w

(4) a → q → y ; b → r → z ; c → p → x
d → s → w

BOP304

19. Match Column-I with Column - II.

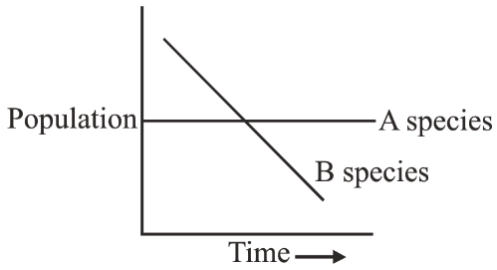
	Column-I		**Column-II**
(a)	Parasite	(i)	Symbiotic association of fungi with plant roots
(b)	Mycorrhiza	(ii)	Interaction in which both the organisms are benefited.
(c)	Mutualism	(iii)	Living on living plants or animals
(d)	Predator	(iv)	Catch, kill and eat the prey

Choose the **correct** answer from the options given below :

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(iii)	(i)	(ii)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

BOP305

20.



Which type of interaction is shown by above graph between species A and B ?

(1) Mutualism (2) Commensalism
(3) Competition (4) Amensalism

BOP306

21. **Assertion:** No population of any species in nature has at its disposal unlimited resources to permit exponential growth.
Reason: Limited resources leads to competition between individuals for limited resources. Eventually, the 'fittest' individual will survive and reproduce.

(1) Both assertion and reason are true and reason is a correct explanation of the assertion.
(2) Both assertion and reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

BOP307

22. Identify the set of correct statements:

(a) Populations evolve to minimize their reproductive fitness, also called Darwinian fitness.
(b) Pacific salmon fish, bamboo breed only once in their lifetime.
(c) Verhulst-Pearl Logistic Growth is described by the following equation: $dN/dt = rN (N-K/N)$.
(d) Oysters, pelagic fishes produce a large number of small-sized offspring.

(1) a, b and c only (2) b, c and d only
(3) b and d only (4) b only

BOP308

23. Both the species benefit in _____ and both lose in _____ in their interactions with each other.

(1) Mutualism, competition
(2) Commensalism, competition
(3) Competition, Mutualism
(4) Mutualism and parasitism

BOP309

24. Which of the following interaction is shown by visiting flamingoes and resident fishes in some shallow South American lakes.

(1) Amensalism (2) Commensalism
(3) Predation (4) Competition

BOP310

AIM : FREE EDUCATION TO ALL APUL

25. Select the incorrect statement from the following?

(1) Nearly 25 per cent of all insects are known to be phytophagous.

(2) Acacia produce and store chemicals that make the herbivore sick when they are eaten.

(3) Some species of insects and frogs are cryptically-coloured (camouflaged) to avoid being detected easily by the predator.

(4) The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body.

BOP311

26. Match List I with List II.

	List I		List II
A.	MacArthur	I	Liver fluke
B.	Endoparasites	II	Resource partitioning
C.	Brood parasitism	III	Cuckoo (koel) and the crow
D.	Barnacles and whale	IV	Commensalism

Choose the correct answer from the options given below:

(1) A-II, B-I, C-III, D-IV (2) A-I, B-II, C-III, D-IV

(3) A-II, B-III, C-II, D-IV (4) A-II, B-I, C-IV, D-III

BOP312

27. How many statements are correct?

(a) Individuals of same species living in groups in a well-defined geographical area, share or compete for similar resources, potentially interbreed and thus constitute a community.

(b) All the cormorants in a wetland forms population.

(c) One of the characteristics of a population is age sex ratio.

(d) A population at any given time is composed of individuals of the same age.

(1) One (2) Two (3) Three (4) Four

BOP313

28. Which of the following is an example of critical link species-

(1) Kangaroo rat in desert

(2) Mycorrhizal fungi

(3) Metasequoia of China

(4) Kangaroo in Australia

BOP314

29. Given below are two statements:

Statement-I: The species which have great influence on the community's characteristics relative to their low abundance or biomass are called key-stone species.

Statement-II: Community is defined as the total number of individual of a species present in a particular area at a given time.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I Statement II is true.

(4) Both Statement I Statement II is false.

BOP315

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	4	1	1	1	1	4	1	3	1	1	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	4	4	3	3	4	1	3	1	4	2	1	2	2	1	

AIM : FREE EDUCATION TO ALL APUL

Exercise - I
Ecosystem

1. The annual net primary productivity of the whole biosphere is approximately :-
(1) 70 billion tons (2) 170 billion tons
(3) 270 billion tons (4) 55 billion tons
BES253

2. GFC is the major conduit for energy flow in :-
(1) Aquatic ecosystem
(2) Terrestrial ecosystem
(3) Grassland ecosystem
(4) Cropland ecosystem
BES254

3. Pyramid of biomass in a pond and sea is :-
(1) Upright (2) Inverted
(3) Irregular (4) None of the above
BES255

4. Total amount of inorganic matter present in particular area at a particular time in an ecosystem in known as :-
(1) Standing state (2) Standing crop
(3) Standing quality (4) Both (2) & (3)
BES256

5. The rate of biomass production is called productivity. It is expressed in terms of ____.
(1) $g\ m^{-2}$ (2) $Kcal\ m^{-2}$
(3) $g\ m^{-2}\ yr^{-1}$ (4) $Kcal\ yr^{-1}$
BES257

6. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as :
(1) Fragmentation (2) Leaching
(3) Humification (4) Catabolism
BES258

7. Match list-I with list-II

List-I	List-II
(a) Producer	(i) Wolf
(b) Primary consumer	(ii) Lion
(c) Secondary consumer	(iii) Grass
(d) Tertiary consumer	(iv) Grass hopper

Choose the correct answer from the options given below :
(1) a-iii, b-ii, c-i, d-iv
(2) a-iii, b-iv, c-i, d-ii
(3) a-i, b-iv, c-ii, d-iii
(4) a-ii, b-iii, c-i, d-iv
BES259

8. Which of the following statement is incorrect?
(1) Detritivores break down detritus into smaller particle this process is called catabolism.
(2) Decomposition is largely an oxygen requiring process.
(3) Decomposition rate is slower if detritus is rich in lignin and chitin.
(4) Temperature and soil moisture are the most important climate factors that regulate decomposition.
BES260

9. Given below are two statements :
Statement-I : Plants capture only 2-10 per cent of the ISR (Incident solar Radiations).
Statement-II : Of the incident solar radiation less than 50 per cent of it is photo-synthetically active radiations (PAR).
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and Statement-II are correct
BES261

AIM : FREE EDUCATION TO ALL ♥ APUL

10. Given below are two statements :

Statement-I : Each trophic level has a certain mass of living material at a particular time called as the standing state.

Statement-II : The standing state is measured as the mass of living organisms or the number in a unit area.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are correct

BES262

11. Given below are two statements:

Statement-I : Dark coloured amorphous substance humus is highly prone to microbial action and undergoes decomposition at very fast rate.

Statement-II : Detritivores help in mineralisation and humification of detritus.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BES263

12. Given below are two statements:

Statement-I : Pyramid of energy is always upright.

Statement-II : A given species may occupy more than one trophic level in the same ecosystem at the same time.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BES264

13. _____% of incident solar radiation or _____% of PAR is captured by plants for the synthesis of organic matter (gross primary production) :-

(1) 2-10, 1-5 (2) 6-8, 12-16

(3) 30-35, 60-75 (4) 1-5, 2-10

BES265

14. The open ocean and tropical rain forest are the largest contributors to Earth's net primary productivity because :-

(1) Both have high rates of net primary productivity

(2) Both cover huge surface areas of the Earth

(3) Nutrient cycle occur fastest in these two ecosystem

(4) The ocean covers a huge surface area and the tropical rain forest has a high rate of productivity

BES266

15. How many statement is/are correct about food chain:-

(a) GFC is the major conduit of energy flow in grassland ecosystem

(b) Source of energy in detritus food chain is not directly sun

(c) Detritus food chain may be connected to grazing food chain somewhere

(d) In grazing food chain size of organism commonly decreases at higher trophic level

(1) One (2) Two

(3) Three (4) Four

BES267

AIM : FREE EDUCATION TO ALL APUL

16. **Assertion :** Complexity of food web is dependent over length of food chains
Reason : Complexity of food webs determine the stability of an ecosystem
(1) Both Assertion & Reason are true, and the reason is the correct explanation of the Assertion.
(2) Both Assertion & Reason are true, but the reason is not the correct explanation of the Assertion.
(3) Assertion is true statement, but Reason is false.
(4) Both Assertion and Reason are false statements.

BES268

17. **Assertion :** Sun is not the immediate source of energy for deep sea hydrothermal vent.
Reason : Around these vents, the food web begins with chemosynthetic bacteria rather than photosynthetic organism.
(1) Both Assertion & Reason are true, and the reason is the correct explanation of the Assertion.
(2) Both Assertion & Reason are true, but the reason is not the correct explanation of the Assertion.
(3) Assertion is true statement, but Reason is false.
(4) Both Assertion and Reason are false statements.

BES269

18. If 20J of energy is trapped at producer level, then how much energy will be available to peacock as food in following chain?
Plant → Mice → Snake → Peacock
(1) 0.02 J (2) 0.2 J
(3) 0.002 J (4) 0.0002 J

BES270

19. Given below are two statements:
Statement-I: Vertical distribution of different species occupying different levels is called stratification.
Statement-II: Interaction of biotic and abiotic components results in a physical structure that is characteristic for each type of ecosystem.
In the light of the above statements, choose the correct answer from the options given below:
(1) Statement I is true but Statement II is false.
(2) Statement I is false but Statement II is true.
(3) Both Statement I Statement II is true.
(4) Both Statement I Statement II is false.

BES271

20. **Assertion:** Of the incident solar radiation less than 50 per cent of it is photosynthetically active radiation (PAR).
Reason: Plants capture only 2-10 per cent of the PAR and this small amount of energy sustains the entire living world.
(1) Both assertion and reason are true, and reason is a correct explanation of the assertion.
(2) Both assertion and reason are true, but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

BES272

21. Given below are two statements:
Statement-I: Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis.
Statement-II: Secondary productivity is defined as the rate of formation of new organic matter by consumers.
In the light of the above statements, choose the correct answer from the options given below:
(1) Statement I is true, but Statement II is false.
(2) Statement I is false, but Statement II is true.
(3) Both Statement I Statement II are true.
(4) Both Statement I Statement II are false.

BES273

22. Which of the following statement is incorrect:
(1) Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.
(2) Secondary productivity is defined as the rate of formation of new organic matter by primary consumers.
(3) Secondary productivity depends on the plant species inhabiting a particular area.
(4) The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter.

BES274

23. The detritus food chain (DFC) begins with _____.
(1) Earthworm
(2) Ants
(3) Dead organic matter
(4) Fungi

BES275

24. Based on the source of their nutrition or food, organisms occupy a specific place in the food chain that is known as:
(1) Food chain (2) Food web
(3) Tertiary level (4) Trophic level

BES276

25. In an ecosystem if the Net Primary Productivity of first trophic level is 400 (kcal m^{-2}) yr^{-1}. What would be the net Productivity of the third trophic level of the same ecosystem?
(1) 40 (kcal m^{-2}) yr^{-1}
(2) 4 (kcal m^{-2}) yr^{-1}
(3) 0.4 (kcal m^{-2}) yr^{-1}
(4) 0.004 (kcal m^{-2}) yr^{-1}

BES277

Exercise - I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	2	1	3	4	2	1	3	1	1	4	4	4	2
Question	16	17	18	19	20	21	22	23	24	25					
Answer	2	1	1	3	2	3	3	3	4	2					

AIM : FREE EDUCATION TO ALL APUL

Exercise - II
Ecosystem

1. Humus is :-
(i) Dark coloured amorphous substance
(ii) Highly resistant to microbial action and undergoes decomposition at an extremely slow rate
(iii) Being colloidal in nature
(iv) Acts as a reservoir of nutrients
(1) i, ii are incorrect
(2) i, ii & iv are correct but iii incorrect
(3) i, ii, iii, iv are correct
(4) All are incorrect

BES278

2. Consider the following statements and choose the correct option :-
(i) Sun is the only source of energy for all ecosystem on earth
(ii) PAR is less than 50 percent of incident solar radiation
(1) only i is correct
(2) only ii is correct
(3) both i & ii are correct
(4) both are incorrect

BES279

3. There are certain limitations of ecological pyramids choose the correct option :-
(A) It assumes a simple food chain
(B) Saprophytes are not given any place in ecological pyramids.
(C) It does not take into account the same species belonging to two or more trophic levels
(1) A, B Only (2) B, C Only
(3) only B (4) All the above

BES280

4. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : The annual net primary producvity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter.
Reason (R) : Majority of this is contributed by the oceans as they occupy larger area of earth.
In the light of the above statement, choose the correct answer from the options given below:
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BES281

5. How many statements are correct?
(A) In an aquatic ecosystem, DFC is the major conduit for energy flow.
(B) The number of trophic levels in the detritus food chain is restricted as the transfer of energy follows to 10 percent law.
(C) The trophic level represents a functional level, not a species as such.
(D) Organism at each trophic level depend on those at the lower trophic level for their energy demands.
(1) One (2) Two
(3) Three (4) Four

BES282

6. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : The available biomass for the consumption to heterotrophs is known as secondary productivity.
Reason (R) : A considerable amount of GPP is utilised by plants in respiration.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is wrong.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BES283

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

7. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : Ecosystems are exempted from the second law of thermodynamics.

Reason (R) : Ecosystems need constant supply of energy to synthesize the molecule they require to counteract the universal tendency towards increasing disorderliness.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is wrong.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BES284

8. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : Number of trophic levels in a grazing food chain is restricted.

Reason (R) : Transfer of energy "from one trophic level to next" follows 10 per cent law.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is wrong.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BES285

9. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : The pyramid of biomass in sea is generally upright.

Reason (R) : Biomass of fishes far exceeds that of phytoplankton in sea.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is wrong.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BES286

10. Identify the correct set of statements :

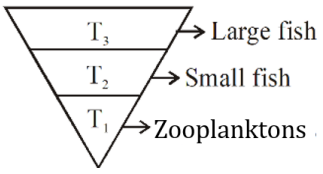
(a) Mass of living material at each trophic level in a particular time is called standing crop.
(b) Number of trophic level in the grazing food chain is restricted.
(c) In an aquatic ecosystem, GFC is the major conduit for energy flow.
(d) Inter connected food chains make food web.
(e) Detritus food chain begins with primary producers.

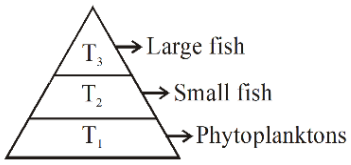
Choose the correct answer from options given below :

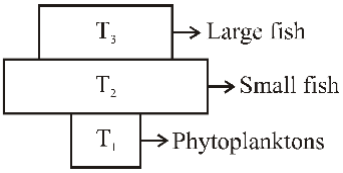
(1) a, b and c only (2) a, c and d only
(3) a, b and e only (4) a, b, c and d

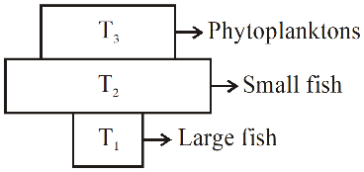
BES287

11. Which of the following option correctly represent pyramid of biomass in lake-

(1)


(2)


(3)


(4)


BES288

AIM : FREE EDUCATION TO ALL APUL

12. Which one of the following options gives the correct categorisation of six organism according to the type of consumer (A,B,C) they give out :-

	A Primary consumer	B Secondary consumer	C Tertiary consumer
(1)	Bird, Wolf	Man, Lion	Cow, Grasshopper
(2)	Man, Lion	Cow, Zooplankton	Bird, Fishes
(3)	Grasshopper, Cow	Fishes, Wolf	Man, Lion
(4)	Zooplankton, Grasshopper	Fishes, Lion	Bird, Cow

BES289

13. 

Above inverted pyramid of biomass also predict that :-

(1) Small standing crop of phytoplanktons supports large standing crop of zooplankton

(2) Organism of high biotic potential supports organism of low biotic potential.

(3) Organism of high energy (productivity)supports organism of low energy(productivity)

(4) All of the above

BES290

14. The base of each pyramid represents _____ while the apex represents _____.

(1) Producers, tertiary consumer

(2) First trophic level, top level consumer

(3) Secondary producers, consumers

(4) Both (1) and (2)

BES291

15. Given below are two statements:

Statement-I: Trophic level represents a functional level, not a species as such.

Statement-II: A given species may occupy more than one trophic level in the same ecosystem at the same time.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true, but Statement II is false.

(2) Statement I is false, but Statement II is true.

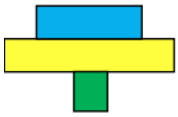
(3) Both Statement I Statement II is true.

(4) Both Statement I Statement II is false.

BES292

16. For the given food chain choose the correct pyramid for biomass:

Phytoplankton→ Zooplankton → Small fishes → Large fishes (Shark)

(1) 

(2) 

(3) 

(4) 

BES293

17. Which of the following is not included in limitations of ecological pyramids-

(1) It assumes a complex food chain.

(2) It does not accommodate a food web.

(3) Saprophytes are not given any place.

(4) All of the above

BES294

18. **Assertion:** Each trophic level has a certain mass of living material at a particular time called as the standing state.

Reason: The standing crop is measured as the mass of living organisms (biomass) or the number in a unit area.

(1) Both assertion and reason are true, and reason is a correct explanation of the assertion.

(2) Both assertion and reason are true, but reason is not a correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

BES295

AIM : FREE EDUCATION TO ALL APUL

19. How many statements are correct?
(a) Pyramid of biomass shows biotic potential of an ecosystem.
(b) Graphical representation of ecological parameters at different trophic levels in ecosystem is called food web.
(c) Those ecosystems which have simple food web are not very stable.
(d) Organisms which use milk or curd as food, are known as secondary consumer.
(1) One (2) Two (3) Three (4) Four

BES296

20. In the process of decomposition, some nutrients get tied up with the biomass of microbes and become temporarily unavailable to other organisms. Such incorporation of nutrient in living microbes (bacteria & fungi) is called _____ .
(1) Leaching
(2) Fragmentation
(3) Nutrient immobilization
(4) Humification

BES297

21. Which trophic level includes organisms that feed directly on producers?
(1) Primary consumers
(2) Secondary consumers
(3) Tertiary consumers
(4) Primary Producers

BES298

22. Which of the following is an example of a primary consumer in a terrestrial ecosystem?
(1) Grasshopper (2) Zooplankton's
(3) Snake (4) Lion

BES299

23. In an ecosystem, which of the following factors mostly favors decomposition?
(1) Warm and moist environment
(2) High pH of soil
(3) Anerobic conditions
(4) Availability of sunlight

BES300

24. Which of the following processes is responsible for the conversion of organic matter into inorganic nutrients in the soil?
(1) Energy flow (2) Decomposition
(3) Transpiration (4) Nitrification

BES301

25. In a food web, which of the following organisms would occupy the highest trophic level?
(1) Producers
(2) Primary consumers
(3) Secondary producers
(4) Tertiary consumers

BES302

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	2	2	3	3	4	3	4	1	3	1	4	3
Question	16	17	18	19	20	21	22	23	24	25					
Answer	2	1	4	2	3	1	1	1	2	4					

Exercise - I
Biodiversity and Conservation

1. Amazon rain forest are considered as lungs of the plant because they contribute _____ of the total oxygen in the earth's atmosphere :-
(1) 10% (2) 20% (3) 30% (4) 25%
BAC212

2. Species diversity in a community depends upon:-
(1) Size of area only
(2) Climatic factors only
(3) Climatic factors, type of soil & altitude
(4) Type & size of area only
BAC213

3. Exploring molecular genetic & species level diversity for products of economic importance like medicine is called :-
(1) Biofortification
(2) Biomagnification
(3) Bioprospecting
(4) Bioprocessing
BAC214

4. How many percent of the earth's land area is covered by all the biodiversity hot spots ?
(1) More than two percent
(2) Less than two percent
(3) More than five percent but less than ten percent
(4) More than two percent but less than five percent
BAC215

5. Which of the following biome has largest number of species ?
(1) Taiga
(2) Deciduous
(3) Tropical forest
(4) Subtropical forest
BAC216

6. According to Robert may, what is the total number of species present on earth ?
(1) 5 million (2) 6 million
(3) 7 million (4) 9 million
BAC217

7. Species area relationship was given by :-
(1) Robert Constranza
(2) Alexander von Humboldt
(3) Alexander Flemming
(4) Stanford ecologist Paul Ehrlich
BAC218

8. The term "The evil quartet" is related with :-
(1) Four major causes of population growth
(2) Four major causes of forest loss
(3) Four major causes of biodiversity losses
(4) Four major causes of sacred groves
BAC219

9. Ex-situ conservation methods for endangered species is :
(1) Wildlife Sanctuaries
(2) Biosphere reserves
(3) Cryopreservation
(4) National parks
BAC220

10. Which of the following represent maximum number of species among global biodiversity?
(1) Lichens (2) Fungi (3) Algae (4) Ferns
BAC221

11. The Earth summit was held in Rio de Janeiro in:
(1) 1987 (2) 1992 (3) 1990 (4) 2002
BAC222

12. Given below are two statements :
Statement-I : More than 70 percent of all the species recorded are animals.
Statement-II : Out of every 10 animals on this planet 7 are insects.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II are correct
(4) Both Statement-I and Statement-II are correct
BAC223

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

13. Given below are two statements :

Statement-I : Hot spots are regions of accelerated habitat loss.

Statement-II : Hot spots are example of ex-situ conservation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II are correct

(4) Both Statement-I and Statement-II are correct.

BAC224

14. Given below are two statements:

Statement-I : The diversity of plants and animals is uniform throughout the world.

Statement-II : In general species diversity decreases as we move away from equator towards the pole.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II both are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BAC225

15. Given below are two statements:

Statement-I : Communities with more species shows less year-to-year variation in total biomass.

Statement-II : Community with higher biodiversity shows high productivity.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BAC226

16. How many statements are correct?

(A) Rauwolfia vomitoria growing in different Himalayan ranges is an example of species diversity.

(B) For many taxonomic groups, species inventories are more complete in tropical than in temperate countries.

(C) Robert May places the global species diversity at about 7 million.

(D) Among vertebrates, insects are the most species-rich taxonomic group, making up more than 70 per cent of the total.

(1) One (2) Three

(3) Two (4) Four

BAC227

17. Match List I with List II.

	List I		**List II**
A.	Alien species invasions	I	Steller's sea cow
B.	Over-exploitation	II	*Lantana*
C.	In situ conservation	III	Firewood
D.	Narrowly utilitarian	IV	Biodiversity hotspots

Choose the correct answer from the options given below:

(1) A-II, B-I, C-III, D-IV

(2) A-I, B-III, C-IV, D-II

(3) A-II, B-I, C-IV, D-III

(4) A-I, B-III, C-II, D-IV

BAC228

AIM : FREE EDUCATION TO ALL APUL

18. Given below are two statements:

Statement-I: The IUCN Red List (2004) documents the extinction of 784 species (including 338 invertebrates, 359 vertebrates and 87 plants) in the last 500 years.

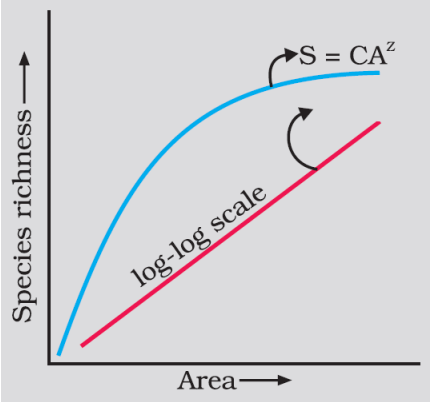
Statement-II: Over- exploitation is the most important cause driving animals and plants to extinction.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.
(2) Statement I is false but Statement II is true.
(3) Both Statement I Statement II is true.
(4) Both Statement I Statement II is false.

BAC229

19. Identify the correct description about the given figure:



(1) Species-Area relationships given by Paul Ehrlich.
(2) On log scale the relationship becomes hyperbola.
(3) The relation between species richness and area for angiosperm plants turns out to be a rectangular hyperbola.
(4) On a logarithmic scale, the relationship is a straight line described by the equation log S = log A + Z log C.

BAC230

20. Given below are two statements:

Statement-I: Communities with more species, generally, tend to be more stable than those with less species.

Statement-II: A stable community should show too much variation in productivity from year to year.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.
(2) Statement I is false but Statement II is true.
(3) Both Statement I Statement II are true.
(4) Both Statement I Statement II are false.

BAC231

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	2	3	3	2	3	3	2	2	4	2	3	4
Question	16	17	18	19	20										
Answer	1	3	4	3	1										

Exercise - II

Biodiversity and Conservation

1. Which is not correct about biodiversity :-
(1) Insects are the most species rich taxonomic group, making up more than 70 % of the total.
(2) India has only 2.4 percent of the world's land area, its share of the global species diversity is an impressive 8.1 percent.
(3) Nearly 45,000 species of plants & twice as many of animals have been recorded from India
(4) India has more than 1200 species of Bird.

BAC232

2. Match list-I with list-II

List-I	**List-II**
(a) Edward Wilson	(i) Species - Area relationships
(b) Paul Ehrlich	(ii) Diversity-productivity
(c) David Tilman	(iii) Rivet-Popper Hypothesis
(d) Alexander von Humbolt	(iv) Biodiversity

Choose the correct answer from the options given below :
(1) a-iv, b-iii, c-ii, d-i (2) a-i, b-ii, c-iii, d-iv
(3) a-iv, b-i, c-ii, d-iii (4) a-ii, b-iii, c-i, d-iv

BAC233

3. Match list-I with list-II

List-I (Taxa)	**List-II (Chances of extinction)**
(a) Gymnosperms	(i) 12%
(b) Birds	(ii) 23%
(c) Amphibians	(iii) 31%
(d) Mammals	(iv) 32%

Choose the correct answer from the options given below :
(1) a-iv, b-i, c-iii, d-ii (2) a-i, b-ii, c-iv, d-iii
(3) a-iii, b-iv, c-ii, d-i (4) a-iii, b-i, c-iv, d-ii

BAC234

4. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
In the light of the above statement, choose the correct answer from the options given below:
Assertion (A) : Tropical regions are more diversity rich in comparison to temperate areas.
Reason (R) : Availability of more solar energy directly affects the presence of more species in these areas.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BAC235

5. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
In the light of the above statement, choose the correct answer from the options given below:
Assertion (A) : Biodiversity hotspots are the regions which possess low levels of species richness and high degree of endemism.
Reason (R) : Total number of biodiversity hotspots in the world is 34 with three of these hotspots found in India.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BAC236

6. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : Loss of tropical forest will exterminate the migratory birds of cold countries.
Reason (R) : Birds of cold countries spends the winter in tropics.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is wrong.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BAC237

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

7. Identify the correct set of statements :
(a) India has more than 50,000 genetically different strains of rice and 1,000 varities of mango.
(b) Eastern ghats have a greater amphibian species diversity than western ghats.
(c) According to IUCN (2004) the total number of plant and animal species described so far is slightly more than 1.5 million.
(d) According to Robert May global species diversity is about 7 million.
Choose the correct answer from options given below :
(1) a, b and c (2) a, c and d
(3) a, b and d (4) b, c and d

BAC238

8. Identify the correct set of statements :
(a) Breakdown of large habitats into small fragments is one of the major cause of loss of biodiversity.
(b) Extinction of Steller's sea cow and Passenger pigeon is an example of co-extinctions.
(c) Introduction of nile perch into lake Victoria in east Africa led to extinction of more than 200 Cichlid fish in the lake.
(d) Loss of the biodiversity may lead to increased variability in some ecosystem processes such as plant productivity.
Choose the correct answer from options given below :
(1) a, b and c (2) a, c and d
(3) a, b and c (4) b, c and d

BAC239

9. Identify the correct set of statements :
(a) Protection of entire forest to save a tiger is an example of in-situ conservation.
(b) Biodiversity hot spot regions shows high degree of endemism and high level of species richness.
(c) The earth summit was held in Rio-de Janeiro in 1992 for the protection of ozone layer.
(d) Zoological parks and botanical gardens serves the purpose of ex-situ conservation.
Choose the correct answer from options given below :
(1) a, b and d (2) a, b and c
(3) b, c and d (4) a, c and d

BAC240

10. Given below are two statements:
Statement-I : Exploring molecular, genetic and species level diversity for products of economic importance is known as bioprospecting.
Statement-II : Aproximetely more than 25% of drugs currently sold in the market worldwide are derived from plants.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BAC241

11. Match the column I with column II

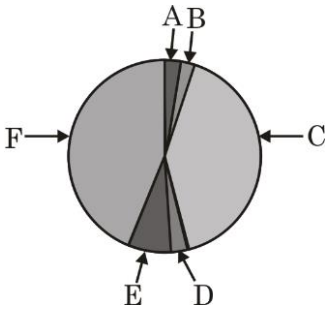
Column-I		**Column-II**	
(a)	Over-exploitation	(i)	Amazon forest
(b)	Co-extinction	(ii)	*Lantana, Eicchornia*
(c)	Habitat loss	(iii)	Plant-Pollinator
(d)	Alien species invasion	(iv)	Stellers sea cow, Passenger pigeon

(1) a – iv, b – ii, c – i, d – iii
(2) a – ii, b – iv, c – i, d – iii
(3) a – iv, b – iii, c – i, d – ii
(4) a – i, b – ii, c – iii, d – iv

BAC242

AIM : FREE EDUCATION TO ALL APUL

12. What does A, B, C, D, E and F represent in this figure?



Pie chart for number of species of major taxa of plants.

(1) A: Ferns and allies, B: Mosses, C: Algae, D: Fungi, E: Lichens, F: Angiosperms

(2) A: Fungi, B: Mosses, C: Lichens, D: Ferns and allies, E: Angiosperms, F: Algae

(3) A: Angiosperms, B: Ferns and allies, C: Fungi, D: Lichens, E: Algae, F: Mosses

(4) A: Mosses, B: Ferns and allies, C: Angiosperms, D: Lichens, E: Algae, F: Fungi

BAC243

13. (A) India has more than 50,000 genetically different strains of rice.

(B) Rauwolfia vomitoria growing in different range of western ghat show variation about concentration of reserpine.

(C) Out of 10 animals on this planet 7 are insects.

(D) Colombia has 1400 species of birds

How many above statement are correct with respect to Biodiversity?

(1) Four (2) Three (3) Two (4) One

BAC244

14. **Assertion :** A stable community should not show too much variation in the productivity from year to year.

Reason : A stable community must be resistant to invasions by the alien species.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BAC245

15. In which of the following both pairs have correct combination :-

(1) In situ conservation : Cryopreservation
Ex situ conservation : Wildlife Sanctuary

(2) In situ conservation : Seed Bank
Ex situ conservation : National Park

(3) In situ conservation : Tissue culture
Ex situ conservation : Sacred groves

(4) In situ conservation : National Park
Ex situ conservation : Botanical Garden

BAC246

16. Match the column I, II, III and IV choose the correct option for resources.

	I Conventional	**II Non conventional**	**III Renewable**	**IV Non Renewable**
a	Coal	Wind energy	Water	Species
b	Solar energy	Wind energy	Plant	Coal
c	Solar energy	Coal	Species	Natural gas
d	Hydropower	Solar energy	Wild life	Minerals

(1) Only a is correct

(2) b and c are correct

(3) a, b, c are correct

(4) a and d are correct

BAC247

17. Match the following :

(a) Ants	(i) Nearly 20,000 species
(b) Beetles	(ii) 28,000 species
(c) Fishes	(iii) 3,00,000 species
(d) Orchids	(iv) > 20,000 species

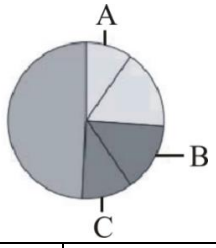
(1) a-iv, b-ii, c-iii, d-i (2) a-ii, b-iii, c-iv, d-i

(3) a-iv, b-iii, c-ii, d-i (4) a-iii, b-ii, c-i, d-iv

BAC248

AIM : FREE EDUCATION TO ALL APUL

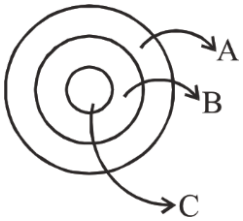
18. Identify A, B and C in the following representation of global biodiversity showing proportionate number of species of major taxa of vertebrates.



	A	B	C
(1)	Mammals	Birds	Reptiles
(2)	Mammals	Birds	Amphibians
(3)	Mammals	Reptiles	Amphibians
(4)	Fishes	Mammals	Birds

BAC249

19. The diagram given below, is of a biosphere reserve :-



Which among the following option is correct for A, B and C

(1) A - Buffer zone - Tribal huts
B - Transition zone - Tourism

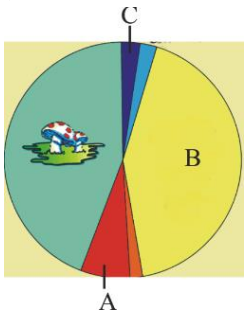
(2) B - Transition zone - Tribal huts
C - Core zone - Tourism

(3) C - Core zone - No human activities are allowed
A - Buffer zone - Tourism

(4) A - Transition zone - Tribal huts
C - Core zone - No human activities are allowed

BAC250

20. Global biodiversity is shown in terms of species in the given groups. Identify the groups - A, B and C :-



	A	B	C
(1)	Angiosperms	Algae	Mosses
(2)	Angiosperms	Algae	Fern
(3)	Algae	Angiosperms	Lichens
(4)	Algae	Angiosperms	Mosses

BAC251

21. Which of the following statement(s) are correct?

In general, loss of biodiversity in a region may lead to:

(a) Decline in plant production

(b) Lowered resistance to environmental perturbations such as drought

(c) Increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles.

(1) Only a and b (2) Only a

(3) All a, b, and c (4) None of the above

BAC252

22. Pollination (without which plants cannot give us fruits or seeds) is another service, ecosystems provide through pollinators layer–bees, bumblebees, birds and bats is an example of _____.

(1) Narrowly utilitarian (2) Broadly utilitarian

(3) In- situ approach (4) Ex-situ approach

BAC253

23. Given below are two statements:

Statement-I: Hotspots are regions of accelerated habitat loss.

Statement-II: Western Ghats and Sri Lanka, Indo-Burma and Himalaya – cover our whole world.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I Statement II is true.

(4) Both Statement I Statement II is false.

BAC254

AIM : FREE EDUCATION TO ALL APUL

24. **Assertion:** India has also a history of religious and cultural traditions that emphasized complete area for protection of nature, these areas are known as sacred groves.
Reason: In Meghalaya, the sacred groves are the last refuges for a large number of rare and threatened plants.
(1) Both assertion and reason are true, and reason is a correct explanation of the assertion.
(2) Both assertion and reason are true, but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.
BAC255

25. **Assertion:** Biodiversity knows no political boundaries and is a collective responsibility of all nations to take appropriate measures for conservation of biodiversity and sustainable utilization.
Reason: The historic Convention on Biological Diversity ('The Earth Summit') held in Rio de Janeiro in 1992.
(1) Both assertion and reason are true, and reason is a correct explanation of the assertion.
(2) Both assertion and reason are true, but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.
BAC256

26. What term describes the number of different species present in a given area?
(1) Genetic diversity
(2) Species richness
(3) Ecological niche
(4) Population diversity
BAC257

27. Which of the following is not a direct threat to biodiversity?
(1) Habitat loss
(2) Over exploitation
(3) Exotic species invasion
(4) Bioprospecting
BAC258

28. Which of the following is an example of in situ conservation method?
(1) Zoo breeding programs
(2) Botanical gardens
(3) Wildlife safari parks
(4) National parks
BAC259

29. **Assertion:** Habitat destruction is the primary cause of biodiversity loss worldwide.
Reason: Human activities such as deforestation, urbanization, and agricultural practices lead to the loss, degradation, and fragmentation of natural habitats, threatening species survival.
(1) Both assertion and reason are true, and reason is a correct explanation of the assertion.
(2) Both assertion and reason are true, but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.
BAC260

30. **Assertion:** Biodiversity is essential for maintaining ecosystem stability and resilience.
Reason: High biodiversity ensures that ecological communities can withstand and recover from environmental changes.
(1) Both assertion and reason are true, and reason is a correct explanation of the assertion.
(2) Both assertion and reason are true, but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason are false.
(4) Assertion is false but reason are true.
BAC261

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	4	2	3	4	2	2	1	4	3	4	2	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	3	4	4	3	2	1	2	2	2	4	4	1	1